Ruukkilabra
UPPER SECONDARY PHYSICS

FY3

Energy and heat

KIVIRUUKIN LUKIO

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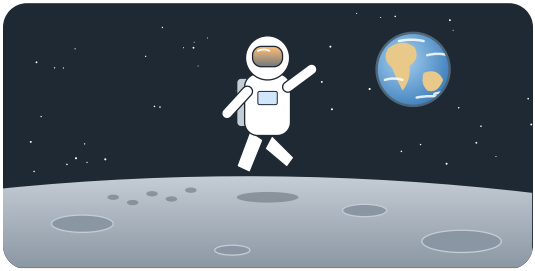
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FY3 · CHAPTER 1

Force as a measure of interaction and weight $G = mg$

Why does an astronaut weigh only one sixth as much on the Moon as on Earth, even though no part of the astronaut has gone missing?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-1.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain why force measures the strength of an interaction and how a force meter measures it in newtons.
- tell weight and mass apart.
- calculate the weight of an object from its mass on different celestial bodies.

PRIOR KNOWLEDGE

- Direct proportionality and a straight line through the origin, for example $s = vt$ (FY1)
- Everyday experience: a heavy and a light object, a stretching spring or rubber band (everyday knowledge, not a previous lesson)

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

A force meter is basically just a spring. Hang a $10\ \text{kg}$ sample box on it on Earth, and the spring stretches a certain distance. Take the same box and the same meter to the Moon: the spring stretches much less. The box contains as much matter as before, and the spring has not changed either. Something else has changed.

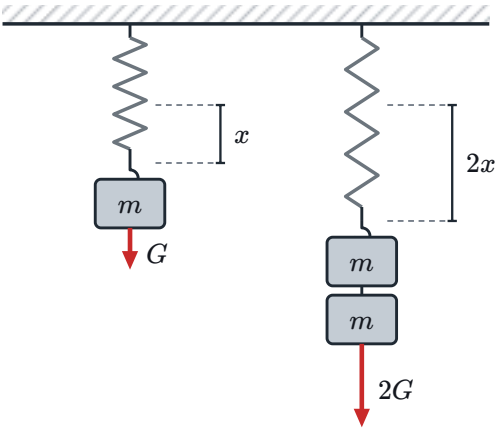
> **What does the spring of a force meter actually measure, and what changes in it when the place changes?**

Theory

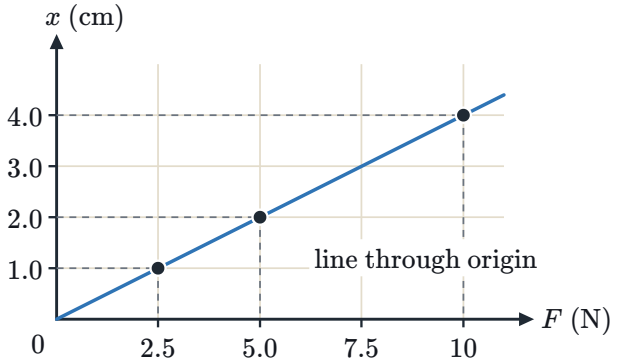
Key idea

When two objects act on each other, they interact. For example, the Earth pulls an apple towards itself. Force tells you how strong the interaction is: the greater the force, the stronger the interaction. The unit of force is the newton (N), and about $1\ \mathrm{N}$ is the weight of a hundred-gram apple on Earth. At this stage only the magnitude of the force matters, and in the figures an arrow shows the direction in which the force points.

You can measure a force with a force meter. It contains a spring that stretches more the greater the force acting on it.



The extension is directly proportional to the force as long as the spring is not stretched too far: double the force stretches the spring double the distance. The graph (force, extension) is therefore a straight line through the origin, like the directly proportional models in FY1, for example uniform motion $s = vt$. If $5.0\ \mathrm{N}$ stretches a spring by $2.0\ \mathrm{cm}$, then $10\ \mathrm{N}$ stretches it by $4.0\ \mathrm{cm}$ and $2.5\ \mathrm{N}$ stretches it by $1.0\ \mathrm{cm}$.



The scale is calibrated by hanging identical small objects on the spring, each with a mass of about $100\ \mathrm{g}$, that is, a weight of about $1\ \mathrm{N}$:

- One object stretches the spring by a certain distance, and the position is marked $1\ \mathrm{N}$.
- Each object pulls on the spring equally hard, so two objects pull with double the force and stretch it double the distance. This position is marked $2\ \mathrm{N}$, and so on.
- After that you can read the force directly in newtons from the extension.

The weight of an object is a measure of the strength of the interaction between the Earth and the object. It is a force, and its unit is the newton. Mass is something quite different: it tells you how much matter an object contains, and its unit is the kilogram. Mass is the same everywhere, but weight depends on where the object is. The Moon is a much smaller celestial body than the Earth, and the interaction between the Moon and the astronaut is weaker than between the Earth and the astronaut, so the astronaut weighs less on the Moon.



Astronaut Jessica Watkins floats in front of the Cupola window of the International Space Station, 2022. The Earth still pulls her there, but the station and the astronaut fall around the Earth together.

Photo: NASA, public domain

Weight

Two identical objects stretch the spring of a force meter twice as much as one, so weight is directly proportional to mass:

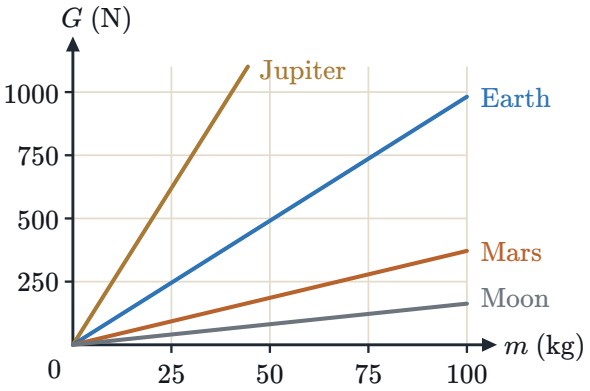
$$G = mg$$

where G is the weight (N), m the mass (kg) and g the acceleration of free fall (m/s²), $9.81\ \mathrm{m/s^2}$ on the surface of the Earth. When mass is in kilograms and g in m/s², the weight comes out in newtons, because $1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m/s^2}$ (this is justified in FY4). So the weight of a one-kilogram mass on Earth is $9.81\ \mathrm{N}$. Elsewhere g is different. The values in the table apply to the surface of the celestial body (for Jupiter, the level of the cloud tops):

Celestial body	g at the surface (m/s²)
Moon	1.62
Mars	3.71
Earth	9.81
Jupiter	24.8

The Moon's g is about one sixth of the Earth's value ($1.62 : 9.81 \approx 0.17 \approx 1 : 6$), so the same mass weighs about one sixth as much on the Moon as on Earth. This answers the question about the astronaut at the beginning.

In the graph, weight is plotted as a function of mass. For each celestial body, weight is directly proportional to mass, so the graph is a straight line through the origin. The greater g is, the steeper the line: for the same mass, weight is greatest on Jupiter and smallest on the Moon.

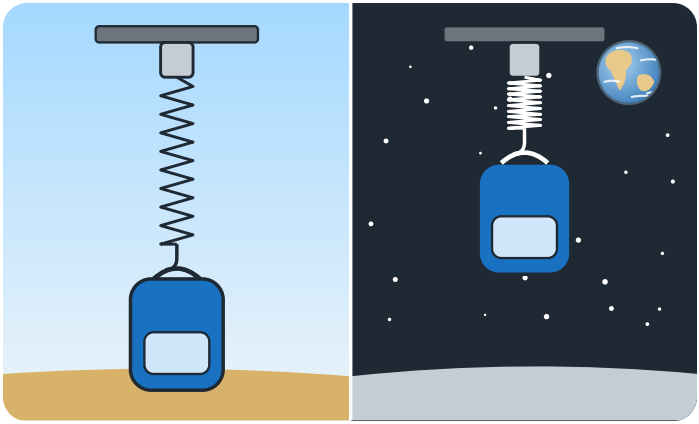


Common misconception

In everyday language people say "I weigh 70 kilos", but the kilogram is a unit of mass, not of weight. In physics the mass is $70\,\mathrm{kg}$ and the weight on Earth is about $690\,\mathrm{N}$. Another mistake is to think that an object's mass changes on the Moon. The mass stays the same, and only the weight decreases. Note also that a force meter measures force, not mass: its reading changes when the same object is taken to another place. A bathroom scale still shows kilograms, because it measures force and its scale has been made on Earth by dividing the force by $9.81\,\mathrm{m/s^2}$. On the Moon such a scale would show only about one sixth of the mass.

Summary

- Force tells you the strength of an interaction, and its unit is the newton (N). The extension of a force meter's spring is directly proportional to the force.
- Weight is a force: $G = mg$. Mass tells you how much matter an object contains, and it is the same everywhere.
- $g = 9.81\,\mathrm{m/s^2}$ on the surface of the Earth, so the weight of a one-kilogram mass on Earth is $9.81\,\mathrm{N}$. On the Moon $g = 1.62\,\mathrm{m/s^2}$ and on Mars $3.71\,\mathrm{m/s^2}$.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A robot's weight on Earth and Mars

A research robot has a mass of $1.20 \cdot 10^{2}\ \mathrm{kg}$. What is its weight on Earth and on Mars?

STEP 1/7

Given and find. The mass is $m = 1.20 \cdot 10^{2}\ \mathrm{kg}$, and from the table $g_{\mathrm{Earth}} = 9.81\ \mathrm{m/s^{2}}$ and $g_{\mathrm{Mars}} = 3.71\ \mathrm{m/s^{2}}$. We want the weights G_{Earth} and G_{Mars}.

STEP 2/7

Principle. Weight is a measure of the strength of the interaction: it is directly proportional to mass, and the constant is the g of the celestial body on whose surface the robot is. The robot's mass is the same in both places, so only g changes.

STEP 3/7

Rearranged formula. The weight we want is already alone on the left-hand side of the formula, so the formula does not need rearranging. The same mass is multiplied by the acceleration of free fall at each place:

$$G_{\text{Earth}} = mg_{\text{Earth}}$$
$$G_{\text{Mars}} = mg_{\text{Mars}}$$

STEP 4/7

Unit conversions. The mass is already in kilograms and g in m/s², so no conversions are needed: $1\,\text{kg} \cdot 1\,\text{m/s}^2 = 1\,\text{N}$, and the result comes out directly in newtons.

STEP 5/7

Substitution. The values are substituted with their units into both formulas:

$$\begin{aligned} G_{\text{Earth}} &= mg_{\text{Earth}} \\ &= 1.20 \cdot 10^2\,\text{kg} \cdot 9.81\,\text{m/s}^2 \\ &= 1177.2\,\text{N} \end{aligned}$$

$$\begin{aligned} G_{\text{Mars}} &= mg_{\text{Mars}} \\ &= 1.20 \cdot 10^2\,\text{kg} \cdot 3.71\,\text{m/s}^2 \\ &= 445.2\,\text{N} \end{aligned}$$

STEP 6/7

Result. To three significant figures, the robot's weight is about $1.18\,\text{kN}$ on Earth and about $445\,\text{N}$ on Mars. The results are given to three significant figures because the mass and the values of the free-fall acceleration are given to that precision.

STEP 7/7

Reasonableness. The weight of a one-kilogram mass on Earth is about $10\,\text{N}$, so the weight of a $120\,\text{kg}$ mass is about $1200\,\text{N}$. Mars's g is a little over one third of the Earth's value ($3.71 : 9.81 \approx 0.38$), and $445\,\text{N}$ is a little over one third of $1.18\,\text{kN}$. On Mars the weight is smaller because the interaction between Mars and the robot is weaker than between the Earth and the robot. The mass is the same in both.

FY3 · CHAPTER 2

Work and energy

> **Why** is carrying a heavy backpack along a level corridor not work in the physics sense, even though your shoulders get tired?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-2.html`



IN THIS LESSON, YOU WILL LEARN TO

- recognise when a force does work in the physics sense.
- calculate work from force and distance, and as the area under a force–distance graph.
- explain what power tells you about doing work.

PRIOR KNOWLEDGE

- Weight $G = mg$ and force as a measure of interaction
  chapter *Force as a measure of interaction and weight* $G = mg$
- Reading a graph: axes, units, slope of a straight line

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

You carry a heavy backpack along a level corridor. Your shoulders get tired and you get out of breath. A lift raises the same backpack up three floors, and it does not seem to make any effort at all. In everyday language, both *do work*.

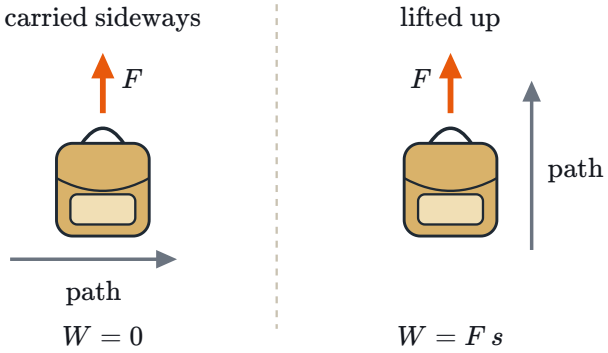
> **What should the word work mean in physics, so that the amount of work can be measured unambiguously?**

Theory

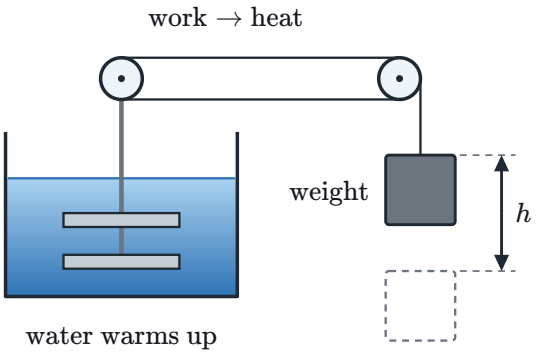
Key idea

In everyday language, work often means effort. In physics, work has a more precise meaning: a force does work when an object is displaced and the force has a component along the displacement.

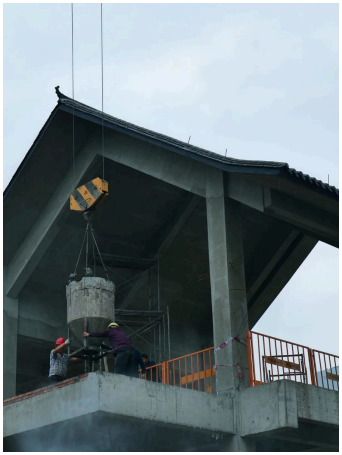
When you carry a backpack horizontally at a steady speed, your force supporting the backpack points upwards, but the backpack is displaced horizontally. The force is perpendicular to the displacement, so the mechanical work done by this force is zero. Your muscles still use energy to hold the backpack up.



Work is a way of transferring energy. James Prescott Joule showed this in the 1840s by letting a falling weight turn a paddle wheel in water. The weight did work, and the water warmed by a measurable amount even though it was not heated: the work increased the internal energy of the water. In the experiments the warming of the water turned out to be proportional to the product of the weight and the falling distance, that is, the product of force and distance. This experimental result justifies the definition of work $W = Fs$ (formula below). The same warming can also be produced by heating, so work and heat are two ways of transferring energy, and both are measured in the same unit, the joule.



When you do work on an object, energy is transferred: the energy of a lifted object increases, a spring stretches and stores energy, and work done against friction warms the surfaces. Energy does not disappear; it changes form. Power tells you how fast work is done. Walking and running up the stairs you do the same work, but when running the power is greater. The $6\,\mathrm{kW}$ power of a sauna heater means that every second it turns 6000 joules of electrical energy into internal energy, which is transferred as heat to the stones and the air.



A crane lifts a concrete bucket on a building site. As the load rises, the lifting force does work and the load's potential energy increases.

Photo: YIMING TANG, Pexels License

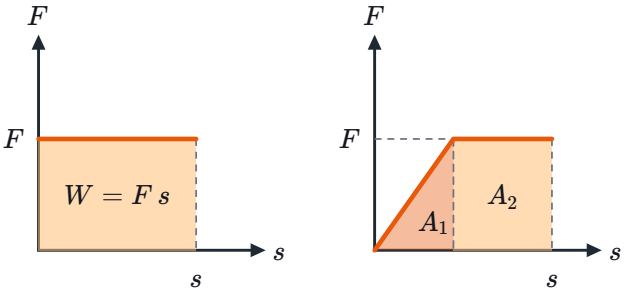
Work and power

When a constant force is in the direction of motion, the work done by the force is the product of force and distance:

$$W = Fs$$

where W is the work (J), F is the magnitude of the force (N) and s is the distance moved by the object (m). One joule is the work done by a force of one newton over a distance of one metre: $1\,\mathrm{J} = 1\,\mathrm{N}\cdot\mathrm{m}$.

If the force changes during the motion, the work is the area between the force curve and the s-axis in an (s, F) graph. For a constant force the area is a rectangle with area $F \cdot s$, so the same rule works in both cases. You can find the area of a complicated graph by dividing it into rectangles, triangles and trapeziums.

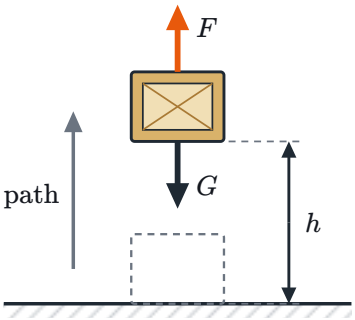


constant force varying force

When you lift an object upwards at a steady speed, the lifting force is equal to the weight of the object $G = mg$ (chapter *Force as a measure of interaction and weight* $G = mg$). The work of lifting to a height h is therefore:

$$W = Gh = mgh$$

where G is the weight (N), m is the mass (kg), $g = 9.81\ \mathrm{m/s^2}$ and h is the lifting height (m). The energy of the lifted object increases by exactly the amount of the lifting work.



$$F = G$$

$$W = F\,h$$

$$W = G\,h$$

$$W = m\,g\,h$$

Power is work divided by the time it takes. In motion at a steady speed, where the force is in the direction of motion, power can also be found as the product of force and speed, because $s/t = v$:

$$P = \frac{W}{t} = Fv$$

where P is the power (W), t is the time (s) and v is the speed (m/s). One watt is one joule per second: $1\ \mathrm{W} = 1\ \mathrm{J/s}$.

Common misconception

Work is not the same thing as effort: getting tired does not mean you have done work in the physics sense. Another common mistake is to mix up work and power: a large power does not mean a large amount of work, but work done quickly.

Summary

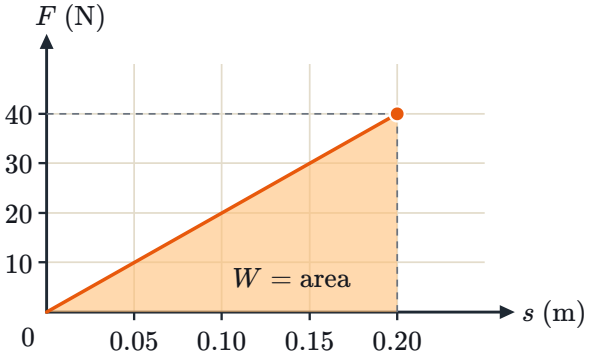
- A constant force in the direction of motion does the work $W = Fs$. For a varying force, the work is the area under the (s, F) graph.
- A force perpendicular to the motion does no work. That is why carrying something horizontally at a steady speed is not work done by the carrier's force.
- Work transfers energy: it can raise the energy of an object ($W = mgh$), stretch a spring or increase the internal energy of surfaces through friction. Joule's experiment showed that work and heat are two ways of transferring energy.
- Power is work per unit time: $P = W/t = Fv$. Units: $1\ \mathrm{J} = 1\ \mathrm{N \cdot m}$ and $1\ \mathrm{W} = 1\ \mathrm{J/s}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Work done stretching a spring

A spring is stretched $20\ \text{cm}$. The stretching force increases steadily from zero to 40 newtons, as the graph shows. How much work does the stretching force do?



STEP 1/7

Given and find. The stretch of the spring, that is, the distance moved by the force, is $s = 20\ \text{cm}$, and the greatest value of the force is $F_{\max} = 40\ \text{N}$. The force increases steadily from zero. We want the work W done by the stretching force.

STEP 2/7

Principle. Work is the product of force and distance only when the force is constant. Here the force increases steadily, so the work is found as the area under the (s, F) graph: the region between the graph and the s-axis is a triangle. The force is in the direction of motion, so the work is positive.

STEP 3/7

Rearranged formula. The force is not constant, so the formula $W = Fs$ cannot be used directly. The work is the area A_{triangle} of the triangle in the (s, F) graph, which is half of the product of the base s and the height $F_{\max}$:

$$\begin{aligned} W &= A_{\text{triangle}} \\ &= \frac{1}{2} F_{\max}\, s \end{aligned}$$

STEP 4/7

Unit conversions. The distance is in centimetres, but work is calculated in joules, so the distance is converted to metres: $s = 20\,\mathrm{cm} = 0.20\,\mathrm{m}$. The force is already in newtons, so the product $\mathrm{N} \cdot \mathrm{m}$ is directly in joules.

STEP 5/7

Substitution. We substitute the values with their units:

$$\begin{aligned} W &= A_{\text{triangle}} \\ &= \frac{1}{2} F_{\max}\, s \\ &= \frac{1}{2} \cdot 40\,\mathrm{N} \cdot 0.20\,\mathrm{m} \\ &= 4.0\,\mathrm{J} \end{aligned}$$

STEP 6/7

Result. The stretching force does $4.0\,\mathrm{J}$ of work, two significant figures like the given values $20\,\mathrm{cm}$ and $40\,\mathrm{N}$. The work does not disappear; the energy is stored in the stretched spring.

STEP 7/7

Reasonableness. The force increases steadily, so its average magnitude is half of the peak value, $F_{\text{avg}} = 20\,\mathrm{N}$. The average force gives the same result:

$$\begin{aligned} W &= F_{\text{avg}}\, s \\ &= 20\,\mathrm{N} \cdot 0.20\,\mathrm{m} \\ &= 4.0\,\mathrm{J} \end{aligned}$$

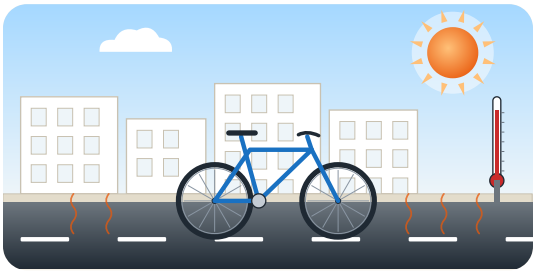
If you calculated $40\,\mathrm{N} \cdot 0.20\,\mathrm{m} = 8.0\,\mathrm{J}$, the result would be twice too large, because the force is not $40\,\mathrm{N}$ over the whole distance.

FY3 · CHAPTER 3

Thermodynamic system, state variables and temperature

Why does a bicycle tyre feel harder on a hot summer day than on a frosty day, even though no more air has been pumped into it?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-3.html`



IN THIS LESSON, YOU WILL LEARN TO

- describe a gas as a thermodynamic system whose state is given by temperature, pressure and volume.
- use the micro model to explain what pressure and temperature are.
- justify why temperatures are compared in kelvins from absolute zero.

PRIOR KNOWLEDGE

- Celsius scale: water freezes at $0\,^{\circ}\mathrm{C}$ and boils at $100\,^{\circ}\mathrm{C}$ at normal pressure (lower secondary school physics)

- Pumping extra air into a tyre makes it harder (everyday experience)

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

A bicycle is left outside in a frosty night, and in the morning the tyre gives way under your thumb. In the afternoon the same bicycle has stood in the sun, and the tyre is hard. The valve has been closed all the time, so the tyre holds the same air, and the rubber stretches so little that the volume of the tyre stays almost the same. The only thing that has changed is the temperature of the air.

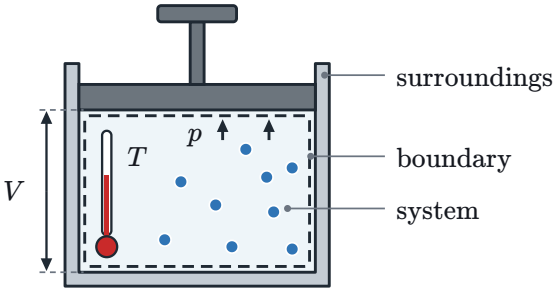
> **What happens to the air inside the tyre when it warms up, and why does the pressure of the air in the tyre then increase?**

Theory

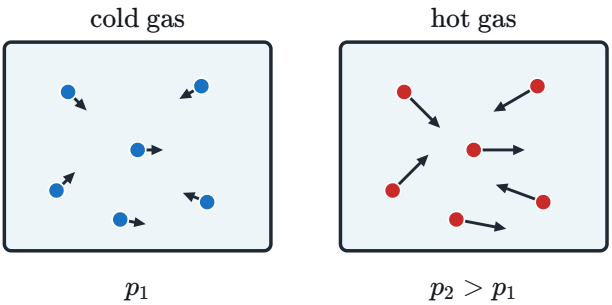
Key idea

When you study a phenomenon, you separate the part of the world you are looking at from the rest. The separated part is the thermodynamic system, and everything outside it is the surroundings. The air in a bicycle tyre is an example of a system. If no matter can leave or enter the system, the system is closed. The state of a system is described by state variables: temperature, pressure and volume, and the amount of substance. Pressure is force per unit area; its unit is the pascal (Pa), and

$1\,\mathrm{kPa} = 1000\,\mathrm{Pa}$. Pressure is studied in more detail later. When you change one state variable of a gas system, at least one of the others changes too.



The micro model of matter explains why this happens. A gas consists of particles that move all the time in all directions and collide with each other and with the walls of the container. The temperature of a gas tells you the average kinetic energy of the particles: the hotter the gas, the faster the particles move on average. Pressure arises when particles hit the wall and each collision pushes the wall a little. When the volume decreases and the temperature stays the same, the particles have less room and hit the wall more often, so the pressure increases, but a single collision does not become harder. When the temperature rises but the volume stays the same, the particles hit the wall both more often and harder, so the pressure increases. This is why the tyre feels harder in summer.



The figure is schematic: the length of an arrow shows the speed of a particle. The containers are the same size and contain the same number of particles.

Temperature is measured on the Celsius scale, where $0\,^\circ\mathrm{C}$ is the freezing point of water and $100\,^\circ\mathrm{C}$ is the boiling point of water at normal pressure, and on the Kelvin scale. The zero of the Kelvin scale is absolute zero, the lowest possible temperature. At absolute zero the average kinetic energy of the particles is at its

smallest, and nothing can be colder. A Celsius degree and a kelvin are the same size, so a temperature difference is the same in kelvins and in degrees Celsius. When the volume and the amount of gas stay the same and the gas is dilute, that is, far from liquefying, the pressure of the gas is directly proportional to the temperature in kelvins. The average kinetic energy of the particles of a dilute gas is also directly proportional to the temperature in kelvins, so ratios of temperatures are compared in kelvins. The connection between pressure, volume and temperature is studied in more detail in the chapter *Changes of state of a gas and Boyle's law*.



A thermometer shows its own temperature. That is why the thermometer must be in thermal equilibrium with the object before the reading tells the object's temperature.

Photo: Maksim Goncharenok, Pexels License



Liquid nitrogen poured from a glass dish at a science festival in Katowice, Poland. Nitrogen boils at a very low temperature, and the white mist is tiny water droplets that condense from water vapour in the cold air.

Photo: Adrian Tync, CC BY-SA 4.0

Limits of the model

The statement "temperature tells you the average kinetic energy of the particles" holds as such for a dilute gas whose particles interact with each other only when they collide. Then the temperature is proportional to the average kinetic energy of the translational motion of the particles. In a liquid and in a solid the particles are close to each other and vibrate, pulled by their neighbours. Temperature describes the strength of the thermal motion of the particles in them too, but the energy of the substance also includes the energy of the interactions between the particles. So temperature is not the same thing as the energy of a substance: ice and water can have the same temperature $0\,^\circ\mathrm{C}$, although water of the same mass has more internal energy than ice (chapters *Internal energy, heat and conservation of energy* and *Changes of state*). Even at absolute zero not all motion of the particles stops, but its energy is the smallest possible.

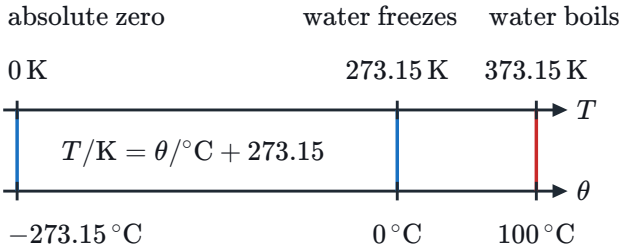
Kelvin scale

The numerical values of the Celsius temperature θ and the kelvin temperature T are related by the equation

$$\frac{T}{\mathrm{K}} = \frac{\theta}{^\circ\mathrm{C}} + 273.15$$

For example, $20\ ^\circ\mathrm{C}$ corresponds to $293.15\ \mathrm{K}$. A temperature difference, on the other hand, has the same numerical value in kelvins and in degrees Celsius: a rise from $20\ ^\circ\mathrm{C}$ to $30\ ^\circ\mathrm{C}$ is $10\ \mathrm{K}$. In practice the number 273 is often enough. For kelvins you do not use the word degree: you say "300 kelvins", not "300 degrees kelvin".

Absolute zero is therefore $0\ \mathrm{K} = -273.15\ ^\circ\mathrm{C}$. A temperature in kelvins cannot be negative, but a Celsius temperature can.



The figure is drawn to scale. The upper scale is the Kelvin scale and the lower scale is the Celsius scale; the interval from $0\ ^\circ\mathrm{C}$ to $100\ ^\circ\mathrm{C}$ is as long as the interval from $273.15\ \mathrm{K}$ to $373.15\ \mathrm{K}$.

Common misconception

Many people think that the temperature doubles when it rises from $10\ ^\circ\mathrm{C}$ to $20\ ^\circ\mathrm{C}$. The physically correct comparison is made in kelvins: from $283\ \mathrm{K}$ to $293\ \mathrm{K}$. The ratio of the temperatures is 293/283 ≈ 1.035, so the increase is about $3.5\ \%$. This is why the average kinetic energy of the particles does not double. Another mistake is to think that $0\ ^\circ\mathrm{C}$ is the lowest possible temperature. It is only the freezing point of water, and the lowest possible temperature is $0\ \mathrm{K}$.

Summary

- A thermodynamic system is a separated part of the world. Its state is described by state variables: temperature, pressure, volume and amount of substance.
- In a gas system, temperature, pressure and volume depend on each other: if you change one, at least one of the others changes too.

- Micro model of a gas: temperature tells you the average kinetic energy of the particles, and pressure arises from the collisions of the particles with the wall. In a liquid and in a solid the energy also includes the energy of the interactions between the particles, so temperature is not the same thing as the energy of a substance.
- When the volume and the amount of substance stay the same, the pressure of a dilute gas is directly proportional to the temperature in kelvins. Continuing the line through the data points, that is, extrapolation, gives an estimate of absolute zero.
- The conversion is $\frac{T}{\mathrm{K}} = \frac{\theta}{^\circ\mathrm{C}} + 273.15$. Absolute zero is $0\ \mathrm{K} = -273.15\ ^\circ\mathrm{C}$.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Where the pressure reaches zero

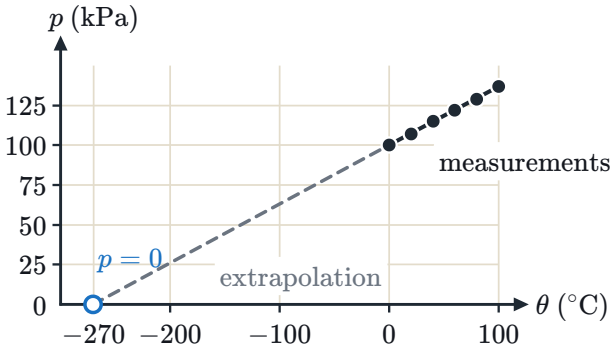
A closed rigid container holds gas, so its volume and the amount of substance stay the same. The pressure of the gas is measured at different temperatures:

θ (°C)	0	20	40	60	80	100
p (kPa)	100	107	115	122	129	137

Use extrapolation to find the temperature at which the pressure of the gas would be zero.

STEP 1/7

Given and find. The table has six data points $(\theta,\ p)$, and the pressure is $p_0 = 100\,\text{kPa}$ at $0\,^\circ\text{C}$. You are asked for the temperature θ_0 at which the pressure would be zero. On a graph the points lie on a straight line: for every $20\,^\circ\text{C}$ the pressure increases by $7\text{–}8\,\text{kPa}$. The line is continued to the left, outside the measured range, that is, extrapolated.



The vertical axis is drawn at the left edge of the figure, at about $-300\,^\circ\text{C}$, and not at $\theta = 0$. The solid line is the measured range and the dashed line is the extrapolation.

STEP 2/7

Principle. The container is closed and rigid, so the amount of gas and the volume stay the same, and the pressure depends only on the temperature. The data points are on a straight line, so the pressure changes steadily with the temperature. The equation of the line gives an estimate of the pressure also outside the measured range, if you assume that the dependence stays the same. The pressure is zero where the line crosses the temperature axis.

STEP 3/7

Rearranged formula. The slope k is calculated from two points of the line that are far apart. Here the data points lie on the line within the reading accuracy, so the end points of the measured range are also points of the line. In the equation of the line, p_0 is the pressure at $0\,^\circ\mathrm{C}$. Set $p = 0$ and solve for θ_0:

$$
\begin{aligned}
k &= \frac{\Delta p}{\Delta \theta} \\
p &= p_0 + k\theta \\
0 &= p_0 + k\theta_0 \\
\theta_0 &= -\frac{p_0}{k}
\end{aligned}
$$

STEP 4/7

Unit conversions. No conversions are needed. The pressures are already in kilopascals, and the temperature difference $\Delta\theta$ is the same in degrees Celsius and in kelvins, so the unit of the slope $\frac{\mathrm{kPa}}{^\circ\mathrm{C}}$ is also $\frac{\mathrm{kPa}}{\mathrm{K}}$. The result comes out directly in degrees Celsius, because the temperatures in the table are in degrees Celsius.

STEP 5/7

Substitution. First calculate the slope and then the zero point:

$$\begin{aligned} k &= \frac{\Delta p}{\Delta \theta} \\ &= \frac{(137 - 100)\,\mathrm{kPa}}{(100 - 0)\,^{\circ}\mathrm{C}} \\ &= 0.37\,\tfrac{\mathrm{kPa}}{^{\circ}\mathrm{C}} \end{aligned}$$

$$\begin{aligned} \theta_0 &= -\frac{p_0}{k} \\ &= -\frac{100\,\mathrm{kPa}}{0.37\,\frac{\mathrm{kPa}}{^{\circ}\mathrm{C}}} \\ &= -270.27\,^{\circ}\mathrm{C} \end{aligned}$$

STEP 6/7

Result. The temperature at zero pressure is about $-270\,^{\circ}\mathrm{C}$. The result is given with two significant figures, because the slope $0.37\,\frac{\mathrm{kPa}}{^{\circ}\mathrm{C}}$ is known only to two figures.

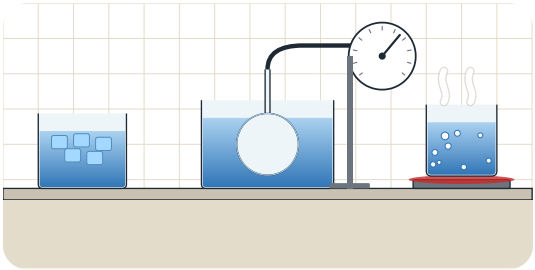
STEP 7/7

Reasonableness. The result is close to absolute zero, $-273.15\,^{\circ}\mathrm{C}$: only about $3\,^{\circ}\mathrm{C}$, or one per cent, away from it. An independent check: if the pressure is proportional to the temperature in kelvins, $p = 100\,\mathrm{kPa}$ corresponds to $273.15\,\mathrm{K}$, so the pressure at $100\,^{\circ}\mathrm{C}$ would be $100\,\mathrm{kPa} \cdot \frac{373.15}{273.15} = 136.6\,\mathrm{kPa}$, and the measured value is $137\,\mathrm{kPa}$. According to the micro model, at zero pressure the motion of the particles would be so small that no pressure would arise any more. Real gases liquefy before this temperature is reached, so the continuation of the line is only a model and describes a limiting value.

Determining absolute zero

Why can the lowest possible temperature be found in a school laboratory, even though a thermometer does not come anywhere near it?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-13.html`



IN THIS LESSON, YOU WILL LEARN TO

- turn measurement data into a straight line and extend the line to zero pressure.
- determine absolute zero experimentally by extrapolation.
- estimate the uncertainty of a result using the steepest and the shallowest line that fit.

PRIOR KNOWLEDGE

- When the volume and the amount of gas stay the same, the pressure of a dilute gas is proportional to the temperature in kelvins, and absolute zero is the zero point of the kelvin scale
  chapter *Thermodynamic system, state variables and temperature*
- The temperature of a gas tells you the average kinetic energy of the particles, and pressure comes from particles colliding with the wall
  chapter *Thermodynamic system, state variables and temperature*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

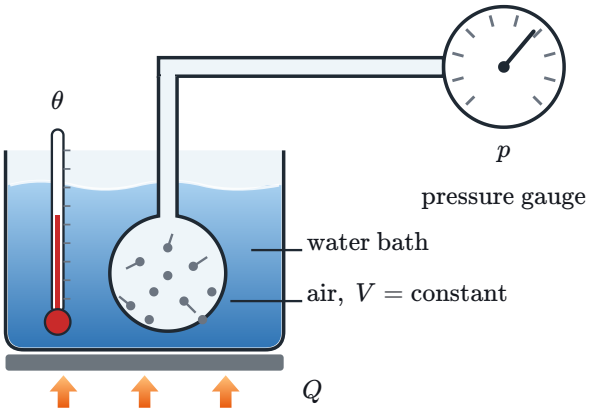
On a frosty night, a car tyre is softer in the morning than it was the previous afternoon: when the air cools, its pressure decreases. Could the pressure keep decreasing if the cooling continued? In a school laboratory, you rarely get below $-20\,^{\circ}\mathrm{C}$, and yet physics knows at which temperature the pressure of a gas would be zero.

> **How can the lowest possible temperature be determined from measurements made at quite ordinary temperatures?**

Theory

Key idea

In a closed rigid container, the volume and the amount of gas do not change, so the pressure depends only on temperature. In a gas thermometer, a glass flask stands in a water bath. As the bath warms up, you read the bath temperature θ from a thermometer and the gas pressure p from a pressure gauge.



When you plot the measurements on a graph, the points lie almost on a straight line: the pressure rises by almost the same amount for every degree Celsius. In real data sets the points are usually not exactly on one line, because of measurement uncertainty. Measurement uncertainty describes the uncertainty attached to a measurement result. It does not usually mean an absolute maximum possible deviation from the true value. Here the uncertainty of the pressure reading is estimated as $\pm 1\ \mathrm{kPa}$. The error limit that a manufacturer states for a gauge is a different thing: it is a property of the device, and the uncertainty of the measurement is also affected by, for example, reading the scale and the bath temperature settling. So you draw the line through the middle of the group of points, so that it passes through the uncertainty bars, not necessarily through every point.



Liquid nitrogen in a glass dish at a science festival in 2025. The white mist is water droplets condensed around the cold nitrogen vapour.

Photo: Adrian Tync, CC BY-SA 4.0

You extend the line beyond the measurement range towards colder temperatures until the pressure is zero. This is called extrapolation: the result comes from a region where nothing has been measured. According to the microscopic model, pressure comes from particles colliding with the wall. In the ideal gas model, the pressure would be zero at the temperature where the average kinetic energy of the particles is at its lowest. The temperature where the line meets zero pressure is the experiment's estimate of absolute zero. It is only an estimate: a real gas liquefies before that point, and the extension is a model.

Temperature at zero pressure

Within the measurement range, the pressure of a gas is a linear function of temperature:

$$p = p_0 + k\theta$$

where p is the pressure (kPa), p_0 is the pressure of the line at $0\ ^\circ\mathrm{C}$ (kPa), k is the slope of the line (kPa/°C) and θ is the temperature in degrees Celsius. You calculate the slope from two points read far apart from the drawn line:

$$k = \frac{p_2 - p_1}{\theta_2 - \theta_1}$$

where (θ_1, p_1) and (θ_2, p_2) are points on the line. Zero pressure is reached when $p = 0$. The temperature is then θ_0, and the equation gives:

$$0 = p_0 + k\theta_0$$
$$\theta_0 = -\frac{p_0}{k}$$

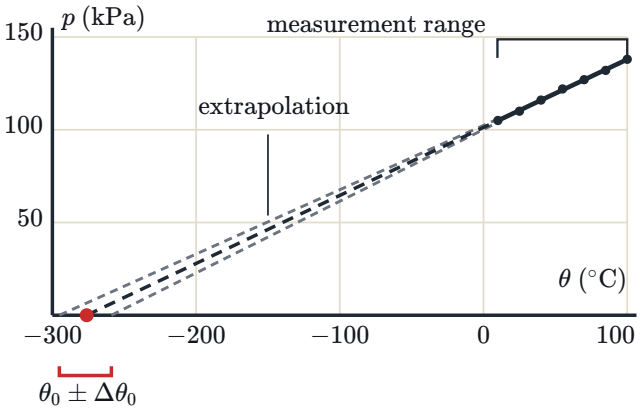
You get the same result from any point (θ_1, p_1) on the line by substituting $p = 0$ into the equation $p = p_1 + k(\theta - \theta_1)$:

$$\theta_0 = \theta_1 - \frac{p_1}{k}$$

You estimate the uncertainty by drawing two limit lines that still pass through all the uncertainty bars: the steepest and the shallowest. For each you calculate the temperature at zero pressure, and the smaller and larger values give an interval. You give the result using the midpoint and half of the interval:

$$\Delta\theta_0 = \frac{\theta_{0,\max} - \theta_{0,\min}}{2}$$

where $\theta_{0,\max}$ is the larger (less negative) and $\theta_{0,\min}$ the smaller temperature at zero pressure. The result has the form $\theta_0 \pm \Delta\theta_0$. The interval is an estimate, not a guaranteed limit: it takes into account the uncertainty of the points but not whether the dependence stays linear outside the measurement range.



The picture is drawn to scale. The limit lines run close to each other in the measurement range but move apart when you extrapolate, so the uncertainty grows as you extend the line. A more precise pressure gauge, more measurement points and a wider measurement range reduce the uncertainty. Once the result is known, the zero of the kelvin scale is placed there: the temperature in kelvins is $T = \theta - \theta_0$.

Common misconception

Many people think that absolute zero has been measured because it is on the graph. Only the seven points in the range 10–$-100\ ^\circ\mathrm{C}$ are measured; the rest is an extension of the line, and the extension is an assumption that the pressure behaves in the same way at colder temperatures. Another mistake is to consider the result a failure if it is not exactly $-273.15\ ^\circ\mathrm{C}$. The result $-276\ ^\circ\mathrm{C}$ differs from the accepted value by about 3 degrees when the uncertainty is 18 degrees, so it agrees with the value. A third mistake is to draw a wiggly line through every point: the variation of the points up and down is caused by measurement uncertainty, and the line should be drawn through the middle of the group of points.

Summary

- In a closed rigid container, the pressure of a dilute gas is a linear function of temperature within the measurement range: $p = p_0 + k\theta$. You turn the measurement data into a straight line by plotting p as a function of temperature and drawing the line through the middle of the group of points.
- Measurement uncertainty describes the uncertainty attached to a measurement result, and it is not usually an absolute maximum possible deviation from the true value. The line does not have to hit every point, only pass through the uncertainty bars.
- Extrapolation extends the line beyond the measurement range: the temperature at zero pressure is $\theta_0 = -p_0/k$.
- You estimate the uncertainty from the limit lines: $\Delta\theta_0 = (\theta_{0,\max} - \theta_{0,\min})/2$. The result $\theta_0 \pm \Delta\theta_0$ agrees with the accepted value $-273.15\ ^\circ\mathrm{C}$ if that value lies in the interval.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Absolute zero from measurements

The example data give the pressure of the air in a sealed flask at seven temperatures, as a gas thermometer would measure them. The values have been calculated from the ideal gas model, and scatter corresponding to measurement imprecision has been added; they are not the results of a real measurement. The uncertainty of the pressure reading is estimated as $\pm 1\,\mathrm{kPa}$.

θ (°C)	p (kPa)
10	105
25	110
40	116
55	122
70	127
85	132
100	138

Determine absolute zero by extrapolation and estimate the uncertainty of the result. The best-fit line through the points is in the figure above; its steepest and shallowest alternatives, which still pass through all the uncertainty bars, are $p = 100.1\,\mathrm{kPa} + 0.387\,\frac{\mathrm{kPa}}{^\circ\mathrm{C}} \cdot \theta$ and $p = 102.3\,\mathrm{kPa} + 0.347\,\frac{\mathrm{kPa}}{^\circ\mathrm{C}} \cdot \theta$. The steepest touches the ends of the uncertainty bars at $10\,^\circ\mathrm{C}$ and $85\,^\circ\mathrm{C}$, the shallowest at $25\,^\circ\mathrm{C}$ and $100\,^\circ\mathrm{C}$.

STEP 1/7

Given and find. Two points far apart are read from the best-fit line: $(\theta_1, p_1) = (0\,^\circ\mathrm{C}; 101.3\,\mathrm{kPa})$ and $(\theta_2, p_2) = (100\,^\circ\mathrm{C}; 138.0\,\mathrm{kPa})$. The temperature $0\,^\circ\mathrm{C}$ is outside the measurement range, so point 1 is read from the extension of the line: it is not a measurement but already a little extrapolation. The limit lines have $p_0 = 100.1\,\mathrm{kPa}$, $k = 0.387\,\frac{\mathrm{kPa}}{^\circ\mathrm{C}}$ and $p_0 = 102.3\,\mathrm{kPa}$, $k = 0.347\,\frac{\mathrm{kPa}}{^\circ\mathrm{C}}$. You are asked for the temperature at zero pressure θ_0 and its uncertainty $\Delta\theta_0$.

STEP 2/7

Principle. The flask is sealed and rigid, so the amount and volume of the gas stay constant, and within the measurement range the pressure is a linear function of temperature, $p = p_0 + k\theta$. So you can extend the line to the point where $p = 0$. The uncertainty comes from the limit lines: they pass through all the uncertainty bars, so they fit the data just as well as the best-fit line, and their zero-pressure points bound the interval where the true absolute zero probably lies.

STEP 3/7

Rearranged formula. The slope, the temperature at zero pressure (with $p_0 = p_1$, because point 1 is at $0\,^\circ\mathrm{C}$) and the uncertainty as half of the interval:

$$k = \frac{p_2 - p_1}{\theta_2 - \theta_1}$$

$$\theta_0 = -\frac{p_0}{k}$$

$$\Delta\theta_0 = \frac{\theta_{0,\max} - \theta_{0,\min}}{2}$$

STEP 4/7

Unit conversions. No conversions are needed: the pressures are in kilopascals, so in the quotient p_0/k the kPa cancels and what is left is the degree Celsius, which is what you are asked for. The temperature difference $100\,^\circ\mathrm{C} - 0\,^\circ\mathrm{C} = 100\,^\circ\mathrm{C}$ is the same in kelvins, and the temperature at zero pressure is given in degrees Celsius.

STEP 5/7

Substitution. First calculate the slope and then the temperature at zero pressure:

$$
\begin{aligned}
k &= \frac{p_2 - p_1}{\theta_2 - \theta_1} \\
&= \frac{(138.0 - 101.3)\,\mathrm{kPa}}{(100 - 0)\,^\circ\mathrm{C}} \\
&= 0.367\ \tfrac{\mathrm{kPa}}{^\circ\mathrm{C}}
\end{aligned}
$$

$$
\begin{aligned}
\theta_0 &= -\frac{p_0}{k} \\
&= -\frac{101.3\,\mathrm{kPa}}{0.367\,\mathrm{kPa/^\circ C}} \\
&= -276.02\,^\circ\mathrm{C}
\end{aligned}
$$

For the limit lines

$$
\begin{aligned}
\theta_{0,\max} &= -\tfrac{100.1}{0.387}\,^\circ\mathrm{C} \\
&= -258.66\,^\circ\mathrm{C} \\
\\
\theta_{0,\min} &= -\tfrac{102.3}{0.347}\,^\circ\mathrm{C} \\
&= -294.81\,^\circ\mathrm{C}
\end{aligned}
$$

so

$$
\begin{aligned}
\Delta\theta_0 &= \tfrac{-258.66-(-294.81)}{2}\,^\circ\mathrm{C} \\
&= 18.08\,^\circ\mathrm{C}
\end{aligned}
$$

STEP 6/7

Result. Absolute zero is $\theta_0 = -276 \pm 18\,^\circ\mathrm{C}$. The uncertainty is given with two significant figures ($18.08 \approx 18$), because the zero points of the limit lines, $-259\,^\circ\mathrm{C}$ and $-295\,^\circ\mathrm{C}$, are known to three figures and their difference $36\,^\circ\mathrm{C}$ to only two. The result is rounded to the same precision, that is, to one degree.

STEP 7/7

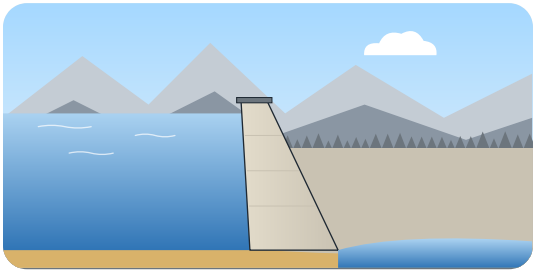
Reasonableness. The accepted value $-273.15\,^{\circ}\mathrm{C}$ lies in the interval $-294\,^{\circ}\mathrm{C} \ldots -258\,^{\circ}\mathrm{C}$, so the result agrees with it; the deviation is about $3\,^{\circ}\mathrm{C}$ and the uncertainty is $18\,^{\circ}\mathrm{C}$. An estimate in your head: the pressure rises by about $0.37\,\mathrm{kPa}$ per degree, so $101\,\mathrm{kPa}$ runs out after about $101/0.37 \approx 270$ degrees. The uncertainty is large because the measurement was made more than $270\,^{\circ}\mathrm{C}$ away from zero pressure, and the uncertainty of the points, $1\,\mathrm{kPa}$, is about one percent of the reading.

FY3 · CHAPTER 5

Pressure and hydrostatic pressure

Why is the wall of a dam much thicker at the bottom than at the top, even though the water pressing on it is the same water all the way down?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-4.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate pressure from force and area and explain how a force is spread over a surface.
- calculate the pressure of a liquid at different depths.
- interpret the constant term and the slope of a graph of pressure against depth.

PRIOR KNOWLEDGE

- Weight is a force, $G = mg$, and $g = 9.81\ \mathrm{m/s^2}$
  chapter *Force as a measure of interaction and weight* $G = mg$

- Pressure is a state variable that arises in the micro model from particles colliding with the wall
  chapter *Thermodynamic system, state variables and temperature*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

You dive towards the bottom of a swimming-pool diving pool. Already at a depth of a couple of metres you feel pressure in your ears, and the lower you go, the more clearly you feel it. At the surface your ears feel nothing, even though the atmosphere presses on them too. The same thing loads the wall of a dam and your eardrum: the water above.

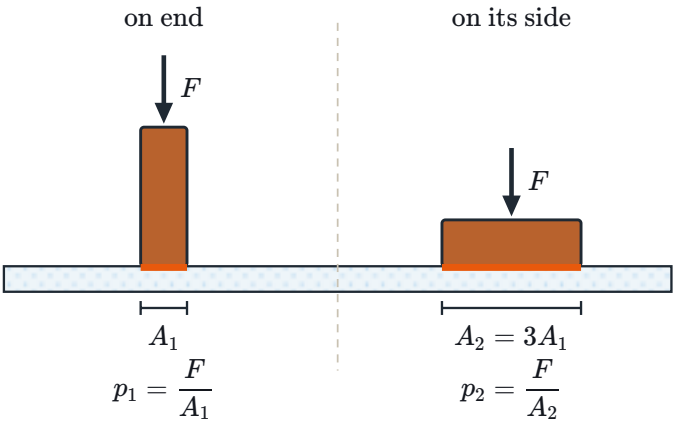
> **How does pressure change as you go deeper in a liquid, and what does it depend on?**

Theory

Key idea

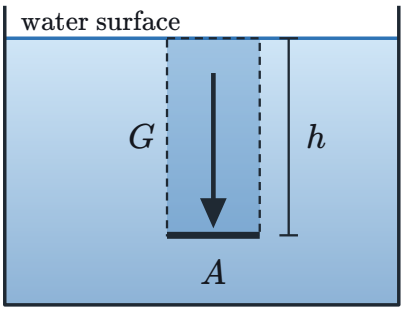
When a force is spread over a surface, its effect depends on how large an area it is spread over. The sharp tip of a drawing pin sinks into a wall with the same force that leaves no mark when applied to the blunt head, and skis carry you on deep snow although boots sink in. The quantity that describes this is pressure: force divided by the area over which the force is spread. The unit of pressure is the pascal (Pa),

and $1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}$. The pascal is a small unit, so in everyday life we use the kilopascal: $1\,\mathrm{kPa} = 1000\,\mathrm{Pa}$.



Where does pressure come from?

According to the micro model, the pressure of a gas comes from particles colliding with the walls of the container. The pressure of a liquid comes from the weight of the liquid itself: above every layer of water there is more water pressing down on it. The deeper you go, the more water there is above you and the greater the pressure. In a liquid at rest, a horizontal surface of area A below the surface carries the weight of the column of water above it. The height of the column is the depth h and its volume is $V = Ah$. Density ρ is mass per volume ($m = \rho V$), so the mass of the column is ρAh and its weight is $G = mg = \rho Ahg$. Pressure is weight divided by area, so the area cancels and what is left is ρgh. The pressure produced by the weight of a liquid is called hydrostatic pressure. It does not depend on the shape of the container or on the total amount of water, only on the depth and the density of the liquid.



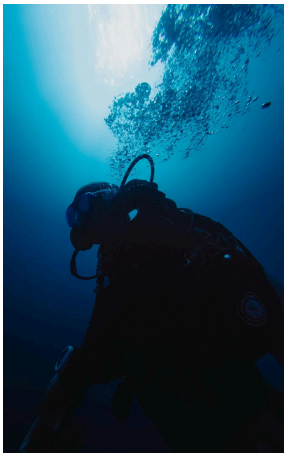
column's weight

$$G = \rho Ahg$$

bottom pressure

$$p = \frac{G}{A} = \rho gh$$

Pressure is not a force but the ratio of force to area, and in a liquid at rest the pressure acts equally in every direction. If the pressure sideways were smaller than downwards, the water would flow sideways and would not stay at rest. If you make holes at several heights in the side of a bottle, the water squirts out hardest from the lowest hole, because the pressure grows with depth. The force caused by pressure is always perpendicular to the surface: on the vertical wall of a dam the force points horizontally, and on a diver's body it acts from all sides. That is why here we can treat the force as just a magnitude; the direction comes from the surface.



A diver deep under water. The bubbles of exhaled air rise towards the surface.

Photo: Mido Makasard, Pexels License

Atmospheric pressure and gauge pressure

Atmospheric pressure, the pressure of the atmosphere, already acts on the surface of the water. At sea level it is about $101\ \mathrm{kPa}$. It is the weight of the column of air per unit area. Atmospheric pressure also acts on the surface of the water and is transmitted through the water to depth. Under water the total pressure is therefore the sum of atmospheric pressure and hydrostatic pressure. At a depth of ten metres the hydrostatic pressure is already about the same as one atmosphere.

This answers the opening question. Atmospheric pressure acts on both sides of the dam wall, so the wall is loaded only by the gauge pressure of the water, that is,

the part of the pressure that is greater than atmospheric pressure. The gauge pressure is zero at the top and grows steadily with depth: at a depth of 30 metres it is about $290\ \text{kPa}$, so every square metre of the wall feels a force of about $290\ \text{kN}$. The lower part of the wall must therefore withstand a much greater force than the upper part, and the wall is built thick at the bottom.

Pressure and hydrostatic pressure

Pressure is the force perpendicular to a surface divided by the area:

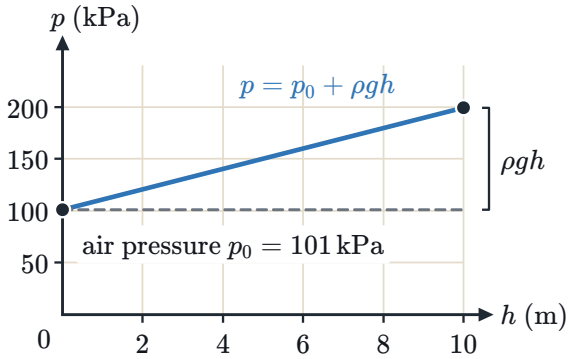
$$p = \frac{F}{A}$$

where p is the pressure (Pa), F the force perpendicular to the surface (N) and A the area (m²). One pascal is a force of one newton on an area of one square metre: $1\ \text{Pa} = 1\ \text{N/m}^2$.

In a liquid at depth h the pressure is

$$p = p_0 + \rho g h$$

where p_0 is the pressure at the surface of the liquid, usually atmospheric pressure (Pa), ρ the density of the liquid (kg/m³), $g = 9.81\ \text{m/s}^2$ and h the depth below the surface (m). The term $\rho g h$ is the hydrostatic pressure, that is, the gauge pressure $\Delta p = p - p_0$, which tells how much greater the pressure is than the pressure at the surface. The graph (h, p) is a straight line whose slope is ρg and whose constant term is p_0, as in the linear model $y = kx + b$ of FY1. For water $\rho = 1000\ \text{kg/m}^3$, so the pressure grows by $9.81\ \text{kPa}$ for every metre.

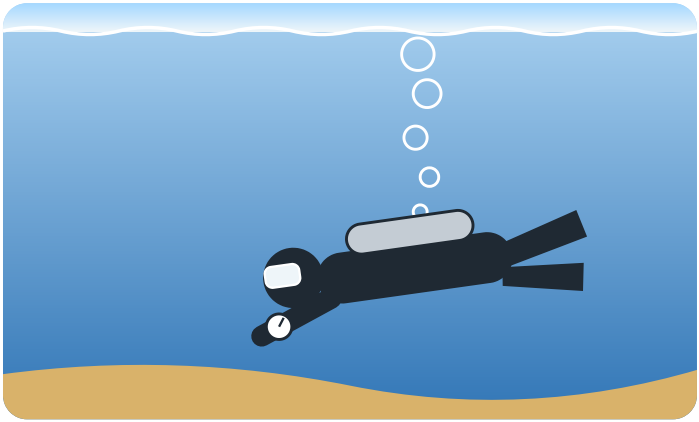


Common misconception

Many people think that the pressure at the bottom depends on how much water the container holds. In a narrow tube and a wide pool, where the water reaches the same height, the pressure at the bottom is the same, even though the pool holds a thousand times as much water. Pressure depends only on the depth, the density and the pressure at the surface. Another mistake is to confuse pressure and force: the sharp tip of a drawing pin does not produce a greater force than the head, but spreads the same force over a smaller area, so the pressure is greater. A third mistake is to forget atmospheric pressure: $\rho g h$ alone gives only the gauge pressure, and the total pressure under water is always greater than that.

Summary

- Pressure is force per unit area: $p = F/A$. The unit is the pascal, $1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}$, and $1\,\mathrm{kPa} = 1000\,\mathrm{Pa}$.
- In a liquid the pressure grows with depth: $p = p_0 + \rho g h$. The term $\rho g h$ is the hydrostatic pressure, and it depends only on the depth and the density, not on the shape of the container.
- The force caused by pressure is perpendicular to the surface and acts in every direction in a liquid.
- Atmospheric pressure at sea level is about $101\,\mathrm{kPa}$. In water the pressure grows by about one atmosphere for every ten metres.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in

the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Pressure on a diver

A diver is in a lake at a depth of $12.0\ \mathrm{m}$, next to the window of an underwater observation room. Atmospheric pressure at the surface is $101\ \mathrm{kPa}$ and the density of water is $1000\ \mathrm{kg/m^3}$. What is the pressure around the diver, and how large is the force on a window of area $150\ \mathrm{cm^2}$ when the room contains air at normal pressure?

STEP 1/7

Given and find. Depth $h = 12.0\ \mathrm{m}$, atmospheric pressure $p_0 = 101\ \mathrm{kPa}$, the density of water from a table or from the problem $\rho = 1000\ \mathrm{kg/m^3}$, $g = 9.81\ \mathrm{m/s^2}$ and window area $A = 150\ \mathrm{cm^2}$. We want the total pressure p around the diver and the force F on the window.

STEP 2/7

Principle. The water is at rest and its density is constant, so the pressure grows steadily with depth: the total pressure is the atmospheric pressure at the surface plus the hydrostatic pressure of the water $\rho g h$. Only the gauge pressure $\rho g h$ acts on the window, because the atmospheric pressure in the room pushes the window from inside just as much as the atmospheric pressure at the surface pushes from outside. The force is pressure times area, because pressure is force per unit area.

STEP 3/7

Rearranged formula. Both unknowns are already alone on the left-hand side:

$$p = p_0 + \rho g h$$
$$F = \rho g h \cdot A$$

STEP 4/7

Unit conversions. Atmospheric pressure in pascals is $101\,\mathrm{kPa} = 101\,000\,\mathrm{Pa}$. The area of the window in square metres is

$$\begin{aligned} 150\,\mathrm{cm}^2 &= 150 \cdot 10^{-4}\,\mathrm{m}^2 \\ &= 0.015\,\mathrm{m}^2 \end{aligned}$$

because $1\,\mathrm{cm}^2 = 10^{-4}\,\mathrm{m}^2$. The other quantities are already in SI units.

STEP 5/7

Substitution. We calculate the hydrostatic pressure, the total pressure and the force one after another:

$$\begin{aligned} &\rho g h \\ &= 1000\,\tfrac{\mathrm{kg}}{\mathrm{m}^3} \\ &\quad \cdot 9.81\,\tfrac{\mathrm{m}}{\mathrm{s}^2} \\ &\quad \cdot 12.0\,\mathrm{m} \\ &= 117\,720\,\mathrm{Pa} \end{aligned}$$

$$\begin{aligned} p &= p_0 + \rho g h \\ &= 101\,000\,\mathrm{Pa} \\ &\quad + 117\,720\,\mathrm{Pa} \\ &= 218\,720\,\mathrm{Pa} \end{aligned}$$

$$\begin{aligned} F &= \rho g h \cdot A \\ &= 117\,720\,\mathrm{Pa} \\ &\quad \cdot 0.015\,\mathrm{m}^2 \\ &= 1765.8\,\mathrm{N} \end{aligned}$$

STEP 6/7

Result. The total pressure is $p \approx 219\,\mathrm{kPa}$ and the force on the window is $F \approx 1.8\,\mathrm{kN}$. The pressure is given with three significant figures, because the depth $12.0\,\mathrm{m}$ and the atmospheric pressure $101\,\mathrm{kPa}$ are given with three figures. The force is given with two, because the area $150\,\mathrm{cm}^2$ is given with two figures.

STEP 7/7

Reasonableness. The total pressure is about 2.2 times atmospheric pressure, which fits the fact that 10 metres of water add about one atmosphere and 12 metres add a little more. Estimate:

$$\begin{aligned} &10\,\mathrm{kPa/m}\cdot 12\,\mathrm{m}+101\,\mathrm{kPa}\\ &\quad\approx 220\,\mathrm{kPa} \end{aligned}$$

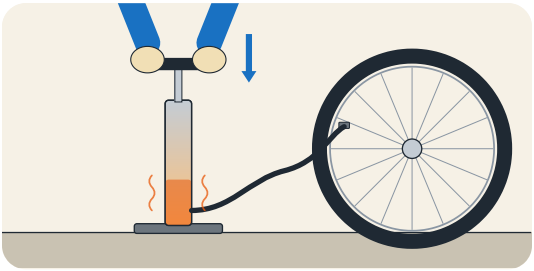
The force equals the weight of a mass of about $180\,\mathrm{kg}$, although the window is small. The pressure does not crush the diver's body, because in water the same pressure acts on the body from all sides and the body fluids transmit it to the inside as well. Only the pressure in the body's air spaces, such as the ears, has to be equalised.

FY3 · CHAPTER 6

Internal energy, heat and conservation of energy

Why does a bicycle pump get hot when you pump up a tyre, even though nothing is heating the pump?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-5.html`



IN THIS LESSON, YOU WILL LEARN TO

- distinguish heat, work and internal energy: heat and work transfer energy, while internal energy is a property of the system.
- apply the first law of thermodynamics.

PRIOR KNOWLEDGE

- Work $W = Fs$ transfers energy, and in Joule's experiment work heated water
  chapter *Work and energy*

- Micro model of a gas: temperature tells you the average kinetic energy of the particles
  chapter *Thermodynamic system, state variables and temperature*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

You pump a bicycle tyre to a high pressure. At the end, the lower end of the pump, where the air is compressed, feels warm in your hand. After the pump has rested for a while, it is at room temperature again. On the other hand, when you let air out of the tyre through the valve, the valve feels cool, even though nothing is cooling it.

> **Where does the air in the pump get extra energy if it is not heated, and where does that energy go when the pump cools down?**

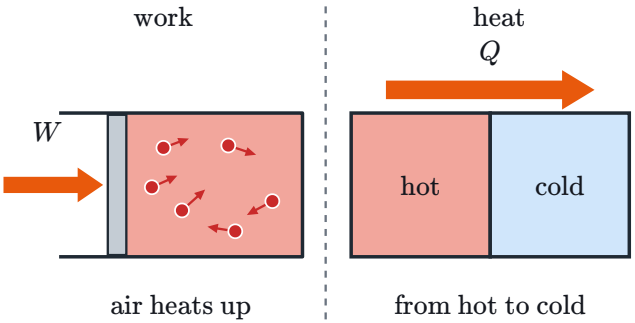
Theory

Key idea

In chapter *Work and energy*, work was described as a way of transferring energy, and Joule's experiment showed that mechanical work can heat water: the work done by the weight increased the internal energy of the water, even though the water was not heated. Water does not contain "heat". The internal energy U of a system is the sum of the kinetic energy of its particles and the energy of the interactions between the particles. In chapter *Forms of energy and conservation of energy*, this energy was called thermal energy; a more precise name is internal

energy. Temperature describes how strongly the particles move because of thermal motion: in a gas, it tells you the average kinetic energy of the particles. When the temperature of a substance rises and its state of matter does not change, the particles move more strongly and the internal energy increases. Even so, temperature and internal energy are not the same thing, because when the state of matter changes, the internal energy changes even though the temperature stays the same: when, for example, pure ice melts at normal pressure, the internal energy increases, but the temperature stays at $0\,^{\circ}\mathrm{C}$. Internal energy also depends on the amount of substance: a big lake contains much more internal energy than a cup of hot tea, even though the lake is colder.

You can change the internal energy of a system in two ways. The first is work: when the piston of a pump compresses air, the surroundings do work on the gas, and the internal energy of the gas increases, because the moving piston hits the particles like a bat hits a ball and they bounce off faster. In the simulation you can see this at the particle level. This is why the pump gets hot. The second is heat: when two objects at different temperatures touch, energy transfers by itself from the hotter one to the colder one until the temperatures are the same. Energy transferred in this way is called heat, and the amount of energy transferred is the quantity of heat Q. So heat is a transfer of energy, not a property of matter: an object has a temperature and internal energy, but it does not have heat.



Energy is conserved. When energy is transferred into a system as heat or work and the kinetic or potential energy of the system does not change, its internal energy increases by exactly the amount of energy transferred, and when energy is transferred out, the internal energy decreases by the same amount. This is the first law of thermodynamics: the law of conservation of energy written for heat and work. An isolated system is a closed system into which, and out of which, no energy is transferred either as heat or as work. Its total energy stays constant, even if energy changes form inside it.



Heated metal glows in a furnace. When the metal is heated, its internal energy increases.

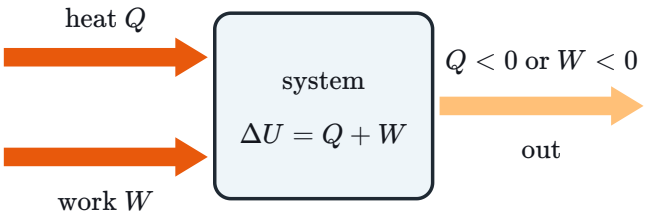
Photo: Nino Rocha, Pexels License

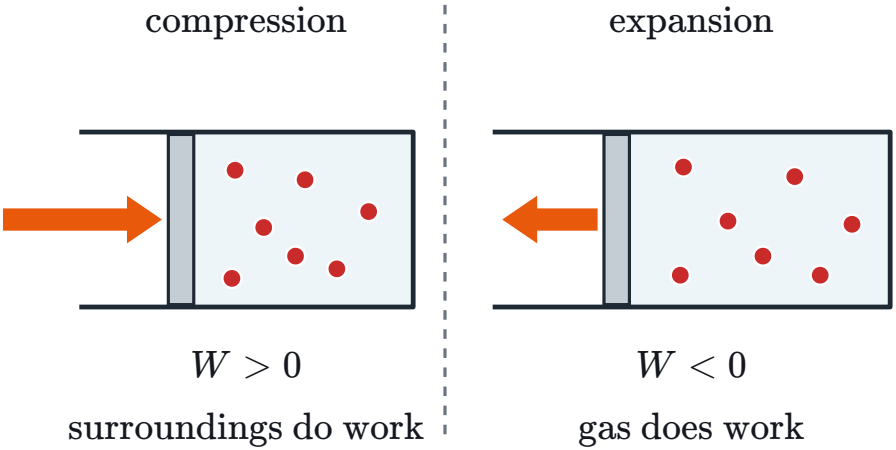
First law of thermodynamics

The change in internal energy is the sum of the quantity of heat transferred into the system and the work done on the system:

$$\Delta U = Q + W$$

where ΔU is the change in internal energy (J), Q is the quantity of heat transferred into the system (J) and W is the work done on the system by the surroundings (J). The signs tell you the direction: Q is positive when heat is transferred into the system and negative when it is transferred out; W is positive when the surroundings do work on the system (for example, compress a gas) and negative when the system does work on the surroundings (the gas expands and pushes the piston). All the terms are in joules, the same unit as work in chapter *Work and energy*, and the work done by a piston on a gas is calculated in the same way as there, as the product of force and distance.





Common misconception

In everyday language we say that an object "has heat" or that heat is "stored". In physics, heat is the transfer of energy because of a temperature difference, and what is stored in an object is internal energy. Another mistake is to treat temperature and internal energy as the same thing: a hot spark has a temperature of thousands of degrees, but its internal energy is small, because it contains very little matter. A third mistake is to think that internal energy can be changed only by heating. The air in a pump gets hot without any heat source, because work is done on it.

Summary

- The internal energy U is the sum of the kinetic energy of the particles of a system and the energy of the interactions between the particles. It increases when the temperature rises while the state of matter stays the same, and it also depends on the amount of substance. Temperature and internal energy are not the same thing.
- Heat is the transfer of energy from hotter to colder because of a temperature difference; the energy transferred is the quantity of heat Q. An object does not "have heat".
- Heat and work change internal energy: $\Delta U = Q + W$. Energy transferred in is positive and energy transferred out is negative.
- Energy is conserved: this is the first law of thermodynamics. The total energy of an isolated system stays constant.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Internal energy of air in a pump

A pump cylinder contains air. The piston is pushed down, so the surroundings do $150\ \mathrm{J}$ of work on the air. At the same time, a quantity of heat of $40\ \mathrm{J}$ is transferred from the warming air through the cylinder walls to the surroundings. By how much does the internal energy of the air change?

STEP 1/7

Given and find. The air is the system. The surroundings do work on the system, so $W = +150\ \mathrm{J}$. Heat is transferred out of the system, so $Q = -40\ \mathrm{J}$. The quantity to find is the change in internal energy ΔU.

STEP 2/7

Principle. Energy is conserved: the internal energy of the system changes by exactly the amount of energy transferred into it as heat and work. This is the first law of thermodynamics, and it applies because the air is a clearly bounded system, and energy crosses its boundary only as work and heat.

STEP 3/7

Rearranged formula. In the first law, the quantity to find ΔU is already alone on the left-hand side, so the equation does not need rearranging:

$$\Delta U = Q + W$$

STEP 4/7

Unit conversions. None needed: both Q and W are given in joules, and the result is wanted in joules. The signs, however, must be decided: energy into the system is positive and energy out of the system is negative.

STEP 5/7

Substitution. Substitute the values with their signs:

$$\begin{aligned}\Delta U &= Q + W \\ &= -40\,\mathrm{J} + 150\,\mathrm{J} \\ &= 110\,\mathrm{J}\end{aligned}$$

STEP 6/7

Result. The internal energy of the air increases by $110\,\mathrm{J}$. The result is given in whole joules, because both the work and the heat were given in whole joules.

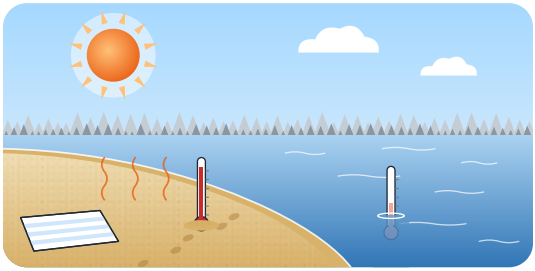
STEP 7/7

Reasonableness. The work brought $150\,\mathrm{J}$ into the air and only $40\,\mathrm{J}$ left as heat, so the internal energy must increase and the temperature of the air must rise. If the same $150\,\mathrm{J}$ of work were done so slowly that all the energy from the work had time to leave as heat, then $Q = -150\,\mathrm{J}$ and $\Delta U = 0$, and the temperature would not change. Now less heat leaves, so the result lies between this limiting case and the work alone, $150\,\mathrm{J}$.

Heating and cooling of matter: $Q = cm\Delta T$

Why does the sand on a beach burn your feet already in the late morning, when the water is still cool, even though the Sun has shone on both for the same time?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-6.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the heat that changes the temperature of a substance.
- explain what the specific heat capacity tells about a substance.
- use conservation of energy to find a calorimeter's equilibrium temperature or an unknown substance's specific heat capacity.

PRIOR KNOWLEDGE

- Heat is energy that is transferred because of a temperature difference, and the energy transferred is the quantity of heat Q
  chapter *Internal energy, heat and conservation of energy*
- Energy is conserved: the total energy of an isolated system stays constant
  chapter *Internal energy, heat and conservation of energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

On a July morning at eight o'clock, the sand on the beach and the shallow water by the shore are about equally warm. By noon the surface of the sand is so hot that you cannot stand on it barefoot, but the water has warmed by only a few degrees. The Sun has shone on the same beach for the same time, and both have received energy as heat.

> **If energy is transferred as heat to two substances, what decides how much the temperature of each rises?**

Theory

Key idea

When heat is transferred to an object and its phase does not change, its internal energy increases and its temperature rises. Experiments show that the quantity of heat needed is directly proportional both to the change in temperature and to the mass of the object: a temperature rise twice as large needs twice the heat, and twice the amount of water needs twice the heat. In addition, the heat depends on the substance. Heating one kilogram of water by one degree takes about $4.2\,\mathrm{kJ}$, but heating one kilogram of iron takes only about $0.45\,\mathrm{kJ}$. The coefficient that is characteristic of a substance is the specific heat capacity c, whose unit is J/(kg·K). It tells how much energy is needed to heat one kilogram by one kelvin, and the same amount is released when the substance cools by one kelvin.

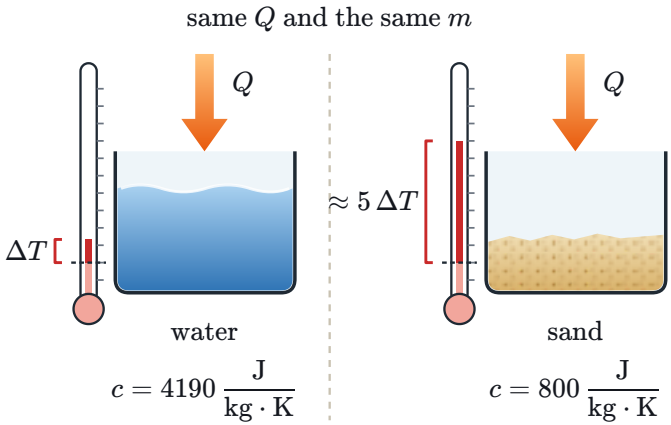


Hot water is poured from a kettle. The visible mist is condensed water droplets.

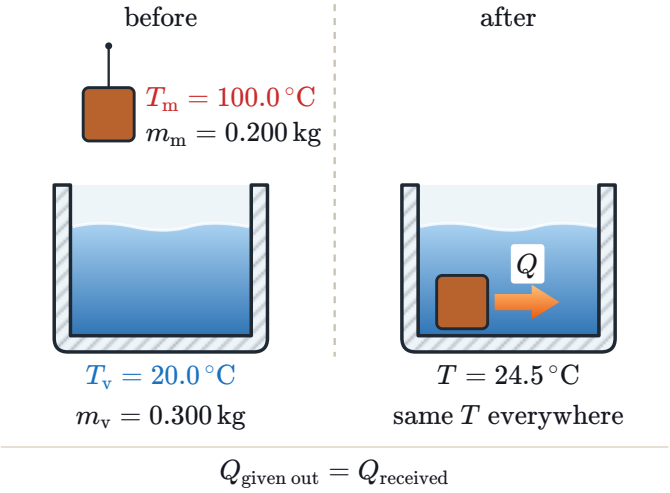
Photo: Jahra Tasfia Reza, Pexels License

An important reason is that the specific heat capacity of water is exceptionally large: water warms slowly in the Sun and cools slowly in the evening, whereas sand, whose c is about one fifth of that of water, gets many times hotter with the same energy. In addition, the Sun heats only a thin surface layer of the sand, whereas in water the warming reaches deeper. Because of their large specific heat capacity,

lakes even out the temperatures of the coast, and water is a good heat-transfer medium in heating pipes.



When a hot and a cold object are put into one insulated container, heat is transferred from the hotter to the colder one until both have the same temperature. This state is called thermal equilibrium. Energy is conserved: the heat given out by the hot object is equal to the heat received by the cold one. An insulated container in which this is measured is called a calorimeter, and the measurement is called a calorimetric measurement. With it you can determine the specific heat capacity of an unknown substance when the specific heat capacity of water is known.



Heat and thermal equilibrium

When the phase of a substance does not change, the quantity of heat needed to heat a mass m, or released when it cools, is

$$Q = cm\Delta T$$
$$\Delta T = T_{\text{final}} - T_{\text{initial}}$$

where Q is the quantity of heat (J), c the specific heat capacity of the substance (J/(kg·K)), m the mass (kg) and ΔT the change in temperature (K), that is, the final temperature T_{final} minus the initial temperature T_{initial}. The change in temperature is the same in kelvins and in degrees Celsius, so ΔT can be calculated directly from Celsius temperatures. When an object warms, ΔT and Q are positive; when it cools, they are negative.

The heat capacity of a whole object

$$C = cm$$

tells how many joules it takes to raise the temperature of the object by one kelvin. C is the heat capacity (J/K), c the specific heat capacity and m the mass. An object with a large heat capacity warms and cools slowly.

In an insulated container, the heat given out by the hot object 1 is equal to the heat received by the cold object 2:

$$c_1 m_1 (T_1 - T) = c_2 m_2 (T - T_2)$$

where T_1 and T_2 are the initial temperatures of the objects and T is the common final temperature, that is, the equilibrium temperature. Both sides of the equation are positive quantities of heat: for the hot object you calculate initial temperature minus final temperature, and for the cold object final temperature minus initial temperature. From the equation you can solve any quantity when the others are known: the final temperature, a mass or an unknown specific heat capacity.

The equilibrium temperature T appears in two terms of the equation. You solve it by multiplying out the brackets, collecting the terms that contain T on the same side and taking T as a common factor:

$$
\begin{aligned}
&c_1 m_1 T_1 - c_1 m_1 T \\
&= c_2 m_2 T - c_2 m_2 T_2
\end{aligned}
$$

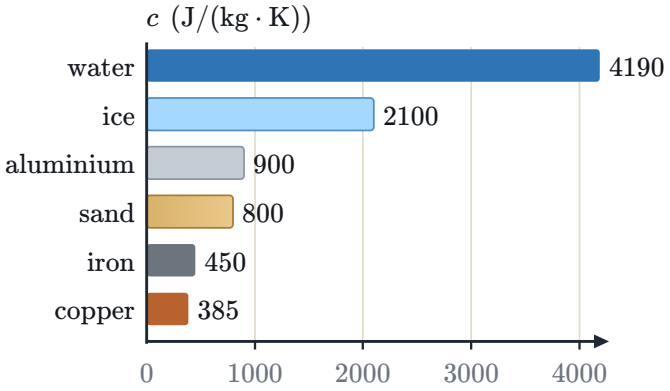
$$
\begin{aligned}
&c_1 m_1 T_1 + c_2 m_2 T_2 \\
&= c_1 m_1 T + c_2 m_2 T \\
&= (c_1 m_1 + c_2 m_2)\, T
\end{aligned}
$$

When the equation is divided on both sides by the sum $c_1 m_1 + c_2 m_2$, you get

$$
T = \frac{c_1 m_1 T_1 + c_2 m_2 T_2}{c_1 m_1 + c_2 m_2}
$$

So the equilibrium temperature is a weighted mean of the initial temperatures, where the weights are the heat capacities $c_1 m_1$ and $c_2 m_2$ of the objects. You get the ordinary mean only when the heat capacities are equal.

Substance	c (J/(kg·K))
water	4190
ice	2100
aluminium	900
sand	800
iron	450
copper	385



Common misconception

Many people think that heat and temperature are the same thing, or that objects heated to the same temperature have received the same amount of energy. Heating a kilogram of water and a kilogram of iron from $20\,^\circ\mathrm{C}$ to $60\,^\circ\mathrm{C}$ takes almost ten times as much energy for the water. Another mistake is to calculate the equilibrium temperature as a mean. When $1.0\,\mathrm{kg}$ of 80-degree water is poured into $3.0\,\mathrm{kg}$ of 20-degree water, the specific heat capacity of water c is on both sides of the equation and cancels:

$$\begin{aligned} &1.0\,\mathrm{kg}\cdot(80\,^\circ\mathrm{C}-T)\\ &\quad=3.0\,\mathrm{kg}\cdot(T-20\,^\circ\mathrm{C})\end{aligned}$$

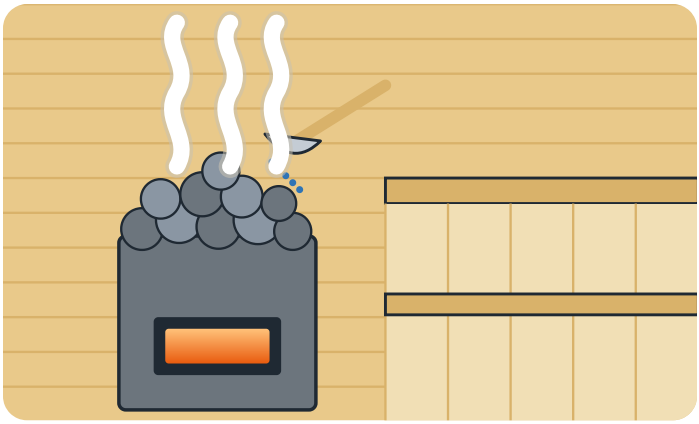
$$\begin{aligned} 80\,^\circ\mathrm{C}-T&=3T-60\,^\circ\mathrm{C}\\ 140\,^\circ\mathrm{C}&=4T\\ T&=35\,^\circ\mathrm{C}\end{aligned}$$

The hot water is on the left side, as in the general equation. On the third line the equation has been divided on both sides by a kilogram and the brackets have been multiplied out. So the final result is not $50\,^\circ\mathrm{C}$ but $35\,^\circ\mathrm{C}$, because there is three times as much cold water as hot. The same result comes from the weighted mean: $T=(3.0\,\mathrm{kg}\cdot 20\,^\circ\mathrm{C}+1.0\,\mathrm{kg}\cdot 80\,^\circ\mathrm{C}):4.0\,\mathrm{kg}=35\,^\circ\mathrm{C}$.

A third mistake is to convert ΔT to kelvins by adding 273: a change in temperature is the same on both scales.

Summary

- The quantity of heat is proportional to the mass and to the change in temperature: $Q = cm\Delta T$. The specific heat capacity c depends on the substance, and for water it is large, $4190\ \mathrm{J/(kg \cdot K)}$.
- The heat capacity $C = cm$ tells how many joules it takes to heat the whole object by one kelvin.
- In an insulated container the heat given out by the hot object is equal to the heat received by the cold one: $c_1 m_1 (T_1 - T) = c_2 m_2 (T - T_2)$. The equilibrium temperature T is solved by collecting the terms with T on the same side; it is a mean weighted by the heat capacities, and usually not the ordinary mean of the initial temperatures.
- A calorimetric measurement determines the specific heat capacity of an unknown substance with the help of water.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Metal block in a calorimeter

A calorimeter contains $0.300\,\mathrm{kg}$ of water at a temperature of $20.0\,^{\circ}\mathrm{C}$. A metal object of mass $0.200\,\mathrm{kg}$, heated in boiling water to a temperature of $100.0\,^{\circ}\mathrm{C}$, is lowered into it. After stirring, the temperature settles at $24.5\,^{\circ}\mathrm{C}$. What is the specific heat capacity of the metal? Ignore the heat capacity of the calorimeter itself and heat losses to the surroundings.

STEP 1/7

Given and find. Water: $m_{\mathrm{w}} = 0.300\,\mathrm{kg}$, $T_{\mathrm{w}} = 20.0\,^{\circ}\mathrm{C}$ and from the table $c_{\mathrm{w}} = 4190\,\frac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}}$. Metal: $m_{\mathrm{m}} = 0.200\,\mathrm{kg}$, $T_{\mathrm{m}} = 100.0\,^{\circ}\mathrm{C}$. Equilibrium temperature $T = 24.5\,^{\circ}\mathrm{C}$. We want the specific heat capacity of the metal c_{m}.

STEP 2/7

Principle. Energy is conserved. When the calorimeter and the air take up no heat, the heat given out by the metal is equal to the heat received by the water. Each one only warms or cools and its phase does not change, so $Q = cm\Delta T$ holds for both. At the end both are at the same temperature T.

STEP 3/7

Rearranged formula. Set the quantities of heat equal and solve for c_{m}:

$$\begin{aligned} & c_{\mathrm{m}} m_{\mathrm{m}} (T_{\mathrm{m}} - T) \\ &= c_{\mathrm{w}} m_{\mathrm{w}} (T - T_{\mathrm{w}}) \end{aligned}$$

$$c_{\mathrm{m}} = \frac{c_{\mathrm{w}} m_{\mathrm{w}} (T - T_{\mathrm{w}})}{m_{\mathrm{m}} (T_{\mathrm{m}} - T)}$$

STEP 4/7

Unit conversions. The masses are already in kilograms. A temperature difference is the same in kelvins and in degrees Celsius, so the metal cools by $T_{\mathrm{m}} - T = 75.5\,\mathrm{K}$ and the water warms by $T - T_{\mathrm{w}} = 4.5\,\mathrm{K}$.

STEP 5/7

Substitution. First calculate the numerator, that is, the heat Q_{w} received by the water, and then divide:

$$\begin{aligned} Q_{\mathrm{w}} &= c_{\mathrm{w}} m_{\mathrm{w}} (T - T_{\mathrm{w}}) \\ &= 4190\ \tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \cdot 0.300\,\mathrm{kg} \cdot 4.5\,\mathrm{K} \\ &= 5656.5\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} c_{\mathrm{m}} &= \frac{Q_{\mathrm{w}}}{m_{\mathrm{m}} (T_{\mathrm{m}} - T)} \\ &= \frac{5656.5\,\mathrm{J}}{0.200\,\mathrm{kg} \cdot 75.5\,\mathrm{K}} \\ &= 374.6\ \tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \end{aligned}$$

STEP 6/7

Result. The specific heat capacity of the metal is about $3.7 \cdot 10^{2}\ \tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}}$. The result is given with two significant figures, because the temperature change of the water, $4.5\,\mathrm{K}$, is known only to two figures: if the final reading were $0.1\,\mathrm{K}$ higher, that is $24.6\,^{\circ}\mathrm{C}$, the result would be $383\,\mathrm{J}/(\mathrm{kg}\cdot\mathrm{K})$.

STEP 7/7

Reasonableness. The value is close to the specific heat capacity of copper, $385\,\mathrm{J}/(\mathrm{kg}\cdot\mathrm{K})$, so the object is probably copper. In a lossless calorimeter a $0.200\,\mathrm{kg}$ copper object would raise the water to $24.6\,^{\circ}\mathrm{C}$ (try it in the simulation); the measured reading, $0.1\,\mathrm{K}$ lower, is explained by the reading precision or by the fact that in reality some of the heat also went into the calorimeter vessel and the air. The water warmed by only $4.5\,\mathrm{K}$ although the metal cooled by $75.5\,\mathrm{K}$: the heat capacity of the water is

$$\begin{aligned} &0.300\,\mathrm{kg} \cdot 4190\ \tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \\ &\quad = 1257\ \tfrac{\mathrm{J}}{\mathrm{K}} \end{aligned}$$

which is much more than the heat capacity of the object,

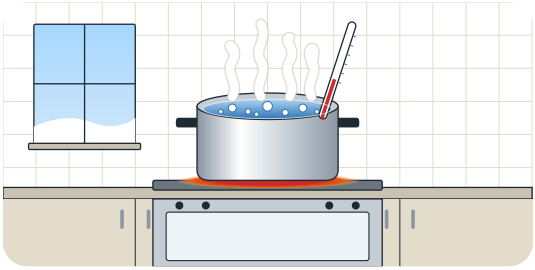
$$0.200\,\text{kg} \cdot 374.6\ \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

$$\approx 75\ \frac{\text{J}}{\text{K}}$$

Changes of state

Why does the temperature of water boiling at normal atmospheric pressure stay at about $100\,^{\circ}\mathrm{C}$, even though energy keeps flowing into the pan?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-7.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain where the energy goes when a pure substance changes state at constant pressure and temperature.
- calculate the quantity of heat needed for melting and vaporisation.

PRIOR KNOWLEDGE

- The quantity of heat $Q = cm\Delta T$ and the specific heat capacity of water, $4190\ \mathrm{J/(kg \cdot K)}$
  chapter *Heating and cooling of matter:* $Q = cm\Delta T$

- Power tells you how fast energy is transferred: $Q = Pt$
  chapter *Work and energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

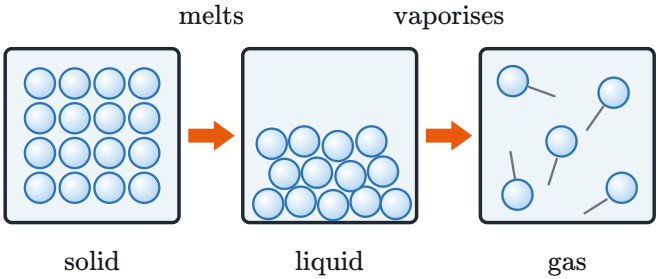
You put ice cubes from the freezer into a pan, switch the hotplate on and push a thermometer in among the ice. The hotplate gives out the same power all the time. The thermometer reading starts below freezing and rises – but does it rise steadily all the time until the pan is empty?

> **Where does the energy from the hotplate go at the moments when the thermometer reading does not change?**

Theory

Key idea

Matter exists in three states: solid, liquid and gas. In the micro model, the particles of a solid vibrate in place, held together by each other. In a liquid, they move past each other but stay together. In a gas, they move freely and only rarely collide. In a change of state, the interactions between the particles change: in melting, the regular order of the solid breaks down, and in vaporisation, the particles move far apart from each other. The particles themselves do not change: when ice melts or water boils, the chemical bonds inside the water molecules do not break. The opposite changes are freezing and condensation. Ice can also sublimate straight into vapour, as in laundry drying in freezing weather.





The eruption on La Palma in 2021. Lava that flowed onto the surface cools and solidifies.

Photo: Eduardo Robaina, CC BY-SA 3.0

Where does the energy go?

Melting and vaporisation need energy but do not raise the temperature. When ice is heated, its temperature rises up to $0\,^{\circ}\mathrm{C}$. Then the temperature stays constant until all the ice has melted, even though energy keeps being transferred. During melting and vaporisation, the energy transferred to the substance changes the interactions between the particles and increases the interaction energy between them. When a pure substance changes state at constant pressure, its temperature stays constant, so the average kinetic energy of the particles does not increase.



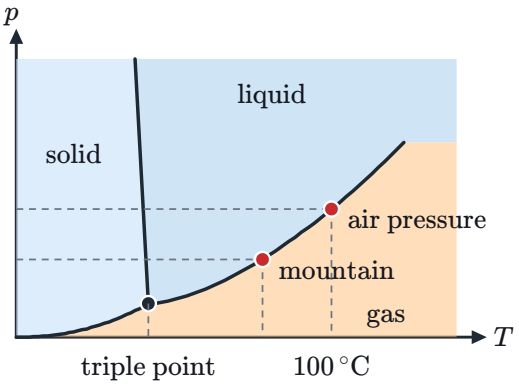
Melting ice photographed close up. Melting ice takes energy.

Photo: Julia Sakelli, Pexels License

For the same reason, boiling water stays at $100\,^{\circ}\mathrm{C}$ even when the hob is on full: the energy transferred vaporises water and does not raise its temperature. Steam takes up about 1700 times as much space as water, so as it expands it pushes the surrounding air. A small part of the energy needed for vaporisation is used for this work, and most of it increases the interaction energy between the molecules. The greater the power, the faster the water boils away, but it does not get any hotter. The energy that was taken up when the water vaporised is released again when the vapour condenses, and that is why 100-degree steam burns worse than 100-degree water.

The quantity of heat needed for melting and vaporisation is directly proportional to the mass. The coefficients are properties of the substance: the specific latent heat of fusion s tells you how much energy it takes to melt one kilogram, and the specific latent heat of vaporisation r how much it takes to vaporise one kilogram. For water, $s = 334\,\mathrm{kJ/kg}$ and $r = 2260\,\mathrm{kJ/kg}$. So vaporising takes almost seven times as much energy as melting, and more than five times as much as heating water from $0\,^{\circ}\mathrm{C}$ to $100\,^{\circ}\mathrm{C}$.

The boiling point depends on pressure. A phase diagram shows the state of a substance at each temperature and pressure: its curves join the melting, boiling and sublimation points at different pressures. On a mountain, where the atmospheric pressure is lower, water boils below $100\,^{\circ}\mathrm{C}$, and in a pressure cooker the pressure raises the boiling point above $100\,^{\circ}\mathrm{C}$, so food cooks faster.



Latent heat of fusion and vaporisation

The quantity of heat needed to melt or vaporise a mass m is

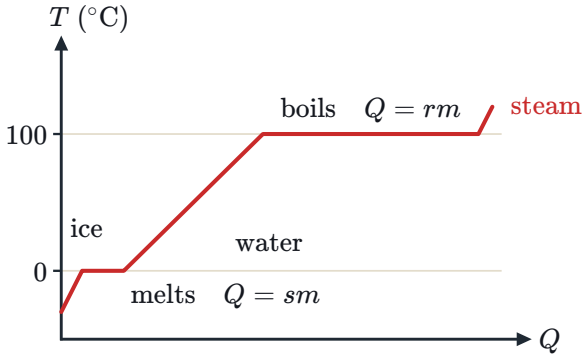
$$Q = sm \qquad \text{and} \qquad Q = rm$$

where Q is the quantity of heat (J), s is the specific latent heat of fusion of the substance (J/kg), r is the specific latent heat of vaporisation (J/kg) and m is the mass of the substance that melts or vaporises (kg). The same quantity of heat is released when the substance freezes or condenses. At constant pressure the temperature does not change during a change of state, so the formula has no temperature change ΔT.

Water	Value
s	$334\ \text{kJ/kg}$
r	$2260\ \text{kJ/kg}$
c (ice)	$2100\ \text{J/(kg}\cdot\text{K)}$
c (water)	$4190\ \text{J/(kg}\cdot\text{K)}$
c (steam)	$2000\ \text{J/(kg}\cdot\text{K)}$

In the table, c is the specific heat capacity, which was introduced in chapter *Heating and cooling of matter*: $Q = cm\Delta T$.

A heating curve shows the temperature as a function of the energy transferred. On the rising parts the substance warms up and $Q = cm\Delta T$ applies. On the horizontal parts the substance changes state and $Q = sm$ or $Q = rm$ applies. The smaller the specific heat capacity, the steeper the temperature rises, so the parts for ice and steam are about twice as steep as the part for water:



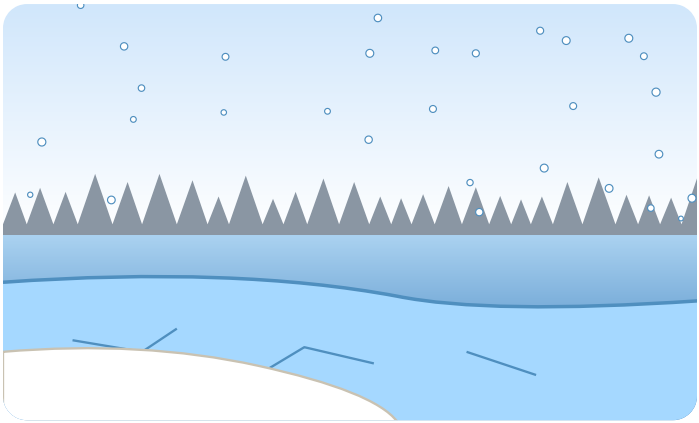
Common misconception

Many people think that adding heat always raises the temperature. When a pure substance changes state at constant pressure, the temperature stays constant even though a lot of energy is transferred: melting ice takes as much energy as heating the same water from $0\,^\circ\mathrm{C}$ to $80\,^\circ\mathrm{C}$. Another mistake is to treat ice at zero degrees and water at zero degrees as equally good coolants. Ice cools a drink much more effectively, because as it melts it takes in a lot of energy before it starts to warm up. A third mistake is to think that water always boils at $100\,^\circ\mathrm{C}$: the boiling point depends on pressure.

Summary

- The states of matter are solid, liquid and gas. In the micro model, a change of state changes the interactions between the particles: in melting and vaporisation the interaction energy between the particles increases, but the chemical bonds inside the molecules do not break.
- At constant pressure, the temperature of a pure substance stays constant during a change of state. Melting needs $Q = sm$ and vaporising needs $Q = rm$; for water, $s = 334\,\mathrm{kJ/kg}$ and $r = 2260\,\mathrm{kJ/kg}$.
- In a heating curve, the rising parts are warming ($Q = cm\Delta T$) and the horizontal parts are changes of state.
- The boiling point depends on pressure; a phase diagram tells you the state of a substance from its temperature and pressure.

- The specific latent heat of vaporisation can be measured by boiling water with an immersion heater on a balance: $r = Pt/\Delta m$, where Δm is the change in the balance reading.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Heat of vaporization of water

The specific latent heat of vaporisation of water is determined experimentally. An open calorimeter containing boiling water stands on a balance, and the water contains a $1000\ \mathrm{W}$ immersion heater. During five minutes ($300\ \mathrm{s}$), the balance reading falls by $128\ \mathrm{g}$ (an invented figure, not the result of a real measurement). What is the specific latent heat of vaporisation of water according to this reading?

STEP 1/7

Given and find. Power $P = 1000\,\mathrm{W}$, time $t = 300\,\mathrm{s}$ and the mass vaporised, that is, the change in the balance reading, $\Delta m = 128\,\mathrm{g}$. The water is boiling, so it is at $100\,^{\circ}\mathrm{C}$. We want the specific latent heat of vaporisation r. Assume that all the energy from the heater goes into the water.

STEP 2/7

Principle. The water boils all the time, so its temperature does not rise and all the energy $Q = Pt$ from the heater is used to vaporise water. The quantity of heat needed for vaporisation is proportional to the mass vaporised, $Q = r\,\Delta m$. Energy is conserved, so these two are equal.

STEP 3/7

Rearranged formula. Set the quantities of heat equal and solve for r:

$$Pt = r\,\Delta m$$
$$r = \frac{Pt}{\Delta m}$$

STEP 4/7

Unit conversions. Convert the change in the balance reading to kilograms: $128\,\mathrm{g} = 0.128\,\mathrm{kg}$. Power and time are already in base units, and $1\,\mathrm{W}\cdot 1\,\mathrm{s} = 1\,\mathrm{J}$. Finally the result is converted to megajoules: $1\,\mathrm{MJ} = 10^{6}\,\mathrm{J}$.

STEP 5/7

Substitution. Substitute the values with their units:

$$\begin{aligned} r &= \frac{Pt}{\Delta m} \\ &= \frac{1000\,\mathrm{W}\cdot 300\,\mathrm{s}}{0.128\,\mathrm{kg}} \\ &= \frac{300\,000\,\mathrm{J}}{0.128\,\mathrm{kg}} \\ &= 2\,343\,750\ \tfrac{\mathrm{J}}{\mathrm{kg}} \end{aligned}$$

STEP 6/7

Result. The specific latent heat of vaporisation is $2.34\ \frac{\mathrm{MJ}}{\mathrm{kg}}$. The result is given to three significant figures, because the mass reading $128\ \mathrm{g}$ is given to three figures and it limits the precision most.

STEP 7/7

Reasonableness. The result $2.34\ \frac{\mathrm{MJ}}{\mathrm{kg}}$ is slightly greater than the table value $2.26\ \frac{\mathrm{MJ}}{\mathrm{kg}}$, by about $3.7\ \%$. This is the expected direction: some of the heater's energy passes through the vessel into the surrounding air and does not vaporise water, so the assumption "all the energy goes into the water" gives too large an r. With the table value,

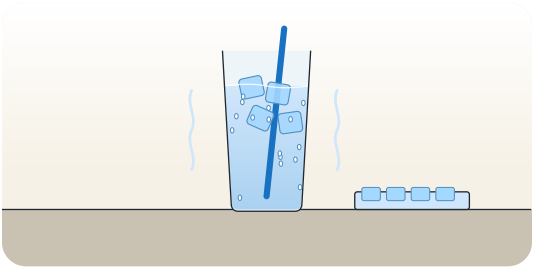
$$\frac{300\ \mathrm{kJ}}{2260\ \frac{\mathrm{kJ}}{\mathrm{kg}}} = 0.133\ \mathrm{kg}$$

or $133\ \mathrm{g}$, would vaporise, only $5\ \mathrm{g}$ more than was measured, so the result is correct to within a few per cent.

Thermal balance during a change of state

Why do ice cubes cool a drink so effectively, even though the ice itself is only at zero degrees? And why can some ice be left at the bottom of the glass, even though the drink was at room temperature?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-14.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain why ice cools a drink so effectively when it melts.
- work out by comparing quantities of heat whether all the ice melts.
- calculate the temperature of thermal equilibrium when part of the substance melts.

PRIOR KNOWLEDGE

- The quantity of heat needed to change a temperature is $Q = cm\Delta T$
  chapter *Heating and cooling of a substance:* $Q = cm\Delta T$
- In an insulated container, the quantity of heat given out by the hot object is equal to the quantity of heat taken in by the cold object
  chapter *Heating and cooling of a substance:* $Q = cm\Delta T$
- During a change of state the temperature stays constant, and melting ice needs $Q = sm$
  chapter *Changes of state*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

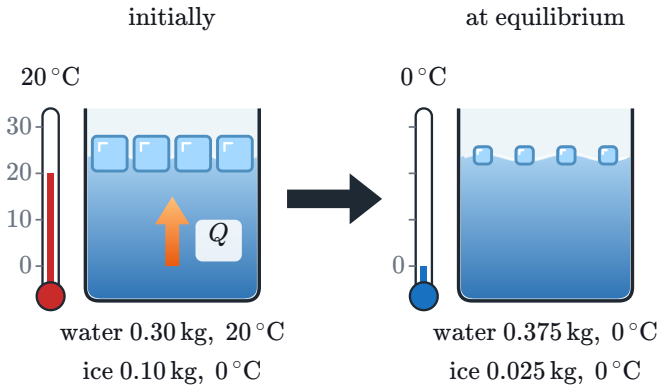
On a summer day you drop a few ice cubes into a glass of juice at room temperature. The ice itself is only at 0 °C, but the juice cools quickly, much more than if you added the same amount of water at 0 °C. When the heat has evened out, there can still be ice floating at the bottom of the glass.

> **Why does ice at 0 °C cool a drink so much more effectively than water at 0 °C, and why does not all the ice always melt?**

Theory

Key idea

When warm water and ice are put in an insulated container, heat is transferred from the water to the ice until the temperatures are equal and the container is in thermal equilibrium. Energy is conserved, so the quantity of heat given out by the water is equal to the quantity of heat taken in by the ice. The energy taken in by the ice does not simply raise its temperature, however: the ice first warms to $0\,^{\circ}\mathrm{C}$ and then melts. During melting the temperature stays constant, but every kilogram of ice takes in $334\,\mathrm{kJ}$ of energy, and the same energy would raise the temperature of a kilogram of water by almost $80\,\mathrm{K}$.



When warm water and cold ice are mixed in an insulated container, the final temperature is generally not the mean of the initial temperatures. The solution is based on conservation of energy.

Thermal equilibrium and melting

An insulated container holds water of mass m_{w} and temperature $T_{\mathrm{w}} > 0\,^{\circ}\mathrm{C}$, and ice of mass m_{i} and temperature $T_{\mathrm{i}} \leq 0\,^{\circ}\mathrm{C}$. The quantity of heat given out by the water as it cools to the melting point is

$$Q_{\mathrm{w}} = c_{\mathrm{w}} m_{\mathrm{w}} \Delta T_{\mathrm{w}}$$

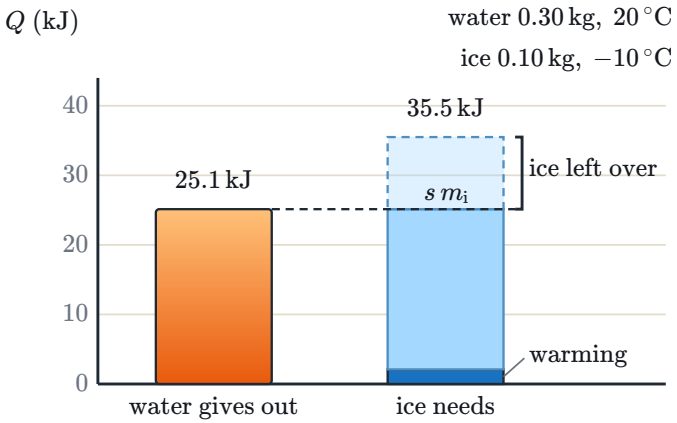
Warming the ice to the melting point needs the energy

$$Q_{\mathrm{i,warm}} = c_{\mathrm{i}} m_{\mathrm{i}} \Delta T_{\mathrm{i}}$$

Melting all the ice needs the additional energy sm_i. So warming and melting the ice need altogether

$$Q_\mathrm{i} = Q_\mathrm{i,warm} + sm_\mathrm{i}$$

where c_w is the specific heat capacity of water and c_i that of ice (J/(kg·K)), $\Delta T_\mathrm{w} = T_\mathrm{w} - 0\,^\circ\mathrm{C}$ is the fall in the temperature of the water to the melting point (K), $\Delta T_\mathrm{i} = 0\,^\circ\mathrm{C} - T_\mathrm{i}$ is the rise in the temperature of the ice to the melting point (K) and s is the specific latent heat of fusion of ice (J/kg).



Quantity	Value
of water c_w	$4190\,\mathrm{J/(kg\cdot K)}$
of ice c_i	$2100\,\mathrm{J/(kg\cdot K)}$
of ice s	$334\,\mathrm{kJ/kg}$

The possible end states are:

1. If $Q_\mathrm{w} > Q_\mathrm{i}$, all the ice melts and the resulting water warms to above $0\,^\circ\mathrm{C}$.
2. If $Q_\mathrm{i,warm} \le Q_\mathrm{w} \le Q_\mathrm{i}$, the final temperature is $0\,^\circ\mathrm{C}$. Some ice may be left, or all the ice may only just melt.
3. If $Q_\mathrm{w} < Q_\mathrm{i,warm}$, the water cooling to zero degrees is not enough on its own to warm all the ice to the melting point. Then part of the water may freeze and give out energy as it solidifies.

4. If even the energy released when all the water freezes is not enough to warm the ice to the melting point, the end state is ice only, at a temperature below $0\,^\circ\mathrm{C}$.

So before calculating the temperature, you must work out which phases are present in the end state.

If $Q_\mathrm{w} \geq Q_\mathrm{i}$, all the ice melts. The heat given out by the water is then used to warm and melt the ice, and the rest warms the whole amount of water $m_\mathrm{w} + m_\mathrm{i}$ to the temperature T. The energy balance is

$$\begin{aligned} &c_\mathrm{w} m_\mathrm{w} (T_\mathrm{w} - T) \\ &= c_\mathrm{i} m_\mathrm{i} \Delta T_\mathrm{i} + s m_\mathrm{i} \\ &\quad + c_\mathrm{w} m_\mathrm{i} T \end{aligned}$$

where the left-hand side is the quantity of heat given out by the water, and on the right are the warming of the ice, the melting of the ice and the warming of the meltwater. Solved:

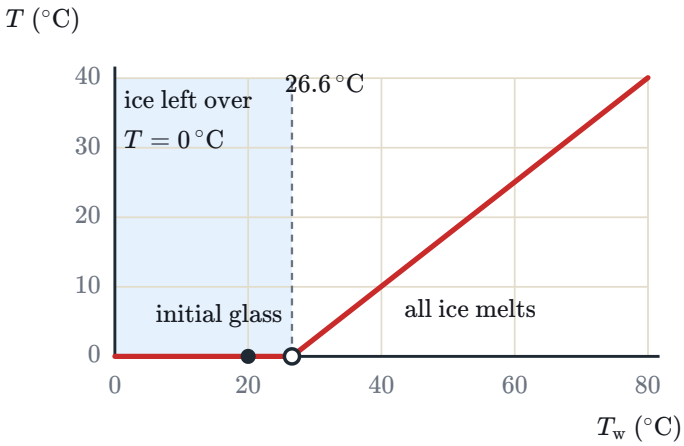
$$T = \frac{Q_\mathrm{w} - Q_\mathrm{i}}{c_\mathrm{w}(m_\mathrm{w} + m_\mathrm{i})}$$

In the borderline case $Q_\mathrm{w} = Q_\mathrm{i}$ the formula gives $T = 0\,^\circ\mathrm{C}$: all the ice has only just melted, but the meltwater does not warm up.

If $Q_\mathrm{w} < Q_\mathrm{i}$ but the water is enough to warm the ice to $0\,^\circ\mathrm{C}$, not all the ice has time to melt. The container then holds ice and water at the same time, so $T = 0\,^\circ\mathrm{C}$. The mass of melted ice m_melt is found from the heat that is left after warming the ice:

$$m_\mathrm{melt} = \frac{Q_\mathrm{w} - c_\mathrm{i} m_\mathrm{i} \Delta T_\mathrm{i}}{s}$$

The mass of ice that is left is $m_\mathrm{i} - m_\mathrm{melt}$. The graph shows how the final temperature depends on the initial temperature of the water when there is $0.30\,\mathrm{kg}$ of water and $0.10\,\mathrm{kg}$ of ice at 0 °C: until the heat given out by the water is enough to melt the ice, the temperature stays at $0\,^\circ\mathrm{C}$, and after that it rises. The black dot is the glass of the first figure, where the water is at 20 °C.



In case 3 the cold ice freezes some of the water. As long as some water is left, ice and water are present together and the temperature is $0\ ^\circ\mathrm{C}$. In case 4 all the water freezes, and the temperature of the ice is found from the same energy balance in the Level 3 exercise.



Ice cubes in a glass of water. The ice melts as heat flows into it from the surrounding water.

Photo: Srattha Nualsate, Pexels License

Common misconception

Many people calculate the final temperature as if the ice were just cold water, which forgets the latent heat of fusion. When $0.30\ \mathrm{kg}$ of water ($20\ ^\circ\mathrm{C}$) and $0.10\ \mathrm{kg}$ of ice ($0\ ^\circ\mathrm{C}$) are mixed, this method would give

$$
\begin{aligned}
T &= \frac{c_{\mathrm{w}} m_{\mathrm{w}} T_{\mathrm{w}} + c_{\mathrm{w}} m_{\mathrm{i}} T_{\mathrm{i}}}{c_{\mathrm{w}} (m_{\mathrm{w}} + m_{\mathrm{i}})} \\
&= \frac{0.30\,\mathrm{kg} \cdot 20\,^{\circ}\mathrm{C}}{0.40\,\mathrm{kg}} \\
&= 15\,^{\circ}\mathrm{C}
\end{aligned}
$$

In reality, melting all the ice would need $33.4\,\mathrm{kJ}$, but the water can give out only $25.1\,\mathrm{kJ}$ as it cools. So the final state is $0\,^{\circ}\mathrm{C}$, and $24.7\,\mathrm{g}$ of ice is left over. Another mistake is to put the temperature below $0\,^{\circ}\mathrm{C}$ when the container still holds ice and water: as long as both are present, the temperature is $0\,^{\circ}\mathrm{C}$. If your calculation gives a temperature below $0\,^{\circ}\mathrm{C}$ with a formula that assumes all the ice melts, the assumption is wrong: some ice is left, and the temperature is $0\,^{\circ}\mathrm{C}$. The exception is very cold ice that freezes all the water: then the container holds only ice, and its temperature is below $0\,^{\circ}\mathrm{C}$.

Summary

- In an insulated container, the quantity of heat given out by the water is equal to the quantity of heat taken in by the ice. The ice first warms to $0\,^{\circ}\mathrm{C}$ and then melts; during melting the temperature does not change.
- First compare the quantities of heat $Q_{\mathrm{w}} = c_{\mathrm{w}} m_{\mathrm{w}} \Delta T_{\mathrm{w}}$ and $Q_{\mathrm{i}} = c_{\mathrm{i}} m_{\mathrm{i}} \Delta T_{\mathrm{i}} + s m_{\mathrm{i}}$. If $Q_{\mathrm{w}} \geq Q_{\mathrm{i}}$, all the ice melts and $T = \frac{Q_{\mathrm{w}} - Q_{\mathrm{i}}}{c_{\mathrm{w}} (m_{\mathrm{w}} + m_{\mathrm{i}})}$; in the borderline case $Q_{\mathrm{w}} = Q_{\mathrm{i}}$ the temperature is exactly $0\,^{\circ}\mathrm{C}$.
- If $Q_{\mathrm{w}} < Q_{\mathrm{i}}$, some ice is left and the temperature is $0\,^{\circ}\mathrm{C}$. The mass of melted ice is $m_{\mathrm{melt}} = \frac{Q_{\mathrm{w}} - c_{\mathrm{i}} m_{\mathrm{i}} \Delta T_{\mathrm{i}}}{s}$. Exception: if the ice is so cold that all the water freezes, the temperature is below $0\,^{\circ}\mathrm{C}$.
- The specific latent heat of fusion of water is $334\,\mathrm{kJ/kg}$: melting one kilogram of ice needs the same energy as warming a kilogram of water by almost $80\,\mathrm{K}$.
- The final temperature is not the mean of the initial temperatures, and when ice and water are together in the same container in equilibrium at normal pressure, the temperature is $0\,^{\circ}\mathrm{C}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in

the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Ice melting in a calorimeter

A calorimeter contains $0.250\ \text{kg}$ of water at a temperature of $24.0\ ^\circ\text{C}$. $0.050\ \text{kg}$ of ice at $-8.0\ ^\circ\text{C}$ is dropped into the water. Does all the ice melt, and what is the temperature of the contents of the calorimeter at thermal equilibrium? The heat capacity of the calorimeter itself and heat losses to the surroundings are ignored.

STEP 1/7

Given and find. Water: $m_\text{w} = 0.250\ \text{kg}$, $T_\text{w} = 24.0\ ^\circ\text{C}$. Ice: $m_\text{i} = 0.050\ \text{kg}$, $T_\text{i} = -8.0\ ^\circ\text{C}$. From the table, $c_\text{w} = 4190\ \frac{\text{J}}{\text{kg·K}}$, $c_\text{i} = 2100\ \frac{\text{J}}{\text{kg·K}}$ and $s = 334\ \frac{\text{kJ}}{\text{kg}}$. We are asked whether all the ice melts, and for the final temperature T.

STEP 2/7

Principle. Energy is conserved, because the calorimeter takes in no heat and no heat escapes to the surroundings: the quantity of heat given out by the water is equal to the quantity of heat taken in by the ice. The ice first warms to $0\ ^\circ\text{C}$, then melts at a constant temperature, and finally the meltwater warms to the final temperature. So we must first check whether the heat given out by the water is enough to warm and melt all the ice. If it is, the ice melts completely and the final temperature is not below $0\ ^\circ\text{C}$.

STEP 3/7

Rearranged formula. When the water cools to $0\ ^\circ\text{C}$ it gives out a quantity of heat Q_w, and to melt, the ice needs a quantity of heat Q_i. When $Q_\text{w} > Q_\text{i}$, the surplus warms all the water $(m_\text{w} + m_\text{i})$ to the temperature T:

$$Q_\text{w} = c_\text{w} m_\text{w} \Delta T_\text{w}$$
$$Q_\text{i} = c_\text{i} m_\text{i} \Delta T_\text{i} + s m_\text{i}$$

$$T = \frac{Q_\text{w} - Q_\text{i}}{c_\text{w}(m_\text{w} + m_\text{i})}$$

STEP 4/7

Unit conversions. The masses are already in kilograms. A temperature change is the same in kelvins and in degrees Celsius: the water cools by

$$\begin{aligned}\Delta T_{\mathrm{w}} &= 24.0\,^{\circ}\mathrm{C} - 0\,^{\circ}\mathrm{C}\\ &= 24.0\,\mathrm{K}\end{aligned}$$

and the ice warms by

$$\begin{aligned}\Delta T_{\mathrm{i}} &= 0\,^{\circ}\mathrm{C} - (-8.0\,^{\circ}\mathrm{C})\\ &= 8.0\,\mathrm{K}\end{aligned}$$

The latent heat of fusion is $334\,\frac{\mathrm{kJ}}{\mathrm{kg}} = 334\,000\,\frac{\mathrm{J}}{\mathrm{kg}}$.

STEP 5/7

Substitution. We calculate the quantities of heat separately and then the temperature:

$$\begin{aligned}Q_{\mathrm{w}} &= c_{\mathrm{w}} m_{\mathrm{w}} \Delta T_{\mathrm{w}}\\ &= 4190\,\tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \cdot 0.250\,\mathrm{kg} \cdot 24.0\,\mathrm{K}\\ &= 25\,140\,\mathrm{J}\end{aligned}$$

$$\begin{aligned}Q_{\mathrm{i}} &= c_{\mathrm{i}} m_{\mathrm{i}} \Delta T_{\mathrm{i}} + s m_{\mathrm{i}}\\ &= 2100\,\tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \cdot 0.050\,\mathrm{kg} \cdot 8.0\,\mathrm{K}\\ &\quad + 334\,000\,\tfrac{\mathrm{J}}{\mathrm{kg}} \cdot 0.050\,\mathrm{kg}\\ &= 840\,\mathrm{J} + 16\,700\,\mathrm{J}\\ &= 17\,540\,\mathrm{J}\end{aligned}$$

$$\begin{aligned}T &= \frac{Q_{\mathrm{w}} - Q_{\mathrm{i}}}{c_{\mathrm{w}}(m_{\mathrm{w}} + m_{\mathrm{i}})}\\ &= \frac{25\,140\,\mathrm{J} - 17\,540\,\mathrm{J}}{4190\,\tfrac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}} \cdot 0.300\,\mathrm{kg}}\\ &= \frac{7600\,\mathrm{J}}{1257\,\tfrac{\mathrm{J}}{\mathrm{K}}}\\ &= 6.05\,^{\circ}\mathrm{C}\end{aligned}$$

STEP 6/7

Result. Because

$$\begin{aligned} Q_{\mathrm{w}} &= 25\,140\,\mathrm{J} \\ &> Q_{\mathrm{i}} \\ &= 17\,540\,\mathrm{J} \end{aligned}$$

all the ice melts. The final state is $0.300\,\mathrm{kg}$ of water at a temperature of $6.0\,^{\circ}\mathrm{C}$. The answer is given to two significant figures, because the mass of the ice, $0.050\,\mathrm{kg}$, is known only to two figures and it determines the precision of the quantity of heat $Q_{\mathrm{i}} = 17\,540\,\mathrm{J}$ (hundreds of joules); the difference $Q_{\mathrm{w}} - Q_{\mathrm{i}} = 7600\,\mathrm{J}$ is therefore a two-figure number.

STEP 7/7

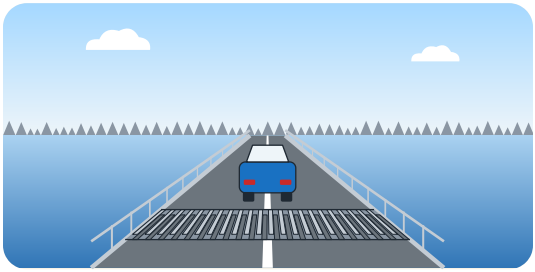
Reasonableness. The result lies between the initial temperatures $-8.0\,^{\circ}\mathrm{C}$ and $24.0\,^{\circ}\mathrm{C}$, but below $8.0\,^{\circ}\mathrm{C}$, the midpoint of the two, even though there is five times as much water: the melting of the ice pulls the result down. If the latent heat were forgotten, the result would be about $19\,^{\circ}\mathrm{C}$, so melting takes two thirds of the heat given out by the water ($16.7\,\mathrm{kJ}$ of $25.1\,\mathrm{kJ}$). The energy balance checks out: the water gives out a total of $4190 \cdot 0.250 \cdot (24.0 - 6.05) \approx 18.8\,\mathrm{kJ}$ and the ice takes in $840 + 16\,700 + 4190 \cdot 0.050 \cdot 6.05 \approx 18.8\,\mathrm{kJ}$. Try the same values in the simulation.

FY3 · CHAPTER 10

Thermal expansion

Why is there a comb-shaped steel joint at the end of a long bridge, which makes the car tyres thump as they cross it?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-8.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain with the particle model why a solid expands when it is heated.
- investigate by experiment what the change in the length of a rod depends on.
- calculate the change in the length and volume of an object when its temperature changes.

PRIOR KNOWLEDGE

- Temperature describes how strongly the particles move in random thermal motion; in a gas it tells you the average kinetic energy of the particles
  chapter *Thermodynamic system, state variables and temperature*
- A change in temperature is the same in kelvins and in degrees Celsius, $T = \theta + 273.15$
  chapter *Thermodynamic system, state variables and temperature*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 7.

Think about it

Steel feels completely rigid: you cannot make a steel railing longer, however hard you pull it. Yet the teeth of the comb-shaped joint at the end of a bridge interlock less on a frosty morning than on a hot afternoon, and nobody has pulled the bridge. Something changes the length of the steel just because of the temperature.

> **What makes rigid steel change its length, and what does the amount the joint has to give depend on?**

Theory

Key idea

Almost all substances expand when they are heated and contract when they are cooled. This phenomenon is called .



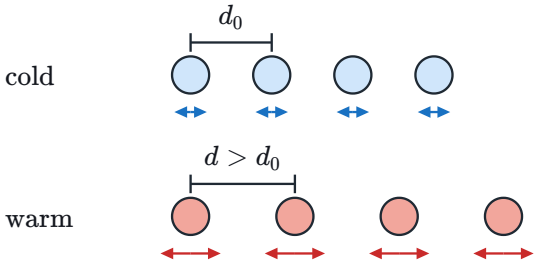
An expansion joint of the Kruunuvuori bridge in Helsinki in August 2026, before renovation. The expansion joint gives the bridge structures room to move, for example when the temperature changes.

Photo: Lasse Keskinen, CC BY 4.0

Why does a substance expand?

The micro model explains thermal expansion. In a solid the particles vibrate around their equilibrium positions. When the temperature rises, the vibration gets stronger.

It is easier for the particles to move apart than to be squeezed closer: particles pressed close together repel each other strongly, but the attraction between particles moving apart weakens gradually. So a particle swings farther from its neighbour than closer to it, and for more strongly vibrating particles the **average distance grows**. When all the gaps between the particles grow, the whole object expands.



particles the same size, gaps exaggerated

Thermal expansion in everyday life

The change is small but powerful. A steel rod gets longer by only about one hundred-thousandth of its length for each kelvin, but if the ends of the rod are fixed, the expanding steel pushes on them with an enormous force. That is why long structures have expansion joints. A steel bridge deck is shorter in a winter frost than in a summer heatwave, and the comb-shaped joint at the end of the bridge lets the deck move freely. For the same reason electric power lines hang lower in a heatwave than in a frost.

An object expands in all directions in the same proportion, as when a photograph is enlarged. A hole in a plate also gets bigger when the plate warms up. This is used in a shrink fit: a heated steel ring is slipped onto an axle, and as it cools it is pressed tightly in place.

Thermal expansion of different substances

Different substances expand differently. A bimetallic strip, in which two different metals are joined together, bends when it warms up, because one metal lengthens more. Such a strip opens and closes an electric circuit in old-style thermostats.

Liquids usually expand clearly more than solids, and gases much more still. The particles of a liquid do not stay in place, and the interactions between them are weaker than in a solid, so a rise in temperature moves them farther apart. The particles of a gas are far apart and almost free, so at constant pressure the volume of a gas grows much more. The volume of a gas also depends on pressure, and we return to this in the chapter *Changes of state of a gas and Boyle's law*. A liquid thermometer is based on the expansion of a liquid: when the temperature rises, the liquid column rises in a thin tube.

Water is an exception: between $0\,^\circ\mathrm{C}$ and $4\,^\circ\mathrm{C}$ it contracts when it warms up, so its density is greatest at about $4\,^\circ\mathrm{C}$. The densest, 4-degree water sinks to the bottom, and colder water stays at the surface and freezes. That is why a lake freezes from the surface, and even in winter there is water at about 4 °C at the bottom, where fish survive.

Thermal expansion of length and volume

When the temperature of an object changes by ΔT, its length changes by

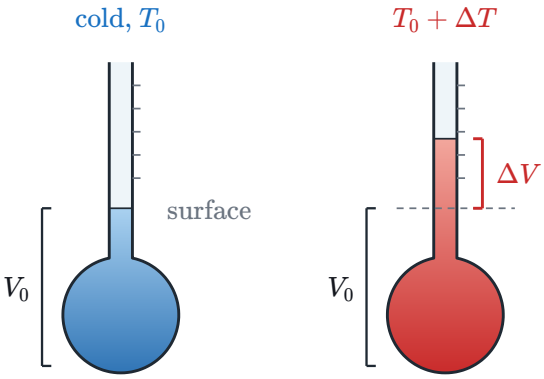
$$\Delta L = \alpha L_0 \Delta T$$

where ΔL is the change in length (m), α is the coefficient of linear expansion (1/K), L_0 is the original length (m) and ΔT is the change in temperature (K). The coefficient of linear expansion α tells you what fraction of its length an object lengthens when its temperature rises by $1\,\mathrm{K}$. So the change is proportional to the original length: a rod twice as long has twice as many gaps between particles, so it lengthens twice as much. The number 12 in the table means that $\alpha = 12 \cdot 10^{-6}\ \frac{1}{\mathrm{K}}$. The new length is $L = L_0 + \Delta L$. When the object cools, ΔT is negative, so ΔL is also negative, that is, the object gets shorter.

The change in the volume of liquids (and of gases at constant pressure) is calculated in the same way:

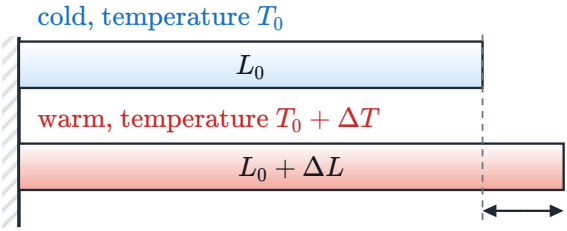
$$\Delta V = \gamma V_0 \Delta T$$

where ΔV is the change in volume (m³), γ is the coefficient of volume expansion (1/K) and V_0 is the original volume (m³). For a solid $\gamma \approx 3\alpha$, because an object expands in all three directions: length, width and height each grow in proportion $\alpha \Delta T$, and when the changes are small, the relative growth in volume is almost the sum of these, $3\alpha\Delta T$. The formulas are good models when the change in temperature is moderate. For solids that means, for example, at most about a hundred kelvins. For liquids, especially water, the coefficient γ changes clearly with temperature, so the values in the table at $20\,^\circ\mathrm{C}$ are used only close to $20\,^\circ\mathrm{C}$.



Substance	α (10^{-6} 1/K)
steel	12
aluminium	23
copper	17
ordinary glass	9
borosilicate glass	3.3

Substance ($20\,^\circ\mathrm{C}$)	γ (10^{-3} 1/K)
water	0.21
ethanol	1.1
petrol	0.95
air (at constant pressure)	3.4



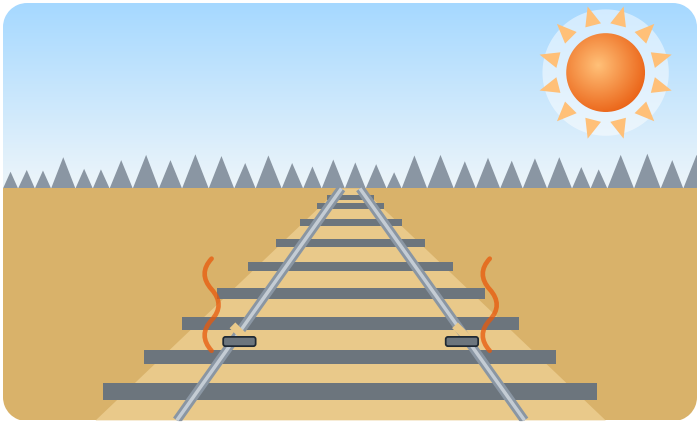
expansion exaggerated

Common misconception

Many people think that when a substance warms up, its particles themselves grow. In the micro model the particles stay the same size, and only their average distances grow. Another mistake is to think that a hole in a plate gets smaller because the plate supposedly expands from its edges towards the hole. The hole gets bigger in the same proportion as the plate does everywhere else. A third mistake is to assume that all substances always expand when they warm up: water contracts between $0\,^\circ\mathrm{C}$ and $4\,^\circ\mathrm{C}$, and when it freezes it expands. That is why ice floats and lakes freeze from the surface.

Summary

- When a substance warms up, the vibration of its particles gets stronger and their average distance grows, so the substance expands.
- The change in length is $\Delta L = \alpha L_0 \Delta T$ and the change in volume is $\Delta V = \gamma V_0 \Delta T$. For a solid $\gamma \approx 3\alpha$.
- Liquids expand more than solids, and gases much more than liquids.
- An object expands in all directions in the same proportion, so holes also get bigger.
- Water is an exception: its density is greatest at about $4\,^\circ\mathrm{C}$.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Thermal expansion of a steel bridge

The deck of a steel bridge is $120\,\mathrm{m}$ long. At the location the temperature varies during the year from a frost of $-30\,^{\circ}\mathrm{C}$ to a heatwave of $+30\,^{\circ}\mathrm{C}$, and the coefficient of linear expansion of steel is $12 \cdot 10^{-6}$ 1/K. By how much does the length of the deck change, that is, how much must the expansion joint give at least?

STEP 1/7

Given and find. The length of the deck is $L_0 = 120\,\mathrm{m}$, $\alpha = 12 \cdot 10^{-6}\,\frac{1}{\mathrm{K}}$ for steel (given in the problem), the lowest temperature is $-30\,^{\circ}\mathrm{C}$ and the highest $+30\,^{\circ}\mathrm{C}$. The question is the change ΔL in the length of the deck over the whole year.

STEP 2/7

Principle. The vibration of the particles of a solid gets stronger when it warms up and their average distance grows, so the deck gets longer. Steel stays solid all year and the deck is free to expand, so the change in length is proportional to the initial length and the change in temperature: $\Delta L = \alpha L_0 \Delta T$.

STEP 3/7

Rearranged formula. The quantity asked for is already alone on the left side of the formula, so the formula does not need rearranging:

$$\Delta L = \alpha L_0 \Delta T$$
$$\Delta T = T_{\max} - T_{\min}$$

STEP 4/7

Unit conversions. The change in temperature is the same in degrees Celsius and in kelvins, so

$$\begin{aligned} \Delta T &= 30\,^\circ\mathrm{C} - (-30\,^\circ\mathrm{C}) \\ &= 60\,\mathrm{K} \end{aligned}$$

The lengths are in metres, so ΔL comes out in metres; finally $1\,\mathrm{m} = 1000\,\mathrm{mm}$.

STEP 5/7

Substitution. Substitute the values with their units:

$$\begin{aligned} \Delta L &= \alpha L_0 \Delta T \\ &= 12 \cdot 10^{-6}\,\tfrac{1}{\mathrm{K}} \cdot 120\,\mathrm{m} \cdot 60\,\mathrm{K} \\ &= 0.0864\,\mathrm{m} \\ &= 86.4\,\mathrm{mm} \end{aligned}$$

STEP 6/7

Result. The length of the deck changes by about $86\,\mathrm{mm}$ during the year. The result is given to two significant figures. The limiting data are the coefficient $12 \cdot 10^{-6}\,\frac{1}{\mathrm{K}}$ and the difference of whole degrees $\Delta T = 60\,\mathrm{K}$ (both have two figures); the length $120\,\mathrm{m}$ has at least two figures and does not limit the result.

STEP 7/7

Reasonableness. A rough estimate:

$$10^{-5}\,\tfrac{1}{\mathrm{K}}\cdot 100\,\mathrm{m}\cdot 60\,\mathrm{K} = 60\,\mathrm{mm}$$

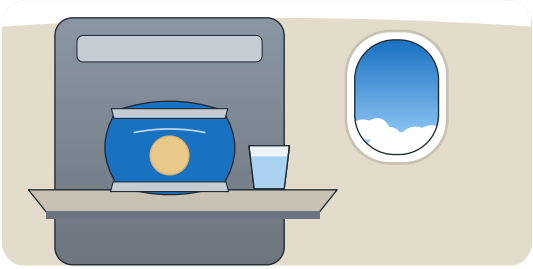
so the result is of the right order of magnitude. The deck gets almost nine centimetres longer by summer, because the temperature rises. That is only $\alpha\Delta T = 0.072\,\%$ of the length of the deck, but without an expansion joint the expanding steel would push on the ends of the bridge with such a large force that the structures would be damaged. That is why there is a comb-shaped joint at the end of the bridge, with teeth that slide between each other.

FY3 · CHAPTER 11

Changes of state of a gas and Boyle's law

Why does a sealed bag of crisps swell up tight on an aeroplane, even though nothing has been added to it?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-9.html`



IN THIS LESSON, YOU WILL LEARN TO

- use Boyle's law: in an isothermal change the product of pressure and volume stays constant.
- show inverse proportionality as a straight line by plotting pressure against the reciprocal of volume.
- explain how the volume of a gas depends on the temperature in kelvins at constant pressure.

PRIOR KNOWLEDGE

- Pressure is force per unit area, $p = F/A$, and atmospheric pressure is about 101 kPa
  chapter *Pressure and hydrostatic pressure*
- When the volume and the amount of gas stay the same, the pressure of a dilute gas is proportional to the temperature in kelvins, and the extension of the line meets absolute zero
  chapter *Thermodynamic system, state variables and temperature*
- Gases expand much more than liquids and solids when heated; for air $\gamma = 3.4 \cdot 10^{-3}\ \frac{1}{\mathrm{K}}$ at constant pressure
  chapter *Thermal expansion*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

At the departure gate the bag of crisps is loose, and you can press its sides in. At cruising altitude the same bag is plump and hard as a drum. The bag has not been opened, so no air has been added, and the cabin is not warmer than the departure gate. After landing the bag is loose again.

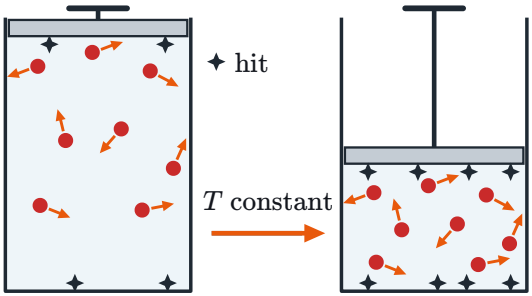
> **What factor affecting the air in the bag changes during the flight, and can the change in the volume of the air in the bag be calculated beforehand?**

Theory

Key idea

The state of a fixed amount of gas is described by its pressure p, volume V and temperature T. When one of them changes, at least one of the others usually changes too. This kind of change is called a change of state. It is easiest to study changes of state by keeping one of the quantities constant. In an isothermal change the temperature is constant, in an isobaric change the pressure, and in an isochoric change the volume.

According to the micro model, the pressure of a gas comes from particles colliding with the walls of the container. If the gas is compressed to half of its original volume and the temperature stays the same, the particles move as fast as before, but they hit the walls twice as often. The pressure doubles.



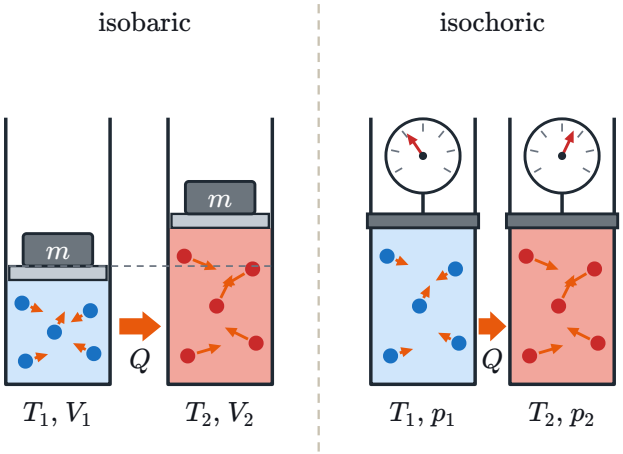
So in an isothermal change the pressure is inversely proportional to the volume, and the product of pressure and volume stays constant. This is called Boyle's law. It answers the opening question: the air pressure in the aeroplane cabin drops to about 78 kPa, so the air in the bag expands until the bag is tight. In a tight bag the pressure of the air stays slightly higher than the cabin pressure, and that is why the bag feels hard.

Boyle's law works well for air and other gases at ordinary pressures and temperatures. The amount of gas must stay the same, so no gas may leak out or come in. The pressure used is the total pressure of the gas, not the gauge pressure shown by a meter.

Boyle's law can be verified experimentally. Air is sealed in a syringe, and the syringe is connected to a pressure gauge. The plunger is pressed in steps, and the pressure is read at each volume. The measurement is done slowly so that the temperature of the air has time to return to room temperature. When the results

are plotted in a (V, p) coordinate system, the points lie on a falling curve, and it is hard to see from it whether the relationship is exactly inverse proportionality. So a new column $1/V$ is calculated in the table, and p is plotted as a function of $1/V$. If the law holds, within the measurement uncertainty the points lie on a straight line through the origin, and the slope of the line is the constant pV. Try it in the simulation: save measurements at different volumes and choose "Plot p as a function of 1/V".

When a gas is heated, its particles move faster. If the piston can move so that the pressure stays constant (an isobaric change), the gas expands. When the amount of gas stays the same and the gas is dilute, its volume is directly proportional to the temperature in kelvins. If the volume is instead kept constant (an isochoric change), the pressure rises, and it is directly proportional to the temperature in kelvins. In both cases the temperature must be given in kelvins, because the proportionality starts from absolute zero.





Filling a NASA BARREL research balloon in Antarctica. Helium fills only the top of the balloon, because the gas expands as the outside pressure falls while the balloon rises.

Photo: NASA/Goddard/BARREL/Nicky Knox, public domain

Boyle's law and changes of state of a gas

In an isothermal change, that is, according to Boyle's law,

$$p_1 V_1 = p_2 V_2$$

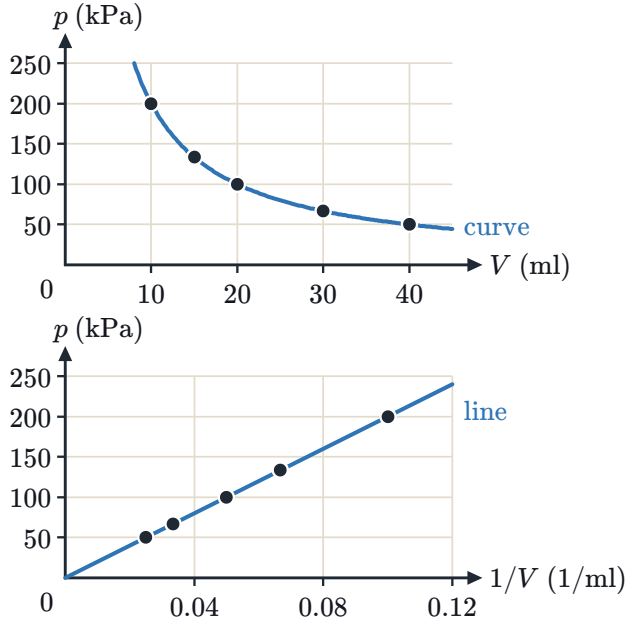
where p_1 and V_1 are the pressure (Pa) and volume (m³) of the gas in the initial state and p_2 and V_2 in the final state. The equation is the same as $pV = \text{constant}$. Dividing it by V gives

$$p = \text{constant} \cdot \frac{1}{V}$$

So the pressure is inversely proportional to the volume. If we write $x = 1/V$ and $y = p$, the equation has the form $y = kx$, like the equation of a straight line in FY1. So the $(1/V, p)$ graph is a straight line through the origin, and its slope is $k = pV$.

The unit of the product pV is $\mathrm{Pa} \cdot \mathrm{m}^3 = \mathrm{N/m^2} \cdot \mathrm{m}^3 = \mathrm{N} \cdot \mathrm{m} = \mathrm{J}$, and $1\,\mathrm{kPa} \cdot 1\,\mathrm{l} = 1\,\mathrm{J}$. Because both sides of the equation $p_1 V_1 = p_2 V_2$ have the same

units, pressure and volume can also be given in, for example, kilopascals and litres, as long as they are the same on both sides.



In an isobaric and in an isochoric change

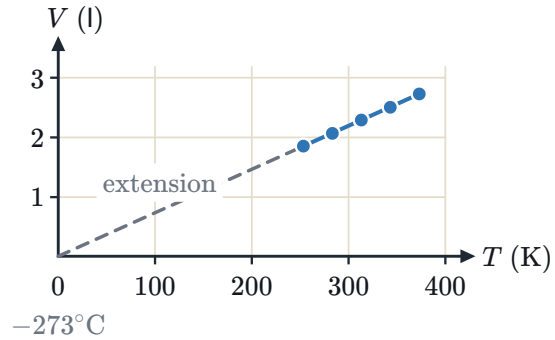
$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad \text{and} \qquad \frac{p_1}{T_1} = \frac{p_2}{T_2}$$

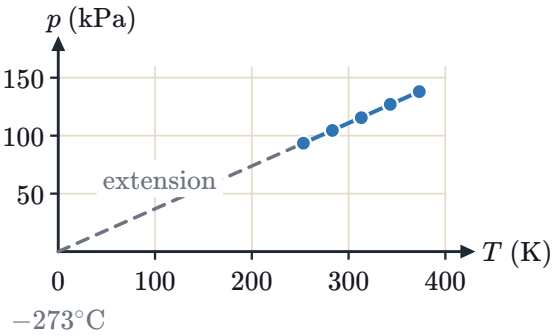
where T_1 and T_2 are the temperatures of the gas in kelvins (K) in the initial and final state. So the volume or the pressure of a closed, dilute gas is directly proportional to the temperature in kelvins, and the extension of the straight line in a (T, V) or (T, p) graph passes through the origin, that is, through absolute zero, $0\,\mathrm{K}$. The extension is a prediction of the model: a real gas condenses into a liquid before $0\,\mathrm{K}$ is reached.

The temperature in kelvins is obtained from the Celsius temperature θ by adding 273.15:

$$T = \theta + 273.15\,\mathrm{K}$$

where θ is the temperature in degrees Celsius as a number; in calculations $273\,\mathrm{K}$ is enough.





Common misconception

Three mistakes are often repeated:

- **Inverse proportionality is not a falling straight line.** When the volume doubles, the pressure halves, so the (V, p) graph is a curve that approaches the axes but never meets them. A straight line is obtained only when p is plotted as a function of $1/V$.
- **Celsius degrees do not work in a ratio of temperatures.** When $1.00\,\mathrm{l}$ of air is heated at constant pressure from $10\,^{\circ}\mathrm{C}$ to $20\,^{\circ}\mathrm{C}$, its volume does not double. Calculated in kelvins, $V_2 = V_1 T_2 / T_1 = 1.00\,\mathrm{l} \cdot 293\,\mathrm{K} / 283\,\mathrm{K} \approx 1.04\,\mathrm{l}$, so the volume increases by only about $3.5\,\%$.
- **The pressure of a gas in a closed container does not come from the weight of the gas.** In a small container the weight of the gas is negligible, and the pressure comes from the collisions of the particles, so it is equal everywhere in the container.

Summary

- In a change of state, the pressure, volume and temperature of a gas change. In an isothermal change T is constant, in an isobaric change p, and in an isochoric change V.
- Boyle's law: in an isothermal change $p_1 V_1 = p_2 V_2$, so the pressure is inversely proportional to the volume.
- Inverse proportionality is verified by calculating a column $1/V$ in the table and plotting p as a function of $1/V$: the points must lie on a straight line through the origin, within the measurement uncertainty.
- When the amount of gas stays the same, the volume of a dilute gas at constant pressure, and the pressure at constant volume, is directly proportional to the temperature in kelvins. In these calculations the temperatures are always given in kelvins.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A crisp packet on a plane

At the airport, a bag of crisps contains $0.80\ \text{l}$ of air at a pressure of 101 kPa. During the flight the cabin pressure is 78 kPa. What is the volume of the air in the bag during the flight, if the temperature stays the same and the bag can expand freely?

STEP 1/7

Given and find. We know $p_1 = 101\ \text{kPa}$, $V_1 = 0.80\ \text{l}$ and $p_2 = 78\ \text{kPa}$. The temperature is the same. We want the final volume V_2 in litres.

STEP 2/7

Principle. The bag is sealed, so the amount of air does not change, and the temperature stays the same. So the change is isothermal and Boyle's law holds: the product pV stays constant. The bag expands freely, so the pressure inside it is always the same as in the cabin.

STEP 3/7

Rearranged formula. Solve V_2 from Boyle's law by dividing both sides by the pressure p_2:

$$\begin{aligned} p_1 V_1 &= p_2 V_2 \\ V_2 &= \frac{p_1 V_1}{p_2} \end{aligned}$$

STEP 4/7

Unit conversions. No units need to be converted. Both pressures are in kilopascals, so they cancel in the quotient, and the volume comes out directly in litres.

STEP 5/7

Substitution. Substitute the values with their units:

$$\begin{aligned} V_2 &= \frac{p_1 V_1}{p_2} \\ &= \frac{101\,\mathrm{kPa} \cdot 0.80\,\mathrm{l}}{78\,\mathrm{kPa}} \\ &= 1.036\,\mathrm{l} \end{aligned}$$

STEP 6/7

Result. The volume of the air is $V_2 \approx 1.0\,\mathrm{l}$, to two significant figures like the initial data $0.80\,\mathrm{l}$ and $78\,\mathrm{kPa}$. The volume increases by about $30\,\%$. If the bag is already almost full, it swells tight.

STEP 7/7

Reasonableness. The pressure decreases, so the volume must increase. The factor of increase is the pressure ratio $101/78 \approx 1.3$, and $0.80\,\mathrm{l} \cdot 1.3 \approx 1.0\,\mathrm{l}$. At the end the product pV is

$$\begin{aligned} &78\,\mathrm{kPa} \cdot 1.036\,\mathrm{l} \\ &\quad \approx 80.8\,\mathrm{kPa} \cdot \mathrm{l} \end{aligned}$$

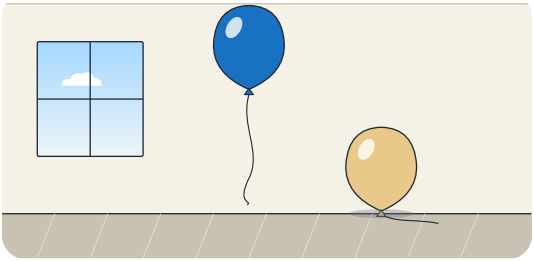
the same as at the start, $101\,\mathrm{kPa} \cdot 0.80\,\mathrm{l} = 80.8\,\mathrm{kPa} \cdot \mathrm{l}$.

FY3 · CHAPTER 12

Ideal gas law $pV = nRT$

Why is a helium balloon much lighter than an air balloon of the same size? Does it contain less gas?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-10.html`



IN THIS LESSON, YOU WILL LEARN TO

- work out what pressure, volume and temperature tell about the amount of gas.
- calculate quantities describing the state of a gas with the equation of state of an ideal gas.
- explain gas pressure as collisions of particles with the walls of the container.

PRIOR KNOWLEDGE

- Boyle's law $p_1 V_1 = p_2 V_2$ at constant temperature
  chapter *Changes of state of a gas and Boyle's law*
- When the amount of gas stays the same, the pressure of a dilute gas at constant volume and its volume at constant pressure are proportional to the temperature in kelvins
  chapter *Changes of state of a gas and Boyle's law*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

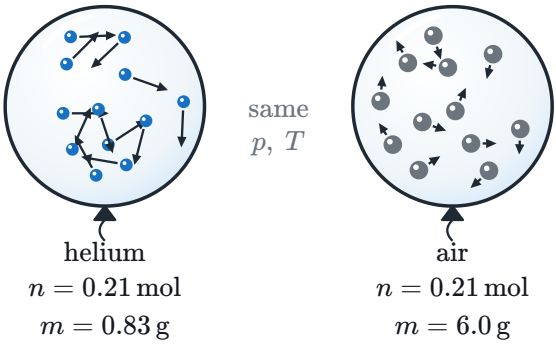
At a party two balloons of the same size are filled: one from a helium cylinder, the other with a pump full of air. Both are at room temperature, and the pressure inside each is almost the same as the outside air. The helium balloon rises to the ceiling, and the air balloon falls to the floor. The gas is invisible, and you cannot count its particles one by one.

> **How could you compare the amounts of gas in the two balloons when you know only the pressure, volume and temperature?**

Theory

Key idea

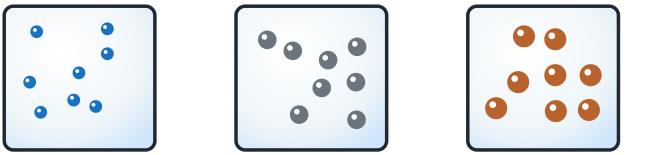
The gas laws in the chapter *Changes of state of a gas and Boyle's law* applied to a fixed amount of gas. Now we compare different amounts of gas and different gases. Fill a rigid 5.00 litre glass flask, first pumped empty, at a pressure of $101\ \mathrm{kPa}$ and a temperature of $20\ ^\circ\mathrm{C}$, one after another, with helium, nitrogen, air and carbon dioxide. In an example weighing (invented values with some scatter of weighing) the masses of the gases are very different: $0.83\ \mathrm{g}$, $5.78\ \mathrm{g}$, $6.02\ \mathrm{g}$ and $9.10\ \mathrm{g}$. But when you divide each mass by the molar mass of the gas, the result is about $0.207\ \mathrm{mol}$ every time. So at the same pressure and temperature, the same volume contains the same amount of gas, whatever the gas is. The helium balloon is lighter because helium particles are lighter, not because there is less gas.



Even a small amount of gas contains an enormous number of particles. So we do not count them one by one but in moles, in the same way as eggs are counted in dozens. One mole is $6.022 \cdot 10^{23}$ particles, and the number $N_\mathrm{A} = 6.022 \cdot 10^{23}\ 1/\mathrm{mol}$ is called the Avogadro constant. The amount of substance n tells you how many moles of particles there are. The mass of one mole is the molar mass M of the substance: $4.00\ \mathrm{g/mol}$ for helium, $29.0\ \mathrm{g/mol}$ for air on average and $44.0\ \mathrm{g/mol}$ for carbon dioxide. So you can get the amount of substance both from the number of particles and from the mass:

$$n = \frac{N}{N_\mathrm{A}} = \frac{m}{M}$$

where N is the number of particles, m is the mass (g or kg) and M is the molar mass (g/mol or kg/mol).



helium
$M = 4.00\,\mathrm{g/mol}$
$m = 4.00\,\mathrm{g}$

air
$M = 29.0\,\mathrm{g/mol}$
$m = 29.0\,\mathrm{g}$

carbon dioxide
$M = 44.0\,\mathrm{g/mol}$
$m = 44.0\,\mathrm{g}$

in each container $n = 1.00\,\mathrm{mol}$, same p, T

The ideal gas is a microscopic model. In it, the particles are thought of as points, and they interact with each other and with the walls of the container only when they collide. So the energy of the interactions between the particles is negligible in the model, and the temperature in kelvins is directly proportional to the mean kinetic energy of the particles. Pressure comes from collisions with the walls. It grows when there are more particles, when they move faster (that is, the temperature is higher) or when the volume is smaller. Real gases follow the model well in ordinary conditions. The model starts to fail at high pressures and low temperatures, when the particles are close together and the attraction between them condenses the gas into a liquid.

Equation of state of an ideal gas

The example weighing gives a number that is the same for all gases: pressure multiplied by volume and divided by the amount of substance and the temperature in kelvins. With the mean amount of substance of the four gases, $0.2071\,\mathrm{mol}$, it is

$$\begin{aligned} R &= \frac{pV}{nT} \\ &= \frac{101\,000\,\mathrm{Pa} \cdot 0.00500\,\mathrm{m}^3}{0.2071\,\mathrm{mol} \cdot 293.15\,\mathrm{K}} \\ &\approx 8.3\,\tfrac{\mathrm{J}}{\mathrm{mol \cdot K}} \end{aligned}$$

Precise measurements give the value $R = 8.31\,\frac{\mathrm{J}}{\mathrm{mol \cdot K}}$ for all gases, as long as the pressure is not very high and the temperature is not very low. This number is called the molar gas constant. From it we get the equation of state of an ideal gas

$$pV = nRT$$

where p is the absolute pressure of the gas, V the volume, n the amount of substance, R the molar gas constant and T the temperature in kelvins. The type of gas does not appear in the equation at all. A temperature θ given in degrees Celsius is converted to kelvins by adding 273.15 to the numerical value.

When the value of the gas constant

$$R = 8.31\ \tfrac{\mathrm{J}}{\mathrm{mol \cdot K}}$$

is used, the unit of the product pV must be the joule. Usually the pressure is converted to pascals and the volume to cubic metres, and then

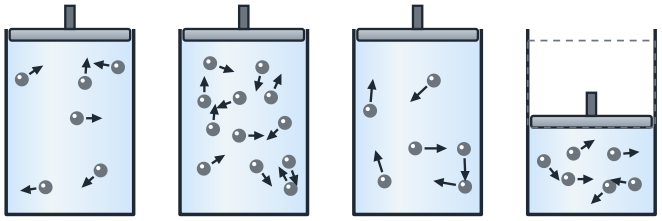
$$1\,\mathrm{Pa} \cdot \mathrm{m}^3 = 1\ \tfrac{\mathrm{N}}{\mathrm{m}^2} \cdot \mathrm{m}^3 = 1\,\mathrm{N} \cdot \mathrm{m} = 1\,\mathrm{J}$$

Kilopascals and litres can also be combined, because

$$1\,\mathrm{kPa} \cdot \mathrm{l} = 1000\,\mathrm{Pa} \cdot 0.001\,\mathrm{m}^3 = 1\,\mathrm{J}$$

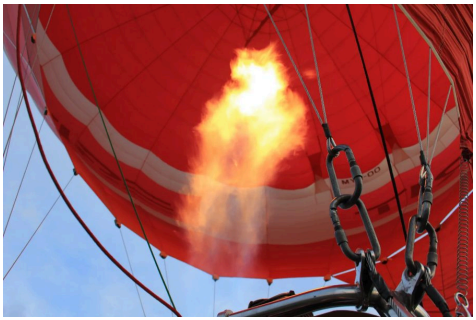
What matters most is to handle the units consistently throughout the calculation.

When you solve the equation of state for pressure, you see that the pressure doubles when the number of particles or the temperature in kelvins doubles, or when the volume is halved. So the pressure is directly proportional to the number of particles and the temperature in kelvins and inversely proportional to the volume, that is, $p \propto NT/V$.



In the simulation you can investigate this at the particle level: change the number of particles, the temperature and the volume, and compare the pressure measured at the walls with the prediction of the equation of state, which you calculate yourself. Collisions between the particles do not change the pressure on average, so they have been left out of the simulation. In the simulation the pressure is shown relative to a reference state (p/p_{ref}), not in pascals. The

reference state is $N = 100$ particles, $T = 300\,\mathrm{K}$ and the full volume ($V = 100\,\%$); in it $p/p_{\mathrm{ref}} = 1$, and the value 2 means twice the reference pressure. The number of particles N represents the amount of gas.



The burner of a hot air balloon heats the air inside the balloon. The mouth of the balloon is open at the bottom, so the pressure of the air inside stays almost equal to the pressure outside.

Photo: Sunridin, CC BY 3.0

When the amount of gas does not change, nR is constant, and the equation of state gives the combined gas law $p_1V_1/T_1 = p_2V_2/T_2$ (subscripts 1 and 2 are the initial and final state). From it you get every gas law of the chapter *Changes of state of a gas and Boyle's law* when one quantity is kept constant.

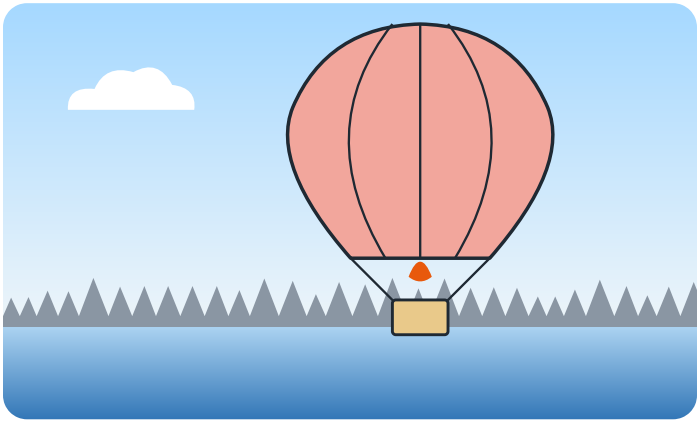
Common misconception

Three mistakes keep appearing in calculations with the equation of state:

- **Degrees Celsius.** In the equation of state the temperature is always in kelvins. For example, $0\,^{\circ}\mathrm{C}$ would give a pressure of zero, but a gas has a pressure at the freezing point too.
- **Mixed units.** If the pressure is in kilopascals but the volume in cubic metres, or the pressure in pascals but the volume in litres, the result is wrong by a factor of a thousand. The safest way is to convert all quantities to base units: pressure to pascals, volume to cubic metres and temperature to kelvins.
- **Heavy gas.** Many people think that fewer moles of a heavier gas fit in the same volume. But the amount of substance depends only on the pressure, the volume and the temperature, not on the type of gas. The mass, on the other hand, depends on the molar mass.

Summary

- The equation of state of an ideal gas is $pV = nRT$, where $R = 8.31\ \frac{\mathrm{J}}{\mathrm{mol \cdot K}}$ and T is in kelvins.
- The amount of substance n tells the amount of gas in moles: $n = N/N_\mathrm{A} = m/M$.
- The pressure is proportional to the expression NT/V: the pressure doubles when the number of particles or the temperature in kelvins doubles, or when the volume is halved.
- When the amount of gas stays the same, $p_1V_1/T_1 = p_2V_2/T_2$, and from it you get the separate gas laws.
- At the same pressure, temperature and volume there is always the same number of moles of gas, whatever the type of gas; weighing gives the same m/M for all gases.
- The ideal gas model works well in ordinary conditions and fails at high pressures and low temperatures.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Amount of helium in a balloon

A party balloon is filled with helium until its volume is $5.0\,\mathrm{l}$. In the balloon the pressure is $101\,\mathrm{kPa}$ and the temperature is $20\,^\circ\mathrm{C}$. How many moles of helium are in the balloon, and what is the mass of the helium? The molar mass of helium is $4.0\,\mathrm{g/mol}$.

STEP 1/7

Given and find. The volume of the balloon is $V = 5.0\,\mathrm{l}$, the pressure is $p = 101\,\mathrm{kPa}$, the temperature is $\theta = 20\,^\circ\mathrm{C}$ and the molar mass of helium is $M = 4.0\,\mathrm{g/mol}$. We want the amount of substance n and the mass m.

STEP 2/7

Principle. The balloon contains a certain amount of helium in a certain volume, at a certain pressure and temperature, and helium behaves as an ideal gas at ordinary pressure and room temperature. So the equation of state $pV = nRT$ gives the amount of substance, and the molar mass converts the amount of substance into mass.

STEP 3/7

Rearranged formula. Solve the amount of substance from the equation of state and the mass from the equation $n = m/M$:

$$n = \frac{pV}{RT}$$
$$m = nM$$

STEP 4/7

Unit conversions. The equation of state needs base units:

- volume

$$\begin{aligned} V &= 5.0\,\mathrm{l} \\ &= 0.0050\,\mathrm{m^3} \end{aligned}$$

- pressure

$$\begin{aligned} p &= 101\,\mathrm{kPa} \\ &= 101\,000\,\mathrm{Pa} \end{aligned}$$

- temperature

$$\begin{aligned} T &= (20 + 273.15)\,\mathrm{K} \\ &= 293.15\,\mathrm{K} \end{aligned}$$

- the molar mass stays as $M = 4.0\,\mathrm{g/mol}$, so the mass comes out in grams

STEP 5/7

Substitution. Substitute the values with their units:

$$\begin{aligned} n &= \frac{pV}{RT} \\ &= \frac{101\,000\,\mathrm{Pa} \cdot 0.0050\,\mathrm{m^3}}{8.31\,\frac{\mathrm{J}}{\mathrm{mol \cdot K}} \cdot 293.15\,\mathrm{K}} \\ &= 0.2073\,\mathrm{mol} \approx 0.21\,\mathrm{mol} \end{aligned}$$

STEP 6/7

Result. The mass of the helium is calculated with the more precise amount of substance:

$$\begin{aligned} m &= nM \\ &= 0.2073\,\mathrm{mol} \cdot 4.0\,\mathrm{g/mol} \\ &\approx 0.83\,\mathrm{g} \end{aligned}$$

The balloon contains $0.21\,\mathrm{mol}$ of helium, and its mass is $0.83\,\mathrm{g}$. The amount of substance is limited by the volume $5.0\,\mathrm{l}$, and the mass by the molar mass $4.0\,\mathrm{g/mol}$ as well, each with two digits, so the results are given with two significant figures. The pressure $101\,\mathrm{kPa}$ and the temperature $293\,\mathrm{K}$ have three digits and do not limit the result.

STEP 7/7

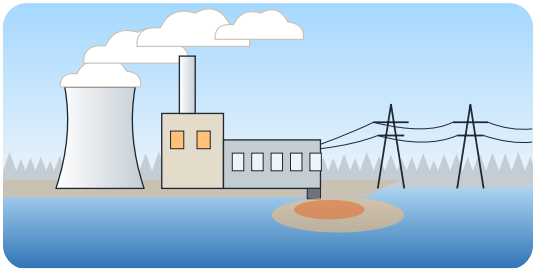
Reasonableness. One mole of gas takes up about $24\,\mathrm{l}$ at room temperature and normal pressure, so $5.0\,\mathrm{l}$ contain about $5.0\,\mathrm{l} : 24\,\mathrm{l/mol} \approx 0.21\,\mathrm{mol}$ of gas. An air balloon of the same size contains the same number of moles of air, but its mass is about $0.2073\,\mathrm{mol} \cdot 29\,\mathrm{g/mol} \approx 6.0\,\mathrm{g}$. The helium balloon is lighter because helium particles are lighter, not because there is less gas.

FY3 · CHAPTER 13

Work done by a gas and the heat engine

Why does a thermal power plant get less than half of the energy of its fuel as electricity, and where does the rest of the energy go?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-11.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the work done by an expanding gas and interpret it as an area in the pressure–volume graph.
- explain why a heat engine turns heat into work only by releasing part of it to something colder.
- calculate the efficiency of a heat engine.

PRIOR KNOWLEDGE

- The first law $\Delta U = Q + W$, where W is the work done on the system by the surroundings, and conservation of energy
  chapter *Internal energy, heat and conservation of energy*
- Isobaric change of state: at constant pressure the volume of a gas grows when its temperature rises
  chapter *Changes of state of a gas and Boyle's law*
- Work is force times distance, and the area under an (s, F) graph
  chapter *Work and energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

In the cylinder of a museum locomotive, hot steam pushes a piston, and the piston turns the wheels of the locomotive. A large condensing power plant does basically the same thing: steam expands in a turbine and turns a generator. The plant is supplied with the energy of fuel at a rate of $1000\ \mathrm{MW}$, but only about $400\ \mathrm{MW}$ goes to the electricity grid. Energy is still conserved, as the first law says.

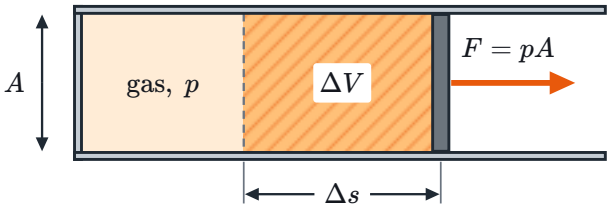
> **Could better technology turn all the energy of the fuel into electricity, and can the part that does not**

become electricity still be put to use?

Theory

Key idea

When a gas expands and pushes a piston, it does work. The gas pushes the piston with a force $F = pA$, where A is the area of the piston. If the piston moves a distance s and the pressure stays the same, the work done by the gas is force times distance, pAs. The product As is the change in the volume of the gas ΔV, so the work done by the gas is $W_{\mathrm{g}} = p\,\Delta V$. In the (V, p) graph it is the area between the graph and the V axis, just like the work in an (s, F) graph. When the gas is compressed, the surroundings do work on the gas, and W_{g} is negative. A gas that does work gives up energy, so in this lesson the first law is written in the form $\Delta U = Q - W_{\mathrm{g}}$.



$$W_{\mathrm{g}} = F\,\Delta s = pA\,\Delta s = p\,\Delta V$$

A heat engine repeats its motion

A heat engine turns heat into work continuously, so it has to repeat its motion. The piston has to be brought back down. If it is pressed down at the same pressure, the return takes all the work done by the gas. The return takes less work if the gas is first cooled to a lower pressure, but then the gas gives up heat to a colder environment. In a steam power plant, water boils in the boiler, the hot steam expands in the turbine and turns the generator, and the steam coming out of the turbine condenses back into water in the condenser, which is cooled by seawater. The working substance goes round the same path again and again, so at the end of a cycle its internal energy is the same as at the start. Energy is conserved: the

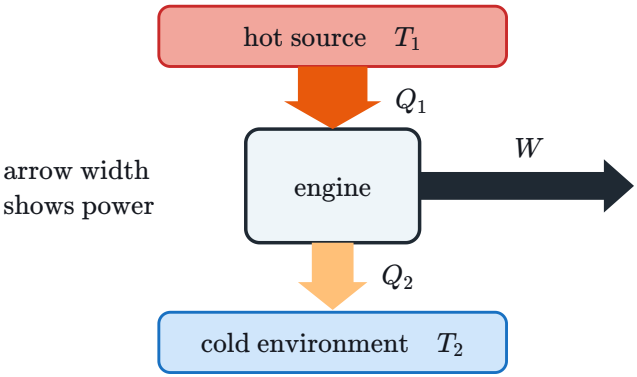
heat Q_1 from the hot source equals the work W done by the engine plus the heat Q_2 given up to the colder environment.



In a steam locomotive the expanding steam pushes a piston, and the connecting rod turns the piston's back-and-forth motion into the rotation of the wheels.

Photo: Masood Aslami, Pexels License

Experience shows that a heat engine cannot turn all the heat it receives into work; part of it must always be given up to a colder environment. This is the content of the second law of thermodynamics. The larger the temperature difference between the hot source and the cold environment compared with the temperature of the hot source, the larger the share of the heat the engine can turn into work. Cyclic processes are not calculated in this course; the simulation only shows why the heat Q_2 cannot be avoided.



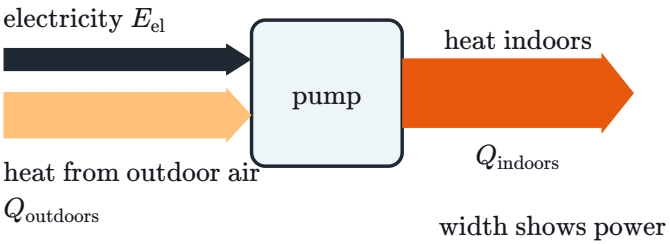
Efficiency

Efficiency tells how large a share of the energy supplied to a machine is put to use, so its value depends on what is counted as useful. In the efficiency of a power plant's electricity production, only the electricity is useful. A Finnish condensing power plant that produces only electricity gets about $40\ \%$ of the energy of its fuel as electricity, and a nuclear power plant about $35\ \%$. Most of the rest is transferred to the sea through the condenser, and in a plant that burns fuel part also leaves into the air with the flue gases. The efficiency of a car engine is about $30\ \%$, and the rest of the energy leaves with the exhaust gases and the radiator.

In combined production of electricity and heat, the heat Q_2 of the condenser is recovered: it heats the water of the district heating network, which carries it to the houses of the city. So the heat engine still gives up heat to something colder, but this heat is put to use. In the overall efficiency, the electricity and the district heating together count as useful, and it is 85–$90\ \%$. The efficiency of electricity production therefore does not rise; it is usually lower than when only electricity is produced. This answers the opening question: less than half becomes electricity, but the rest does not disappear; it is transferred as heat to the environment or to the district heating network.

Refrigerator and heat pump

A refrigerator and a heat pump work the opposite way to a heat engine: they use electrical energy to transfer heat from a cold place to a warm one. A refrigerator transfers heat from inside the cabinet to the room, and an air-source heat pump from the outdoor air to the inside. The heat given out indoors is the sum of the heat taken from outside and the electricity used, so a pump can heat a house with, for example, three times as much heating power as the electric power it uses. The ratio of the heating power to the electric power is called the coefficient of performance. The coefficient of performance is not an efficiency: it can be greater than 1, because part of the heat that comes indoors is transferred from the outdoor air and does not come from the electricity. The pump does not create energy; it transfers it.



Work done by a gas and efficiency

When the pressure of a gas stays constant, the work done by the gas is

$$W_{\text{g}} = p\,\Delta V$$

where W_{g} is the work done by the gas (J), p the pressure of the gas (Pa) and ΔV the change in the volume of the gas (m^3). The work is positive when the gas expands and negative when the gas is compressed. The change in the internal energy of the gas is

$$\Delta U = Q - W_{\text{g}}$$

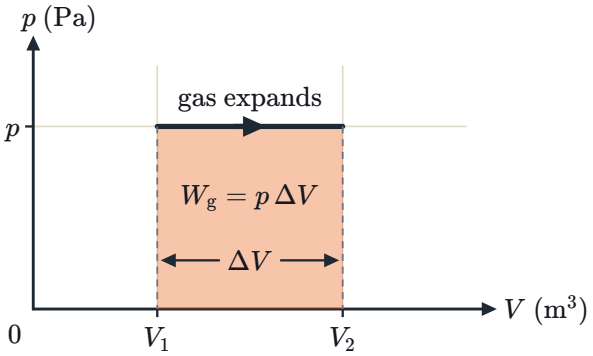
where Q is the quantity of heat transferred to the gas (J). This is the first law $\Delta U = Q + W$ presented in the chapter *Internal energy, heat and conservation of energy*, where W is the work done on the gas by the surroundings: $W = -W_{\text{g}}$.

During one cycle of a heat engine, the work done and the efficiency are

$$W = Q_1 - Q_2$$

$$\eta = \frac{W}{Q_1}$$

where Q_1 is the heat transferred to the engine from the hot source (J), Q_2 the heat given up to the colder environment (J), W the work done by the engine during the cycle (J) and η the efficiency, which is often given as a percentage. Efficiency can be calculated in the same way from powers, for example the electric power divided by the fuel power.



A heat pump gives out indoors a quantity of heat that is the sum of the heat taken from the outdoor air and the electrical energy used by the pump:

$$Q_{\text{indoors}} = Q_{\text{outdoors}} + E_{\text{el}}$$

The coefficient of performance tells how many times as much heat the pump gives out indoors as the electrical energy it uses:

$$\varepsilon = \frac{Q_{\text{indoors}}}{E_{\text{el}}}$$

where Q_{indoors} is the heat given out indoors (J), Q_{outdoors} the heat taken from the outdoor air (J), E_{el} the electrical energy used by the pump (J) and ε the coefficient of performance. The coefficient of performance can be calculated in the same way from powers.

Common misconception

Many people think that better technology could make a heat engine almost perfect, so that all the heat would turn into work. Reducing friction and other losses helps, but part of the heat must always be given up to a colder environment, so the efficiency of a heat engine stays clearly below $100\ \%$. The frictionless piston of the simulation shows the reason: for the piston to be brought down with less work than it took to raise it, the gas has to be cooled, and in the cooling heat is transferred to the cold water. The limit applies to a machine that turns heat into work, not to all devices: for example, in an electric heater practically all the electrical energy increases the internal energy of the resistor and is transferred as heat to the room.

Another mistake is to think of a heat pump as a perpetual motion machine, because it gives out more heat than it uses electricity. The extra heat comes from the outdoor air, and energy is not created out of nothing. A third mistake is to think

that a pressurised gas does work just because its pressure is high. Work is done only when the volume changes and the piston moves.

Summary

- A gas that expands at constant pressure does the work $W_{\mathrm{g}} = p\,\Delta V$, which is the area under the (V, p) graph. The internal energy changes by $\Delta U = Q - W_{\mathrm{g}}$.
- A piston machine can repeat its work only if the gas is cooled before the piston is returned; then heat is transferred to something colder.
- A heat engine takes heat Q_1 from a hot source, does work W and gives up heat Q_2 to a colder environment: $W = Q_1 - Q_2$.
- Efficiency is $\eta = W / Q_1$. A heat engine cannot turn all the heat into work.
- The efficiency of electricity production of a condensing power plant is about $40\ \%$. In combined production of electricity and heat the heat of the condenser is used as district heating, so the overall efficiency, in which electricity and heat both count as useful, is 85–$90\ \%$.
- A refrigerator and a heat pump use work to transfer heat from cold to warm. They do not create energy. The coefficient of performance of a heat pump, $\varepsilon = Q_{\mathrm{indoors}} / E_{\mathrm{el}}$, is not an efficiency, and it can be greater than 1, because part of the heat comes from the outdoor air.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Efficiency of a biomass plant

Wood chip energy is fed into the boiler of a biopower plant at a power of $500\ \mathrm{MW}$. When the plant produces only electricity, the electric power is $190\ \mathrm{MW}$. In winter the same plant works as a combined heat and power plant: it produces $150\ \mathrm{MW}$ of electricity and $290\ \mathrm{MW}$ of district heating. What is the efficiency in each case, and how much power is transferred to the environment as waste heat?

STEP 1/7

Given and find. Fuel power $P_1 = 500\,\mathrm{MW}$. When only electricity is produced, the electric power is $P_\mathrm{e} = 190\,\mathrm{MW}$; in combined production $P_\mathrm{e} = 150\,\mathrm{MW}$ and the district heating power is $P_\mathrm{dh} = 290\,\mathrm{MW}$. The efficiencies η and the power of the waste heat P_2 are asked for in both cases.

STEP 2/7

Principle. Efficiency is the useful power divided by the power supplied; it is calculated from powers just like from energies, because both are energy in the same time. Energy is conserved, so the fuel power is divided into the useful power and the waste heat. In electricity production alone only the electricity is useful; in combined production both the electricity and the district heating are.

STEP 3/7

Rearranged formula. Solve for the quantities asked for, with symbols before numbers:

$$\eta = \frac{P_\mathrm{useful}}{P_1}$$
$$P_2 = P_1 - P_\mathrm{useful}$$

For electricity alone $P_\mathrm{useful} = P_\mathrm{e}$, and for combined production $P_\mathrm{useful} = P_\mathrm{e} + P_\mathrm{dh}$.

STEP 4/7

Unit conversions. All powers are given in megawatts, so no conversions are needed: in the efficiency the unit cancels, and the waste heat comes out in megawatts. A decimal number is changed to a percentage by multiplying by one hundred.

STEP 5/7

Substitution. For electricity alone and for combined production separately:

$$
\begin{aligned}
\eta_1 &= \frac{P_\mathrm{e}}{P_1} \\
&= \frac{190\,\mathrm{MW}}{500\,\mathrm{MW}} \\
&= 0.380 \\
P_{2,1} &= P_1 - P_\mathrm{e} \\
&= 500\,\mathrm{MW} - 190\,\mathrm{MW} \\
&= 310\,\mathrm{MW}
\end{aligned}
$$

$$
\begin{aligned}
\eta_2 &= \frac{P_\mathrm{e} + P_\mathrm{dh}}{P_1} \\
&= \frac{150\,\mathrm{MW} + 290\,\mathrm{MW}}{500\,\mathrm{MW}} \\
&= 0.880 \\
P_{2,2} &= P_1 - (P_\mathrm{e} + P_\mathrm{dh}) \\
&= 500\,\mathrm{MW} - 440\,\mathrm{MW} \\
&= 60\,\mathrm{MW}
\end{aligned}
$$

STEP 6/7

Result. With electricity production alone the efficiency is $\eta_1 = 0.380 = 38.0\,\%$ and $310\,\mathrm{MW}$ of waste heat is transferred to the environment, most of it through the condenser to the sea. In combined production the overall efficiency is $\eta_2 = 0.880 = 88.0\,\%$ and there is only $60\,\mathrm{MW}$ of waste heat. The powers are given in whole megawatts, so the differences $310\,\mathrm{MW}$ and $60\,\mathrm{MW}$ are known to the nearest megawatt. The efficiencies are quotients, so they have as many figures as the least precise power: $190\,\mathrm{MW}$, $150\,\mathrm{MW}$, $290\,\mathrm{MW}$ and $500\,\mathrm{MW}$ all have three figures, so the efficiencies are given with three figures.

STEP 7/7

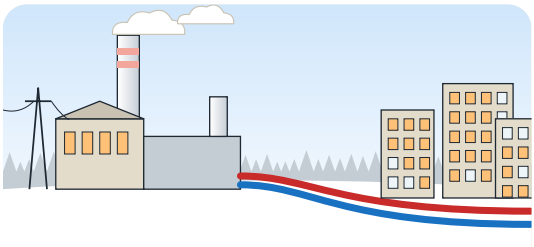
Reasonableness. Energy is conserved: $190\,\mathrm{MW} + 310\,\mathrm{MW} = 500\,\mathrm{MW}$ and $150\,\mathrm{MW} + 290\,\mathrm{MW} + 60\,\mathrm{MW} = 500\,\mathrm{MW}$. The efficiency of electricity production stays clearly below $100\,\%$, because a heat engine must give up part of the heat to something colder. Even the overall efficiency is not $100\,\%$, because part of the heat leaves with the flue gases and other losses. Less electricity is obtained in combined production because the steam is condensed at a higher temperature so that its heat is suitable for district heating. The energy of the fuel is still put to much better use.

FY3 · CHAPTER 14

Energy production and the efficiency of power plants

Why does a thermal power plant need about 2.5 times as much fuel energy as the electrical energy it produces, and how could the rest of the energy be put to use?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-15.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the efficiency of an energy production chain as the product of the efficiencies of its steps.
- identify the biggest energy loss of a condensing power plant.
- explain why the overall efficiency rises in combined heat and power production but less electricity is produced.

PRIOR KNOWLEDGE

- Efficiency $\eta = W/Q_1$ and the energy balance of a heat engine $Q_1 = W + Q_2$
  chapter *Work done by a gas and the heat engine*
- Quantity of heat $Q = cm\,\Delta T$ and specific heat capacity
  chapter *Heating and cooling of a substance:* $Q = cm\Delta T$
- Power is energy per unit time, and energy is conserved
  chapter *Work and energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

The fuel power of a condensing power plant is $500\ \mathrm{MW}$, but only $200\ \mathrm{MW}$ goes to the electricity grid. Energy is still conserved: it cannot disappear anywhere. At the same time the cooling water leaving the condenser of the plant is clearly warmer than the sea water coming in.

> **In which step of the energy chain does the most energy of the fuel end up as something other than electricity, and could it be put to use?**

Theory

Key idea

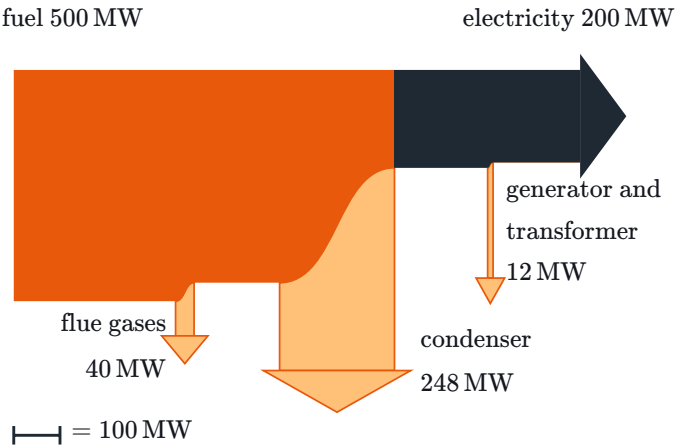
A power plant does not produce energy but converts it. The chemical energy of fuel, nuclear energy, the kinetic energy of flowing water or wind and the radiation of the Sun turn into electricity step by step, and this series of steps is called an energy chain. Energy is conserved in every step, but part of it is transferred to the surroundings as heat and is no longer available as electricity. Efficiency tells what fraction of the power coming into a step comes out in a useful form.



The turbine and generator of the waste-to-energy unit of the Iru power plant near Tallinn, Estonia. The turbine turns the energy of steam into the kinetic energy of rotation, and the generator turns that into electrical energy.

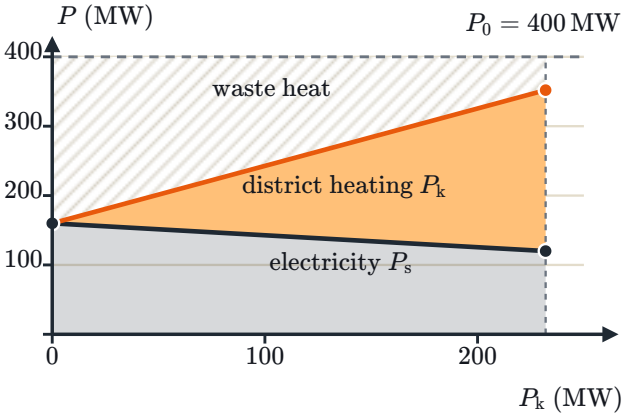
Photo: Lauri Veerde, CC BY-SA 4.0

In a condensing power plant the chain is fuel, boiler, turbine and condenser, generator and transformer, electricity grid. In the boiler the energy of the fuel is transferred to steam, and part of it leaves with the flue gases. In the turbine the steam expands and turns the generator, and in the condenser the steam condenses back into water. About half of the energy of the fuel is transferred to the sea through the condenser. The figure shows an example plant whose band widths are drawn to scale.



The biggest loss is not mainly caused by friction or poor technology. Most of it is the waste heat of the steam cycle, which a heat engine cannot avoid completely: part of the heat must be released to a colder environment, as was seen in the chapter Work done by a gas and the heat engine. A hydropower plant, a wind turbine and a solar panel are not heat engines, so they do not have this compulsory loss: the efficiency of a hydropower plant is about $90\ \%$, of a wind turbine $40–45\ \%$ and of a solar panel about $20\ \%$. However, the denominator of these numbers is the power of flowing water, wind or radiation and not bought fuel, so they cannot be compared directly with the efficiency of a combustion plant.

In combined heat and power production the heat of the condenser is led into the district heating network. For the water to warm up enough, the steam is condensed at a higher temperature, and less electricity is obtained from the turbine. The graph shows an example plant with a fuel power of $400\ \mathrm{MW}$: as the district heating power grows, the electric power decreases a little, but the waste heat decreases much more. In the graph the vertical distance across a layer is power, and the total height of the three layers is always $400\ \mathrm{MW}$.



The overall efficiency rises in combined production to as much as 85–90 %, but electricity and heat are not equally valuable. Electricity can be used for anything, and it can be transmitted hundreds of kilometres. District heating is good only for heating, and its network reaches only a town. That is why the ratio of electricity to heat in a combined heat and power plant is chosen according to the need for heat: in summer little heat is needed and the plant produces almost only electricity, in winter in frosty weather a lot of heat is needed.



The Nesjavellir geothermal power plant in Iceland produces electricity and district heat. The cloud rising from the plant is condensed water vapour that carries waste heat to the surroundings.

Photo: Michal Klajban, CC BY-SA 4.0

Efficiency of a power plant

The efficiency of one step or of the whole plant is the ratio of the useful power to the power brought in, and the power of waste heat is the part that is not put to use:

$$\eta = \frac{P_{\text{useful}}}{P_0}$$

$$P_{\text{h}} = P_0 - P_{\text{useful}}$$

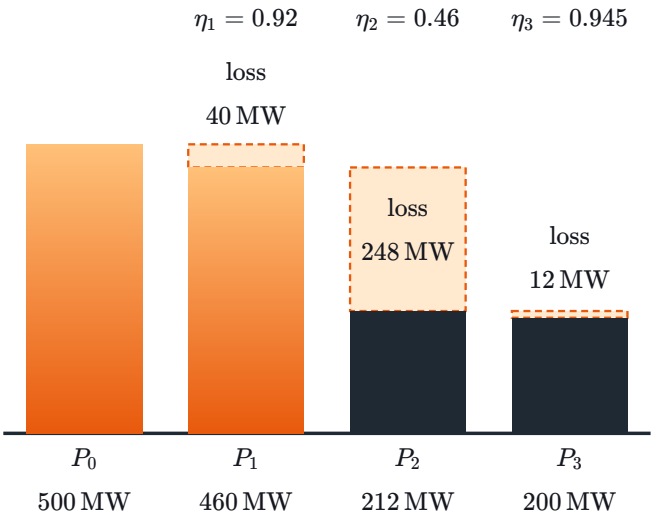
where η is the efficiency (often as a percentage), P_{useful} is the useful power obtained (W), P_0 is the power brought in, for example the fuel power (W), and P_{h} is the power of waste heat (W). In a power plant the steps are in series. The output power of each step is the product of its efficiency and the power coming in:

$$P_1 = \eta_1 P_0$$
$$P_2 = \eta_2 P_1$$
$$P_3 = \eta_3 P_2$$

Substituting the upper equation into the lower one gives the efficiency of the whole chain:

$$\eta = \frac{P_3}{P_0} = \eta_1 \, \eta_2 \, \eta_3$$

where η_1, η_2 and η_3 are the efficiencies of the steps (boiler, steam cycle, generator and transformer) and P_3 is the electric power. The efficiencies are multiplied, not added. In a real power plant every factor is less than 1, so the efficiency of the chain is less than that of the weakest step. In the figure the bars P_1, P_2 and P_3 are obtained from the previous bar by multiplying it by the efficiency of the step, η_1, η_2 or η_3.



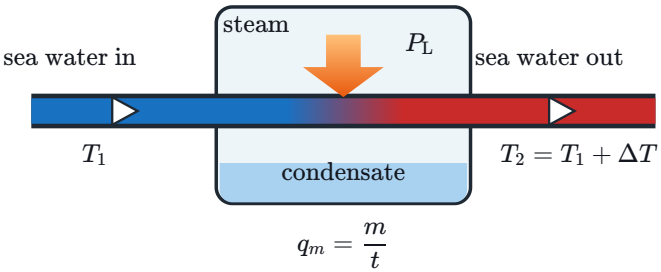
In a combined heat and power plant both electricity and district heating are obtained as useful output, so the overall efficiency is

$$\eta_{\text{tot}} = \frac{P_{\text{s}} + P_{\text{k}}}{P_0}$$

where P_{s} is the electric power (W) and P_{k} is the district heating power (W). The heat of the condenser warms the cooling water. If a mass m of water flows in a time t and its temperature rises by ΔT, then with the quantity of heat $Q = cm\,\Delta T$ the heat power of the condenser is

$$P_{\text{L}} = c\, q_m\, \Delta T$$

where P_{L} is the heat power transferred from the condenser to the water (W), c is the specific heat capacity of water ($4190\ \text{J}/(\text{kg}\cdot\text{K})$), $q_m = m/t$ is the water flow (kg/s) and ΔT is the rise in the temperature of the water (K). The heat power of the condenser is most of the waste heat but not all of the waste heat P_{h}: part of the waste heat leaves with the flue gases and as the losses of the generator and transformer.



Common misconception

Many people think that waste heat is energy that has been lost. Energy does not disappear, however: it is transferred, for example, to the cooling water or the surrounding air.

If heat is transferred to the surroundings at a temperature only slightly above that of the surroundings, it is hard to obtain much mechanical work or electricity from it. The smaller the temperature difference, the lower the maximum possible efficiency of a heat engine.

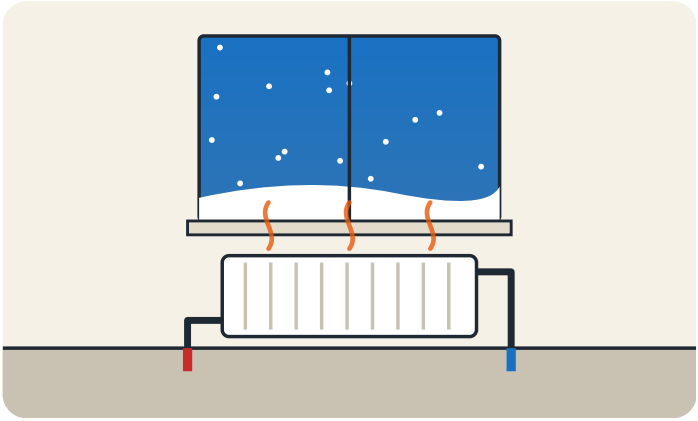
In combined heat and power production, part of the energy that would otherwise be left as waste heat can be used as district heating.

Another mistake is to add the efficiencies of the steps or to take their mean. When the efficiencies of three steps are 0.92, 0.46 and 0.945, the efficiency of the chain is their product 0.40, not the mean 0.78. A third mistake is to take the overall efficiency of $88\ \%$ of combined production as a sign that all the energy would be equally valuable: electricity can be used for everything, district heating only for heating.

Summary

- A power plant converts energy step by step in an energy chain, and in each step part of it is transferred to the surroundings as heat.
- Efficiency is $\eta = P_\mathrm{useful}/P_0$, and the power of waste heat is $P_\mathrm{h} = P_0 - P_\mathrm{useful}$.
- The efficiency of the chain is the product of the efficiencies of the steps, $\eta = \eta_1\,\eta_2\,\eta_3$, and it is smaller than that of the weakest step.
- The biggest loss of a condensing power plant is the heat transferred to the condenser of the steam cycle, which a heat engine cannot avoid completely. The efficiency of electricity production is about $40\ \%$ (typically $38\text{–}42\ \%$).

- In combined heat and power production the heat of the condenser is led into the district heating network, and the overall efficiency is $\eta_{\text{tot}} = (P_{\text{s}} + P_{\text{k}})/P_0$, about 85–90 %. Then less electricity is obtained.
- A hydropower plant, a wind turbine and a solar panel are not heat engines, but the denominator of their efficiencies is the power of flowing water, wind or radiation, not bought fuel.
- The cooling water of the condenser receives the heat power $P_{\text{L}} = c\, q_m\, \Delta T$, which is most of the waste heat.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Electric power of a condensing plant

The fuel power of a condensing power plant is $600\ \text{MW}$. The efficiency of the boiler is 0.910, of the steam cycle 0.440 and of the generator and transformer 0.950. What is the electric power of the plant, what is the efficiency of the whole plant, and how much power is transferred to the surroundings as waste heat?

STEP 1/7

Given and find. The fuel power is $P_0 = 600\,\mathrm{MW}$ and the efficiencies of the steps are $\eta_1 = 0.910$ (boiler), $\eta_2 = 0.440$ (steam cycle) and $\eta_3 = 0.950$ (generator and transformer). We are asked for the electric power P_3, the efficiency of the whole chain η and the power of waste heat P_h.

STEP 2/7

Principle. The steps are in series: the output power of each step is the input power of the next, and each step lets through only its own efficiency's share of the power coming into it. The plant runs steadily and energy is conserved, so the fuel power is either electricity or waste transferred to the surroundings as heat: $P_0 = P_3 + P_\mathrm{h}$.

STEP 3/7

Rearranged formula. The efficiencies are multiplied, and the waste heat is the part that does not turn into electricity:

$$\begin{aligned} P_3 &= \eta_1\,\eta_2\,\eta_3\,P_0 \\ \eta &= \frac{P_3}{P_0} \\ P_\mathrm{h} &= P_0 - P_3 \end{aligned}$$

STEP 4/7

Unit conversions. The efficiencies are dimensionless numbers and all powers are in megawatts, so no conversions are needed. Only the efficiency η is also given as a percentage at the end by multiplying it by a hundred.

STEP 5/7

Substitution. The product of the efficiencies is $0.910 \cdot 0.440 \cdot 0.950 = 0.38038$. The electric power is

$$\begin{aligned} P_3 &= \eta_1\,\eta_2\,\eta_3\,P_0 \\ &= 0.38038 \cdot 600\,\mathrm{MW} \\ &= 228.228\,\mathrm{MW} \end{aligned}$$

From this the efficiency is

$$\eta = \frac{228.228\,\mathrm{MW}}{600\,\mathrm{MW}} = 0.38038$$

and the waste heat is

$$P_\mathrm{h} = 600\,\mathrm{MW} - 228.228\,\mathrm{MW} = 371.772\,\mathrm{MW}$$

STEP 6/7

Result. The electric power is $P_3 \approx 228\,\mathrm{MW}$, the efficiency $\eta \approx 0.380 = 38.0\,\%$ and the power of waste heat $P_\mathrm{h} \approx 372\,\mathrm{MW}$. The results are given to three significant figures, because the efficiencies 0.910, 0.440 and 0.950 are given with three figures and the fuel power $600\,\mathrm{MW}$ to the nearest megawatt.

STEP 7/7

Reasonableness. Estimate: $0.9 \cdot 0.45 \cdot 0.95 \cdot 600\,\mathrm{MW} \approx 231\,\mathrm{MW}$, which matches the result. The efficiency is smaller than that of the weakest step, 0.440, as a product should be, and fits the usual efficiency of a condensing power plant of about $40\,\%$ (typically 38–$42\,\%$). The balance works out: the electricity $228\,\mathrm{MW}$ and the waste $372\,\mathrm{MW}$ add up to $600\,\mathrm{MW}$. The power transferred to the condenser is

$$P_1 - P_2 = 546\,\mathrm{MW} - 240.24\,\mathrm{MW} \approx 306\,\mathrm{MW}$$

which is over half of the energy of the fuel.

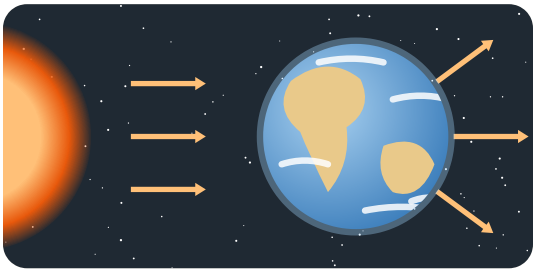
The loss of the boiler is only $54\,\mathrm{MW}$ and that of the generator and transformer $12\,\mathrm{MW}$, so improvements should be aimed at the steam cycle and after that at the use of the heat of the condenser.

FY3 · CHAPTER 15

Heat transfer, energy balance and climate change

Why is the mean temperature of the Earth about $+15\,^{\circ}\mathrm{C}$ and not below freezing, even though the Earth radiates as much energy into space as it absorbs from the Sun?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-12.html`



IN THIS LESSON, YOU WILL LEARN TO

- recognise the ways heat is transferred: conduction, convection and thermal radiation.
- calculate the thermal power conducted through a building part.
- use energy balance to explain the greenhouse effect and climate change.

PRIOR KNOWLEDGE

- Heat and work change internal energy, $\Delta U = Q + W$, and energy is conserved
  chapter *Internal energy, heat and conservation of energy*
- Heat is the transfer of energy from hotter to colder because of a temperature difference
  chapter *Internal energy, heat and conservation of energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

The Moon is on average as far from the Sun as the Earth, and the same amount of sunlight hits each square metre of its surface. However, the Moon has no atmosphere, and the mean temperature of its surface is well below freezing. The same difference can be seen on a small scale on Earth: on a clear winter night the frost gets harsher quickly, but on a cloudy night the temperature falls only a little.

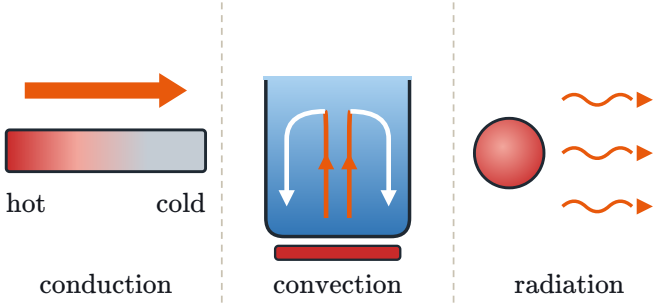
> **The atmosphere does not produce energy itself. How can it still keep the ground warmer than it would be without it?**

Theory

Key idea

Because of a temperature difference, energy transfers by itself, as a net amount, from a warmer object to a colder one. Energy that transfers in this way is called heat, and it transfers in three ways. In conduction, energy passes through matter from one particle to the next, as faster vibrating particles nudge their slower neighbours. Metals conduct heat well, because in them free electrons also carry energy, and they carry most of it; air and wool conduct it poorly. In convection, a warm liquid or gas flows and carries energy with it: warm air rises above a radiator, because it has expanded and is lighter than the surrounding air.

In thermal radiation, an object emits electromagnetic radiation, which needs no medium. This is how the Sun's energy reaches the Earth through empty space, and how the glow of a fireplace warms your face. All objects radiate, and the hotter an object is, the more radiation it emits and the shorter the wavelengths of that radiation are. Thermal radiation therefore leaves both objects, so the colder object also radiates towards the warmer one. The warmer one radiates more, however, so the net transfer of energy is to the colder object.



Heat losses of a building

A large part of the heating energy of a house is used because heat is conducted out through the walls, roof, windows and floor. The heat insulation of a building part is described by its U-value, or thermal transmittance, which tells you how many watts of thermal power pass through one square metre when the difference between the indoor and outdoor temperature is one kelvin. The smaller the U-value, the better the structure insulates. The U-value of a modern wall is about $0.17\,\mathrm{W/(m^2 \cdot K)}$, but even a good window has about $1.0\,\mathrm{W/(m^2 \cdot K)}$, so a small

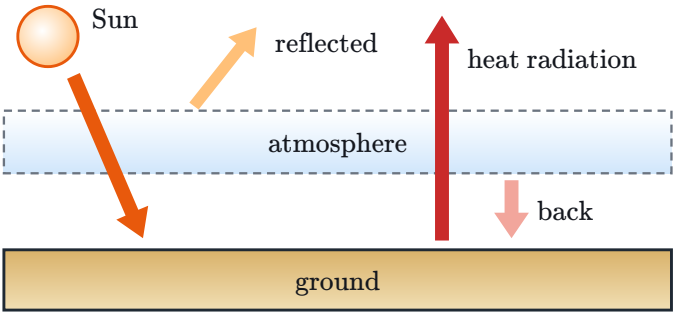
window lets through as much heat as many times its area of wall. Good insulation saves energy and reduces the emissions from heating.

Energy balance

The temperature of an object stays constant when the power coming into it is equal to the power leaving it. This is called energy balance. The Earth receives energy as radiation from the Sun, and about $30\ \%$ of it is reflected straight back into space by clouds, ice and light surfaces. The rest is absorbed, and the Earth radiates the same power away as long-wavelength thermal radiation. Without an atmosphere, the balance would settle at a mean temperature of about $-18\ ^\circ\mathrm{C}$.

Greenhouse effect

Water vapour, carbon dioxide and methane in the atmosphere let sunlight mostly through, although about a fifth of it is absorbed in the atmosphere. They absorb the thermal radiation emitted by the ground and radiate part of it back down. The ground therefore receives more energy than from the Sun alone, and the balance settles at about $+15\ ^\circ\mathrm{C}$. This is called the greenhouse effect, and it answers the question at the start.



Climate change

Human activity, mainly the use of fossil fuels, has raised the carbon dioxide concentration of the atmosphere from about 280 parts per million to over 420 parts per million. When greenhouse gases are added, for a while less energy escapes into space than arrives from the Sun. Today the difference is about $1\ \mathrm{W}$ for every square metre of the Earth. The extra energy warms the oceans, the land and the air until a new balance settles at a higher temperature. This is called climate change. A power difference that looks small is huge over the whole area of the

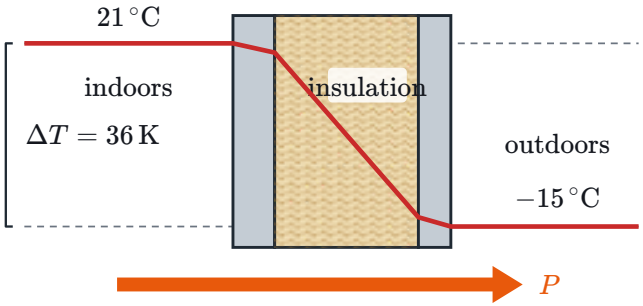
Earth, and most of it is stored in the oceans, whose large heat capacity slows the warming down but also keeps it going for a long time. Energy choices such as insulation, heat pumps and zero-emission power plants are therefore also ethical choices that affect future generations.

Thermal power and U-value

The thermal power conducted through a building part is

$$P = UA\Delta T$$

where P is the thermal power (W), U is the U-value of the building part (W/(m²·K)), A is the area (m²) and ΔT is the difference between the indoor and outdoor temperature (K). The formula holds when the temperatures on both sides of the building part stay constant. In energy balance the power coming into an object and the power leaving it are equal, and the temperature stays constant.



Common misconception

Many people mix up the greenhouse effect and ozone depletion, although they are different things: ozone depletion lets more ultraviolet radiation through, but it hardly warms the Earth. Another mistake is to think that the greenhouse effect is entirely harmful. The natural greenhouse effect keeps the Earth habitable, and the problem is its strengthening because of human emissions. A third mistake is to believe that a woollen sweater produces heat. It only slows down the conduction and convection of heat away from the body, just as the insulation in a wall slows down the escape of heat from a house.

Summary

- Heat transfers by conduction, convection and thermal radiation, and because of a temperature difference the net transfer of energy is from the warmer object to the colder one. All objects emit thermal radiation, the colder one too.

- The thermal power conducted through a building part is $P = UA\Delta T$. A small U-value means good heat insulation.
- In energy balance the power coming in and the power going out are equal, and the temperature stays constant.
- Greenhouse gases absorb the thermal radiation of the ground and radiate part of it back, which raises the equilibrium temperature of the ground from about $-18\,^{\circ}\mathrm{C}$ to about $+15\,^{\circ}\mathrm{C}$.
- An increase in greenhouse gases breaks the balance for a while, and the Earth warms until a new balance settles at a higher temperature.



Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Heat loss of a house

The walls of a detached house have an area of 120 m² and a U-value of $0.17\,\mathrm{W/(m^2\cdot K)}$, and the windows have an area of 20.0 m² and a U-value of $1.0\,\mathrm{W/(m^2\cdot K)}$. It is $21\,^{\circ}\mathrm{C}$ indoors and $-15\,^{\circ}\mathrm{C}$ outdoors. How large is the thermal power conducted through the walls and windows, and how much energy does it transfer in one day?

STEP 1/7

Given and find. Walls: $A_\mathrm{s} = 120\,\mathrm{m}^2$, $U_\mathrm{s} = 0.17\,\frac{\mathrm{W}}{\mathrm{m}^2\cdot\mathrm{K}}$. Windows: $A_\mathrm{i} = 20.0\,\mathrm{m}^2$, $U_\mathrm{i} = 1.0\,\frac{\mathrm{W}}{\mathrm{m}^2\cdot\mathrm{K}}$. Temperatures $T_\mathrm{in} = 21\,^\circ\mathrm{C}$ and $T_\mathrm{out} = -15\,^\circ\mathrm{C}$. You are asked for the thermal power P and the energy E in one day.

STEP 2/7

Principle. Heat is conducted through a structure from warm to cold, and when the temperatures stay constant, the power is proportional to the area and the temperature difference: $P = UA\Delta T$. The walls and windows conduct heat in parallel, so their powers are added. The power is constant, so the energy is the power multiplied by the time.

STEP 3/7

Rearranged formula. First solve the power and then the energy:

$$P = (U_\mathrm{s}A_\mathrm{s} + U_\mathrm{i}A_\mathrm{i})\,\Delta T$$

$$E = Pt$$

STEP 4/7

Unit conversions. The temperature difference is the same in kelvins and in degrees Celsius:

$$\begin{aligned}\Delta T &= 21\,^\circ\mathrm{C} - (-15\,^\circ\mathrm{C})\\ &= 36\,\mathrm{K}\end{aligned}$$

The time is $t = 24\,\mathrm{h}$, and the power is converted to kilowatts so that the energy comes out in kilowatt-hours.

STEP 5/7

Substitution. Calculate the power and then the energy:

$$\begin{aligned} P &= (0.17 \cdot 120 \\ &\quad + 1.0 \cdot 20.0)\,\tfrac{\mathrm{W}}{\mathrm{K}} \cdot 36\,\mathrm{K} \\ &= (20.4 + 20.0)\,\tfrac{\mathrm{W}}{\mathrm{K}} \cdot 36\,\mathrm{K} \\ &= 1454.4\,\mathrm{W} \\ E &= 1.4544\,\mathrm{kW} \cdot 24\,\mathrm{h} \\ &= 34.9\,\mathrm{kWh} \end{aligned}$$

STEP 6/7

Result. The power conducted through the walls and windows is $P \approx 1.5\,\mathrm{kW}$, and in one day it transfers the energy $E \approx 35\,\mathrm{kWh}$. The result is given to two significant figures. The limiting data are the U-values 0.17 and 1.0 and the temperature difference $36\,\mathrm{K}$ (a difference of whole degrees), all with two figures; the areas $120\,\mathrm{m^2}$ and $20.0\,\mathrm{m^2}$ are more precise.

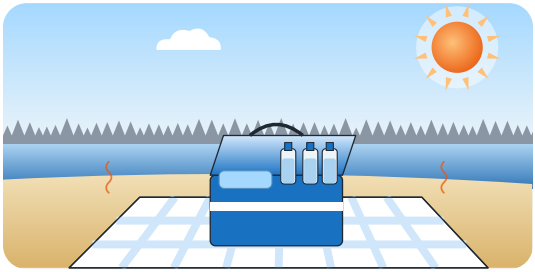
STEP 7/7

Reasonableness. The area of the windows is only a sixth of the area of the walls, but the same amount of heat goes through them ($20.0\,\mathrm{W/K}$ and $20.4\,\mathrm{W/K}$), because their U-value is almost six times as large. With rounded numbers, $40\,\mathrm{W/K} \cdot 36\,\mathrm{K} \approx 1.4\,\mathrm{kW}$ and $1.4\,\mathrm{kW} \cdot 24\,\mathrm{h} \approx 34\,\mathrm{kWh}$. The heating of the house replaces this power, and the roof, floor and ventilation add to the need.

Applied lesson: heat phenomena in everyday life

Why does the ice pack in a cool bag keep drinks cold for hours in hot weather, but once the ice has melted, the drinks warm up quickly?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy3-16.html`



IN THIS LESSON, YOU WILL LEARN TO

- combine heat, latent heats of fusion and vaporisation, and conducted power into one energy balance.
- justify whether the incoming energy is used for heating or for a change of state.
- solve multi-step problems about everyday heat phenomena.

PRIOR KNOWLEDGE

- The quantity of heat when heating is $Q = cm\Delta T$
  chapter *Heating and cooling of matter:* $Q = cm\Delta T$

- Melting and vaporisation need energy (freezing and condensation release it), but the temperature does not change, $Q = sm$ and $Q = rm$
  chapter *Changes of state*
- The power conducted through a building part is $P = UA\Delta T$
  chapter *Heat transfer, energy balance and climate change*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

On a trip, a cool bag contains an ice pack and bottles of drinks. It is hot outside, and the walls of the bag let in a little heat all the time. For a few hours the drinks are ice cold. Then there is nothing left of the ice but water, and a couple of hours later the drinks are no longer cold.

> **At the start, the same power comes through the wall all the time. Why do the drinks stay cold for hours but then warm up quickly?**

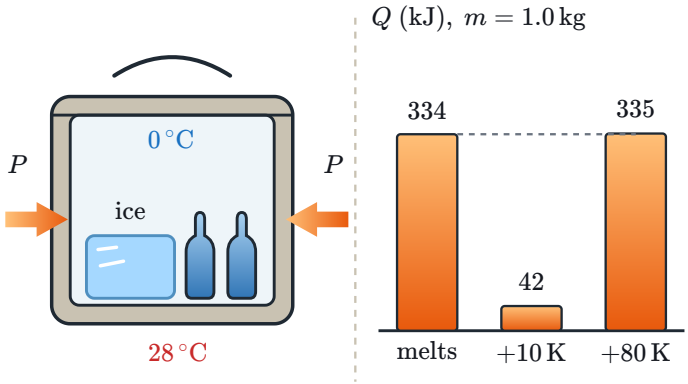
Theory

Key idea

Everyday heat phenomena, such as heating a sauna, a cool bag staying cold and cooling an engine, are solved with the same principle: energy bookkeeping.

Energy is conserved, so every joule needs a source and a destination. The source is a power, which means energy per second, or another object that gives out heat. There are two destinations. The energy raises the temperature, $Q = cm\Delta T$, or it changes the state of matter, $Q = sm$ or $Q = rm$. During a change of state the temperature does not change. When ice melts or water boils, all the energy goes into the change, and the temperature rises only after the change is over.

A power $P = UA\Delta T$ flows into the cool bag through its walls, even though the insulation is good. As long as there is ice left, this energy melts ice. When the ice and the meltwater are in thermal equilibrium with each other, the temperature of the ice–water mixture stays at about $0\,^\circ\mathrm{C}$. Elsewhere in the cool bag the temperature can be higher. Melting takes a lot of energy: melting one kilogram of ice takes $334\,\mathrm{kJ}$, the same as heating the same amount of water by 80 kelvins. When the ice has melted, the same power starts to raise the temperature of the water, and the drinks warm up. This is why it is worth putting ice in the bag and not just cold water.



A sauna stove heats stones with a specific heat capacity of about $800\,\mathrm{J/(kg\cdot K)}$. When you throw water on the hot stones, the water warms up and evaporates, and the energy is taken from the stones. Most of it goes into vaporisation: $0.20\,\mathrm{kg}$ of water warms from $20\,^\circ\mathrm{C}$ to $100\,^\circ\mathrm{C}$ with $67\,\mathrm{kJ}$ of energy, but vaporising it takes $452\,\mathrm{kJ}$. This is why the stones cool down at once and the stove has to heat them again. When the steam condenses on your skin, the same energy is released again, and this is why the steam feels hot.

A car's coolant carries the waste heat of the engine to the radiator. Power is energy per second, so the power transferred depends on the mass of liquid carried per second and on its rise in temperature in the engine. With the same power, a larger mass flow means a smaller rise in temperature. Water has a large specific heat capacity, $4190\,\mathrm{J/(kg\cdot K)}$, but it freezes in frost and boils easily, so glycol is mixed

with it. The mixture withstands frost and heat, but its specific heat capacity is lower, about $3500\,\mathrm{J/(kg\cdot K)}$.

Energy balance

When energy is transferred into a system with power P during a time t, the energy Pt is shared between the heating of the parts of the system and the changes of state:

$$\begin{aligned} Pt &= Q_1 + Q_2 + \dots \\ Q_{\text{heating}} &= c\,m\,\Delta T \\ Q_{\text{melting}} &= s\,m \\ Q_{\text{vaporisation}} &= r\,m \end{aligned}$$

where P is power (W), t is time (s), Q is quantity of heat (J), c is specific heat capacity (J/(kg·K)), m is mass (kg), ΔT is change in temperature (K), s is specific latent heat of fusion and r is specific latent heat of vaporisation (J/kg). For water $s = 334\,\mathrm{kJ/kg}$ and $r = 2260\,\mathrm{kJ/kg}$. You calculate each stage separately, and melting and vaporisation involve no change in temperature.

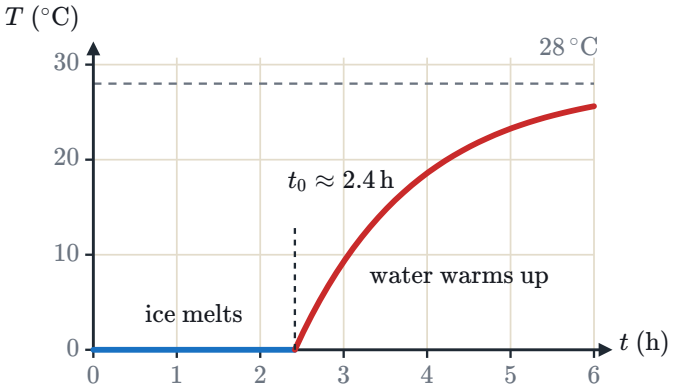
The power conducted through a building part, such as the wall of a cool bag, is

$$P = U A \Delta T$$

where U is the U-value (W/(m²·K)), A is the total area of the walls (m²) and ΔT is the difference between the outside and inside temperatures (K). In the model, the inside of the bag is taken to be at $0\,^\circ\mathrm{C}$ as long as there is ice, so ΔT is the outside temperature in degrees Celsius. Then all the energy coming through the wall melts ice, $U A \Delta T\, t = s m$, and the time the ice lasts is

$$t = \frac{s\,m}{U A \Delta T}$$

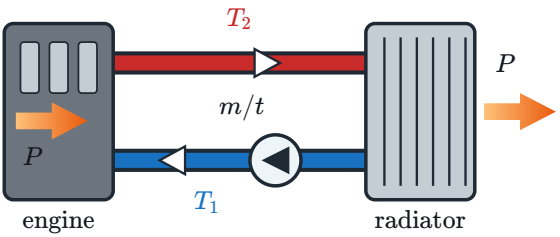
The graph shows an example of the inside temperature of a cool bag calculated with the model. The temperature stays at $0\,^\circ\mathrm{C}$ for a time t_0, until the ice has melted. After that the water warms up towards the outside temperature. The rise is steep at first and slows down, because the power $U A \Delta T$ becomes smaller as the temperature difference becomes smaller. This is the bag of the example; the bag in the Investigate step has a smaller $U A$.



In the cooling circuit of an engine, the liquid takes the heat power P from the engine, so the heating formula $Q = cm\Delta T$ is divided by the time t:

$$P = c\frac{m}{t}\Delta T$$

where m/t is the mass of liquid carried per second (kg/s) and $\Delta T = T_2 - T_1$ is the rise in the temperature of the liquid in the engine (K). In steady running the radiator gives the same power P out to the air, so the temperatures T_1 and T_2 of the liquid stay constant.



$$P = c\frac{m}{t}\Delta T$$

When two objects or substances are in an insulated container, the quantity of heat given out by the hot one is equal to the quantity of heat taken in by the cold one:

$$Q_{\text{released}} = Q_{\text{absorbed}}$$

The problem goes like this:

1. Define the system and decide where the energy comes from and where it goes.
2. Write the energy of the source: Pt or $cm\Delta T$ of the substance that gives out heat.
3. Go through the stages in order of temperature: warming up to the melting point, melting, warming up. During melting the temperature is constant.
4. Check whether the energy is enough: compare the energy available with what the next change of state needs. If it is not enough, the change stays unfinished and the temperature is the melting point. If it is just enough, the change finishes and the temperature is exactly the melting point.
5. Check the units and the order of magnitude.

Common misconception

Many people think that ice produces cold that flows into the drinks. Cold does not flow: because of the temperature difference, energy moves on net from the warmer object to the colder one, and ice cools things because it takes in energy. Another mistake is to assume that the temperature rises steadily all the time. During melting the temperature stays constant, even though energy keeps coming in.

A third mistake in a mixing problem is to assume that the ice always melts completely and that the end result is an average. When $1.0\ \mathrm{kg}$ of ice and $1.5\ \mathrm{kg}$ of water at 20 °C are mixed, the water can give out at most

$$\begin{aligned} Q_{\text{water}} &= c\,m\,\Delta T \\ &= 4190\ \tfrac{\mathrm{J}}{\mathrm{kg\cdot K}} \cdot 1.5\,\mathrm{kg} \cdot 20\,\mathrm{K} \\ &\approx 126\,\mathrm{kJ} \end{aligned}$$

but melting all the ice takes $334\,\mathrm{kJ}$. The ice melts only partly, and the temperature of the mixture is $0\,^{\circ}\mathrm{C}$.

Summary

- An everyday heat phenomenon is solved with energy bookkeeping: the incoming energy (Pt or the quantity of heat given out by another object) is used for heating, $Q = cm\Delta T$, or for a change of state, $Q = sm$ or $Q = rm$.
- During a change of state the temperature stays constant. Melting ice takes as much energy as heating the same amount of water by 80 kelvins.
- The power coming through the wall of a cool bag is $P = UA\Delta T$, and the time the ice lasts is $t = sm/(UA\Delta T)$.

- The power transferred by coolant is $P = c\,(m/t)\,\Delta T$. A lower specific heat capacity needs a larger mass flow to transfer the same power.
- In a mixing problem, first check whether the energy given out is enough to melt all the ice. When the ice is at $0\,^{\circ}\mathrm{C}$ at the start and the energy is not enough, the temperature is $0\,^{\circ}\mathrm{C}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Ice pack in a cool bag

On a trip, a cool bag contains a $1.0\,\mathrm{kg}$ ice pack at $0\,^{\circ}\mathrm{C}$ and $0.50\,\mathrm{kg}$ of drinks in bottles at $20\,^{\circ}\mathrm{C}$. The total area of the walls of the bag is $0.80\,\mathrm{m}^2$ and the U-value is $1.5\,\mathrm{W/(m^2\cdot K)}$, and it is $28\,^{\circ}\mathrm{C}$ outside. Assume that the drinks cool to $0\,^{\circ}\mathrm{C}$ at once, melting ice, and that the inside stays at $0\,^{\circ}\mathrm{C}$ until the ice has melted. How long does the ice last? The specific heat capacity of the drinks is the value for water.

STEP 1/7

Given and find. The mass of the ice is $m_{\mathrm{ice}} = 1.0\,\mathrm{kg}$, the mass of the drinks is $m_{\mathrm{j}} = 0.50\,\mathrm{kg}$ and their drop in temperature is $\Delta T_{\mathrm{j}} = 20\,\mathrm{K}$. For the walls $UA = 1.5\,\frac{\mathrm{W}}{\mathrm{m^2K}}\cdot 0.80\,\mathrm{m}^2$ and the temperature difference from outside to inside is $\Delta T_{\mathrm{u}} = 28\,\mathrm{K}$. From the table $c = 4190\,\frac{\mathrm{J}}{\mathrm{kg\cdot K}}$ and $s = 334\,\frac{\mathrm{kJ}}{\mathrm{kg}}$. You are asked for the time t after which the ice has run out.

STEP 2/7

Principle. Energy is conserved: the energy $s\,m_{\mathrm{ice}}$ needed to melt the ice comes from two sources, the cooling drinks and the heat coming through the wall. While there is ice, the inside stays at $0\,^{\circ}\mathrm{C}$, so the temperature difference ΔT_{u} and the power through the wall $P = U A \Delta T_{\mathrm{u}}$ stay constant, and the energy from the wall is $P\,t$. The ice melts but stays at $0\,^{\circ}\mathrm{C}$, so melting takes $s\,m_{\mathrm{ice}}$ and nothing goes into raising the temperature.

STEP 3/7

Rearranged formula. The cooling drinks give part of the energy $s\,m_{\mathrm{ice}}$ needed to melt the ice, $c\,m_{\mathrm{j}}\Delta T_{\mathrm{j}}$. The rest comes through the wall, $U A \Delta T_{\mathrm{u}}\,t$:

$$U A \Delta T_{\mathrm{u}}\,t = s\,m_{\mathrm{ice}} - c\,m_{\mathrm{j}}\Delta T_{\mathrm{j}}$$

$$t = \frac{s\,m_{\mathrm{ice}} - c\,m_{\mathrm{j}}\Delta T_{\mathrm{j}}}{U A \Delta T_{\mathrm{u}}}$$

STEP 4/7

Unit conversions. Changes in temperature are the same in kelvins and in degrees Celsius: the drinks cool by $20\,^{\circ}\mathrm{C} - 0\,^{\circ}\mathrm{C} = 20\,\mathrm{K}$ and the difference across the wall is $28\,^{\circ}\mathrm{C} - 0\,^{\circ}\mathrm{C} = 28\,\mathrm{K}$. The latent heat of fusion is converted to joules per kilogram:

$$\begin{aligned} s &= 334\ \tfrac{\mathrm{kJ}}{\mathrm{kg}} \\ &= 334\,000\ \tfrac{\mathrm{J}}{\mathrm{kg}} \end{aligned}$$

This way the energies can be calculated in joules and the power in watts. Finally the time is converted from seconds to hours: $1\,\mathrm{h} = 3600\,\mathrm{s}$.

STEP 5/7

Substitution. First calculate the energy given out by the drinks and the power through the wall:

$$\begin{aligned} & c\,m_{\mathrm{j}}\Delta T_{\mathrm{j}} \\ &= 4190\,\tfrac{\mathrm{J}}{\mathrm{kg\cdot K}} \cdot 0.50\,\mathrm{kg} \cdot 20\,\mathrm{K} \\ &= 41\,900\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} P &= U A \Delta T_{\mathrm{u}} \\ &= 1.5\,\tfrac{\mathrm{W}}{\mathrm{m^2 K}} \cdot 0.80\,\mathrm{m^2} \cdot 28\,\mathrm{K} \\ &= 33.6\,\mathrm{W} \end{aligned}$$

STEP 6/7

Result. Melting the ice takes $s\,m_{\mathrm{ice}} = 334\,000\,\mathrm{J}$. Substitute into the formula:

$$\begin{aligned} t &= \frac{s\,m_{\mathrm{ice}} - c\,m_{\mathrm{j}}\Delta T_{\mathrm{j}}}{U A \Delta T_{\mathrm{u}}} \\ &= \frac{334\,000\,\mathrm{J} - 41\,900\,\mathrm{J}}{33.6\,\mathrm{W}} \\ &= \frac{292\,100\,\mathrm{J}}{33.6\,\mathrm{W}} \\ &\approx 8.7 \cdot 10^{3}\,\mathrm{s} \\ &\approx 2.4\,\mathrm{h} \end{aligned}$$

The ice lasts about $2.4\,\mathrm{h}$. The result is given with two significant figures, because all the initial values have two figures: the mass of the ice $1.0\,\mathrm{kg}$, the mass of the drinks $0.50\,\mathrm{kg}$, their drop in temperature $20\,\mathrm{K}$, the U-value 1.5, the area $0.80\,\mathrm{m^2}$ and the temperature difference $28\,\mathrm{K}$.

STEP 7/7

Reasonableness. Without the warm drinks the ice would last

$$\begin{aligned} \tfrac{334\,000\,\mathrm{J}}{33.6\,\mathrm{W}} &\approx 9940\,\mathrm{s} \\ &\approx 2.8\,\mathrm{h} \end{aligned}$$

so the result is moderately shorter. Cooling the drinks melts a share

$$\frac{41\,900\,\mathrm{J}}{334\,000\,\mathrm{J}} \approx 0.13$$
$$= 13\,\%$$

of the ice, and the time becomes shorter by the same share: $2.8\,\mathrm{h} \cdot 0.87 \approx 2.4\,\mathrm{h}$.

When the ice has melted, the bag contains $1.0\,\mathrm{kg}$ of meltwater and $0.50\,\mathrm{kg}$ of drinks. Assuming that the specific heat capacity of the drinks is the same as that of water, the total heat capacity of the contents is

$$\begin{aligned} C &= (m_{\mathrm{ice}} + m_{\mathrm{j}})\,c \\ &= 1.50\,\mathrm{kg} \cdot 4190\ \tfrac{\mathrm{J}}{\mathrm{kg \cdot K}} \\ &\approx 6285\ \tfrac{\mathrm{J}}{\mathrm{K}} \end{aligned}$$

When heat flows into the bag at a power of $33.6\,\mathrm{W}$, the model says the temperature of the contents starts to rise by about $19\,\mathrm{K}$ per hour:

$$\begin{aligned} \frac{\Delta T}{\Delta t} &= \frac{P}{C} \\ &= \frac{33.6\,\mathrm{W}}{6285\ \tfrac{\mathrm{J}}{\mathrm{K}}} \\ &\approx 0.00535\ \tfrac{\mathrm{K}}{\mathrm{s}} \\ &= 0.00535 \cdot 3600\ \tfrac{\mathrm{K}}{\mathrm{h}} \\ &\approx 19\ \tfrac{\mathrm{K}}{\mathrm{h}} \end{aligned}$$

This estimate holds at the start of the temperature rise and assumes that the heat flow stays constant.

Theory and worked examples from Ruukkilabra

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